

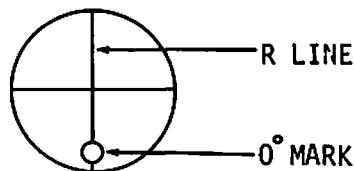
|                        |      |
|------------------------|------|
| SL-1/2                 |      |
| CSM<br>RENDEZVOUS BOOK |      |
| PART NO.               | S/N  |
| SKC32100150-802        | 1001 |

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NOMINAL M=5

NOTES  
1.) BACKUP GDC ALIGN STAR PAIR  
IS VISIBLE FROM ~~SS+6 MIN TO~~  
~~SR~~ ~~SS TO SR~~



DEFINITIONS  
H = AVERAGE ALTITUDE  
θ<sub>p</sub> = ORDEAL FOAI PITCH  
TA = TRUNNION ANGLE

### NC1 TARGETING PAD

| NOMINAL       |     |   | NOMINAL UPDATE |   |         | PRELAUNCH UPDATE |   |  |
|---------------|-----|---|----------------|---|---------|------------------|---|--|
| N95           | HR  | + | +              | + | 0 0 2   | +                | + |  |
| TIG NC1       | MIN | + | +              | + | 2 3     | +                | + |  |
|               | SEC | + | +              | + | 3 3 0 0 | +                | + |  |
| N57 HALF REVS |     |   | +              | + | 0 3     |                  |   |  |
| N37           | HR  | + | +              | + | 0 0 7   | +                | + |  |
| TIG TPI       | MIN | + | +              | + | 0 2     | +                | + |  |
|               | SEC | + | +              | + | 5 3 0 0 | +                | + |  |

### TALIGN PAD

| NOMINAL |     |   | NOMINAL UPDATE |   |         | PRELAUNCH UPDATE |   |  |
|---------|-----|---|----------------|---|---------|------------------|---|--|
| N34     | HR  | + | +              | + | 0 0 6   | +                | + |  |
| GET     | MIN | + | +              | + | 5 4     | +                | + |  |
| ALIGN   | SEC | + | +              | + | 4 5 0 0 | +                | + |  |

### BACKUP GDC ALIGN PAD

| NOMINAL    |   |  | NOMINAL UPDATE |   |         | UPDATE |   |  |
|------------|---|--|----------------|---|---------|--------|---|--|
| 0°/R STARS |   |  | 2 5            | / | 3 4     |        | / |  |
| ASCP tw    | R |  | +              | + | 2 0 1 8 | +      | + |  |
| (RNDZ      | P |  | +              | + | 3 5 8 1 | +      | + |  |
| REFSMAT)   | Y |  | +              | + | 3 5 9 5 | +      | + |  |

### STAR ACQUISITION PAD

| NOMINAL      |    |   | NOMINAL UPDATE |   |           | UPDATE |   |     |
|--------------|----|---|----------------|---|-----------|--------|---|-----|
| N22 (NC1)    | R  | + | +              | + | 1 8 0 0 0 | +      | + | 0 0 |
|              | P  | + | +              | + | 0 0 9 0 0 | +      | + | 0 0 |
| (RNDZ        | Y  | + | +              | + | 0 0 2 0 0 | +      | + | 0 0 |
| REFSMAT)     |    |   |                |   |           |        |   |     |
| N71 1ST STAR |    |   | +              | + | 2 5       | +      | + |     |
| TPAC         | SA |   | +              | + | 2 2 7 9 0 | +      | + | 0   |
| ANGLES       | TA |   | +              | + | 0 2 9 4 0 | +      | + | 0   |
| N71 2ND STAR |    |   | +              | + | 3 3       | +      | + |     |
| TPAC         | SA |   | +              | + | 0 7 4 6 0 | +      | + | 0   |
| ANGLES       | TA |   | +              | + | 0 2 5 7 0 | +      | + | 0   |
| N71 3RD STAR |    |   | +              | + | 3 4       | +      | + |     |

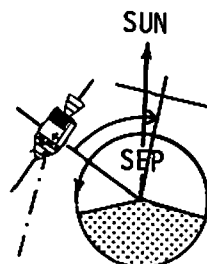
DATE 5/21/73  
5/22/73

0+00

SL-2 LIFT OFF

0+10

SL-2 INSERTION



SL-2 SEPARATION (180,0/317,0)

- ✓ COMPLETE LAUNCH C/L THRU STEP 2, Pg L/3-1
- ✓ VERIFY PRELAUNCH NC1 TARGETING PAD AND
- ✓ ALIGN PAD COPIED (Pg 1-a)

0+20

- ✓ BACKUP GDC ALIGN PAD (COPY Pg 1-a)
- ✓ LAUNCH/RNDZ REALIGN PAD (COPY Pg 5-a)
- ✓ PERFORM OPTICS DUST COVER
- ✓ JETTISON CHECKLIST (Pg 1-1)
- ✓ V48E (LOAD 2 JET, 0.5°/SEC): V62E 3
- ✓ V49E (TWEAK MNVR IF REQ TO 180,163,8)
- (LAUNCH REFSMMAT)

UNSTOW &amp; MOUNT ORDEAL BOX (U3)

(11002)  
(01111)

0+30

## PLT SYSTEMS CHECKS

- ✓ MAIN REG CHECK
- ✓ SEC RAD LEAK CHECK
- ✓ RCS POST INSERTION CONFIG
- ✓ RCS MONITORING CHECK
- ✓ RCS MONITORING CHECK
- ✓ C/W OPERATIONAL CHECK
- ✓ PG BURGE CHECK
- ✓ EPS MONITORING CHECK
- ✓ RCS MONITORING CHECK
- ✓ SECONDARY GLYCOL LOOP CHECK
- ✓ SPS MONITORING CHECK
- ✓ EXTEND DOCKING PROBE

## OPTICS DUST COVER JETTISON CHECKLIST

- ✓ INSTALL OPTICS EYEPIECES (LEB)
- ✓ OPTICS PMR-ON(up)
- ✓ OPT ZERO-OFF, THEN ZERO (15 SEC)
- ✓ OPT ZERO-OFF
- ✓ OPT MODE-MAN
- ✓ OPT COUPLING CONT-DIRECT
- ✓ OPT SPEED CONT-HI
- ✓ OHC-MAX RIGHT (OBS EJECT THRU SCT)
- (SXT 40°, SCT 150° SHAFT ANGLE)

## PS2 (OPTION 3)

|               |     |   |     |    |    |
|---------------|-----|---|-----|----|----|
| N71 1ST STAR  | ×   | 0 | 0   | 0  | 25 |
| N71 2ND STAR  | ×   | 0 | 0   | 0  | 33 |
| N05(R1) } ERR | ×   |   |     |    |    |
| N93           | X   | - | 000 | 10 |    |
| GYRO          | Y   | + | 000 | 71 |    |
| TORQUING      | Z   | + | 000 | 56 |    |
| ANGLES        |     |   |     |    |    |
|               | HR  | + | 0   | 0  | 59 |
| TIME OF       | MIN | + | 0   | 0  | 05 |
| GYRO          |     |   |     |    |    |
| TORQUE        | SEC | + | 0   | 00 | 00 |

NOMINAL M=5

DATE 5/21/73  
5/22/73



NOMINAL M=5

DATE 5/21/73  
5/22/73

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(01111)

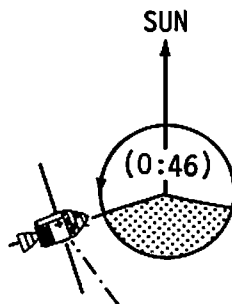
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✓P52 (OPT 3)(RECORD Pg 1-1)  
✓REF STAR ACQ PAD STARS (Pg 1-a)

✓P52 (OPT 2; COARSE ALIGN) (RECORD)  
✓REF TALIGN PAD (Pg 1-a)  
(NOMINAL N22 = 180,9,2)

✓START SCS DRIFT CHECK (BMAG 2)  
✓ALIGN GDC  
✓RESET AND START DET COUNTING UP  
✓PERFORM COAS LOS DETERMINATION  
CHECKLIST (Pg 1-2)

BORESIGHT STAR  
NUNKI-37  
(180, 26, 347)

37

P52 (OPTION 2)

|                      |     |   |   |   |    |
|----------------------|-----|---|---|---|----|
| N71 1ST STAR         | X   | 0 | 0 | 0 | 25 |
| N71 2ND STAR         | X   | 0 | 0 | 0 | 33 |
| N05(R1) & ERR        | X   |   |   |   |    |
| N93                  | X   | + |   |   | 66 |
| GYRO TORQUING ANGLES | Y   | + |   |   | 75 |
|                      | Z   | + |   |   | 59 |
| TIME OF GYRO TORQUE  | HR  | + | 0 | 0 | 0  |
|                      | MIN | + | 0 | 0 | 0  |
|                      | SEC | + | 0 |   |    |

COAS LOS DETERMINATION CHECKLIST

- ✓COAS PHR-ON(up)  
✓SWR TO BORESIGHT ATTITUDE  
✓CHC MODE-FREE  
V37E52E  
F 04 06 R1 00001  
R2 00003  
PRO  
F 50 25 00015  
ENTER  
F 01 70 000DE STAR CODE  
LOAD BORESIGHT STAR CODE  
✓OPT MODE-CHC(verify)  
✓OPT ZERO-OFF  
PRO  
06 92 SHAFT,TRUN
- BORESIGHT ON STAR AND  
MARK with VERB key  
RECORD SHAFT,TRUN  
(NOMINAL = 0°, +57.47°)  
TO VERIFY: KEY RLSE  
AND REPEAT STEP 2
- V37E00E  
CHC MODE-AUTO  
OPT MODE-MAI  
OHC-Drive trun<10°  
OPT ZERO-ZERO

COAS LOS DETERMINATION

|     |    |   |   |   |   |   |   |
|-----|----|---|---|---|---|---|---|
| N92 | SA | + | 3 | 5 | 9 | 5 | 3 |
|     | TA | + | 5 | 7 | 3 | 3 | 5 |

+00001 2 359.53  
+57329 57.340

-100.0 98.8 +1.2 13:30

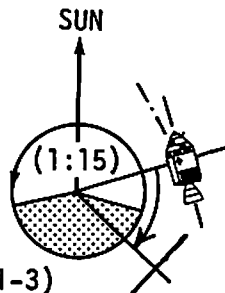
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2.5

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(11002)  
(01111)PERFORM WASTE WATER DUMP  
CHECKLIST

- ✓ P00
- ✓ PERFORM EMS ΔV TEST & NULL
- ✓ BIAS CHECK CHECKLIST (Pg 1-3)

- ✓ P31; BYPASS MINKEY OPT & AUTO MNVR
- ✓ LOAD NC1 TARGETING PAD (Pg 1-a)
- (N57 REG 1 LOAD +3)
- \*\*LOAD +4 IF NC1 AT 1ST PERIGEE\*\*
- ✓ P31 (FINAL COMP: RECORD)
- ✓ V56E

- ✓ P20 (OPT 5)
- (N78 = +0, +9000, +18000)
- (N78 EARTH N70 = 47)
- ✓ **EMERGENCY CABIN PRESSURE SEL-OFF**
- PERFORM STOWAGE CHECKLIST (Pg 1-3)
- DIR O<sub>2</sub> - OPEN (until cabin press 5.7 psia)
- Side hatch dump vlv - open (until
- cabin press 4.0 psia)
- DIR O<sub>2</sub> - OPEN (until cabin press 5.7 psia)
- ✓ **EMERGENCY CABIN PRESSURE SEL-BOTH**

PERFORM HOUSEKEEPING CHECKLIST (Pg 1-3)  
(IN PREPARATION FOR UNSTOWING CAMERA EQUIP. (Pg 1-6))

- ✓ END SCS DRIFT CHECK (BMAG 2); PERFORM
- ✓ GDC/IMU COMPARISON CHECKLIST (Pg 1-3)
- \*\*\*\*\*IF BMAG 2 > 10°/HR/AXIS\*\*\*\*\*
- \* START SCS DRIFT CHECK (BMAG 1) \*
- \* ALIGN GDC \*
- \* BMAG MODE(3)-RATE 1 \*
- \* RESET AND START DET COUNTING UP \*
- \*\*\*\*\*

+30:00

0:00

1+30

## WASTE WATER DUMP CHECKLIST

BAT VENT VLV-CLOSED (252)  
H2O QTY IND-WASTE (PANEL 2)  
WATER CONT PRESS REL VLV-DUMP A (352)  
MONITOR WASTE H2O QTY IND-DECREASING  
(5 % PER MIN)  
WHEN WASTE H2O QTY IND-15 %  
WATER CONT PRESS REL VLV-2  
BAT VENT VLV-VENT  
POT TK IN VLV-CLOSED (VERIFY) (352)

## P31 FINAL COMP

|     |                   |   |   |   |   |   |   |
|-----|-------------------|---|---|---|---|---|---|
| N84 | ΔV <sub>NC2</sub> | + | 0 | 0 | 4 | 6 | 8 |
|     | ΔH <sub>NC2</sub> | + | 0 | 0 | 2 | 9 | 2 |
|     | ΔV <sub>NCC</sub> | + | 0 | 0 | 3 | 0 | 9 |
| N81 | ΔV <sub>X</sub>   | + | 0 | 2 | 1 | 4 | 0 |
|     | ΔV <sub>Y</sub>   | + | 0 | 0 | 0 | 0 | 0 |
|     | ΔV <sub>Z</sub>   | + | 0 | 0 | 0 | 0 | 0 |

## HOUSEKEEPING CHECKLIST

- ✓ FOLD CENTER COUCH UP
- ✓ REMOVE 3 UTILITY POLE RODS, STOW
- ON RIGHT WIRE RUN
- ✓ STOW MSFC RODS ON LEFT WIRE RUN
- CUT LOOSE MSFC BASE PLATE & TEMP
- STOW
- CUT OFF SSC SAIL, STOW NEAR SOMAS
- ✓ STOW CENTER COUCH UNDER LEFT
- COUCH, STRAPS IN RS
- CLEAN OFF A1, A5 ETC., FOR
- ACCESS TO STOWAGE LOCKERS
- ✓ STOW CLOTHES LINE UNDER STOWED
- COUCH
- CUT LOOSE MSFC SAIL, STOW
- BELOW CDR'S FEET
- CUT LOOSE T027 STOW UNDERNEATH
- PLTS COUCH

## GDC/IMU COMPARISON CHECKLIST

V16H20E  
FDAI SELECT-1  
FDAI SOURCE-ATT SET  
ATT SET-GDC  
ZERO FDAI 1 err needles with ASCP tw  
Key VERB when ZERO  
RECORD H20 values  
RECORD ASCP tw values  
RECORD DET  
FDAI SELECT-1/2

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
X TOTAL ATTITUDE DIFFERENCE GREATER X  
X THAN 5 DEG (10 DEG/HR) PER AXIS X  
X IS UNACCEPTABLE. X  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

## EMS ΔV TEST &amp; NULL BIAS CHECK CHECKLIST

- ✓ EMS MODE-STBY
- ✓ EMS FUNC-ΔV SET/VHF RING
- ✓ SET ΔV ind to 1586.8 fps
- ✓ EMS MODE-NORMAL
- ✓ EMS FUNC-ΔV TEST
- ✓ SPS THRUST Lt-on/off (10 sec)
- ✓ ΔV ind stops at -0.1 to -41.5
- ✓ EMS MODE-STBY
- ✓ EMS FUNC-ΔV SET/VHF RING
- ✓ SET ΔV ind to -100.0 fps
- ✓ EMS MODE-FREE
- ✓ EMS FUNC-ΔV (wait 5 sec)
- ✓ EMS MODE-NORMAL (for 1 min 40 sec)
- ✓ EMS MODE-STBY
- ✓ If ΔV < 1 fps, do not bias
- ✓ If ΔV > 1 fps but < 10 fps,
- bias if desired
- ✓ If ΔV > 10 fps, EMS is HD-GO
- ✓ CMC MODE-AUTO

## PGA DOFFING AND STOWAGE CHECKLIST

- ✓ SNAP 2 TSB's (U1) ON AFT BKH & UEB
- ✓ STOW LIFE VESTS (F1)
- ✓ STOW WRIST DAMS & NECK DAMS (F2)
- ✓ STOW C/L POCKETS (F2) & SUNGLASSES
- ✓ STOW SCISSORS, PROB'S, SUIT POCKET EQUIP.,
- ✓ WATCHES & BANDS (TOP POCKET-TSB)
- ✓ STOW COMFORT GLOVES & WRISTLETS (ACCESSORY BAGS)
- ✓ UNSTOW 02 UMBILICAL SCREEN CAPS (B2)
- ✓ CONFIGURE GCU & 02 UMBILICALS (PER 5/1-7, STEP 6)
- ✓ STOW GCU (ACCESSORY BAGS)
- ✓ DOFF PGA's (REF EVA C/L (R1) sect. 3.2)
- ✓ INSTALL HELMETS & IV GLOVES (PGA BAG) ON PGA's
- ✓ CONNECT ELECTRICAL DUST COVERS (B2) TO PGA's
- ✓ STOW PGA's (UNDER COUCHES)
- ✓ DOFF UGTA's (WHEN FULL)
- ✓ PLACE UGTA CLAMPS (AG) ON UGTA's
- ✓ STOW UGTA's (TSO ON VED)
- ✓ DON UGTA's (AG)
- ✓ STOW OBS/DIODELTS (ACCESSORY BAGS)
- ✓ DON CHG HARNESSSES & LT-WT HEADSETS (U2)
- ✓ DON TROUSERS, JACKETS & BOOTS (U2)
- ✓ UNSTOW CUE CARDS (DATA CARD KIT-R1)

BMAG 2 GDC/IMU COMPARISON  
RESULTS

|            |    |   |   |   |   |   |         |
|------------|----|---|---|---|---|---|---------|
| H20        | R  | + | 1 | 8 | 0 | 4 | 4       |
|            |    | + | 2 | 1 | 3 | 6 | 3       |
|            |    | + | 3 | 5 | 9 | 9 | 7       |
| ASCP<br>tw | R  | X | 1 | 8 | 1 | 5 | X       |
|            |    | X | 2 | 1 | 5 | 5 | X       |
|            |    | X | 0 | 0 | 0 | 2 | X       |
| DET        | ΔT | X | 3 | 2 | 0 | 0 | (30:00) |
|            |    | X | 3 | 2 | 0 | 0 |         |

NOMINAL M=5

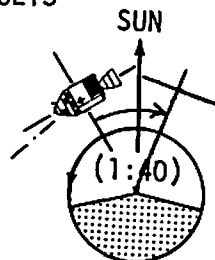
DATE 5/21/73  
5/22/73

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~~TO~~ ENABLE REFSMMAT DOWNLINK(AT STDN AOS):  
~~GMC~~ MODE-FREE  
~~V37E52E~~ (HOLD FOR 30 SECONDS)  
~~V37E20E~~  
~~GMC~~ MODE-AUTO  
~~NC1~~ PRELIMINARY PAD (COPY Pg 1-5)  
~~STAR~~ ACQ PAD (COPY Pg 1-a)  
~~TRANSMIT~~ GYRO TORQUE ANGLES  
~~LAND~~ TIME OF GYRO TORQUE (Pgs 1-1 & 1-2)  
~~TRANSMIT~~ EMS ΔV TEST RESULTS



*EMS RESULTS.  
GYRO DRIFT.*

1+40

N  
F  
D

1+50

M  
A  
D

STDN UPLINK (P27) [CSM & SWS SV'S, PIPA  
 BIAS UPDATE]  
 NC1 FINAL PAD (COPY Pg 1-5)  
 NC1 BURN ATT CHECK PAD (Copy Pg 1-5)  
 NPC ADVISORY (COPY) ⊗  
 V48E LOAD NC1 VALUES IF REQ'D  
 P00

\*\*\*\*\*IF BMAG2>10°/HR/AXIS\*\*\*\*\*  
 \* END SCS DRIFT CHECK (BMAG 1); PERFORM \*  
 \* GDC/IMU COMPARISON CHECKLIST (Pg 1-3)\*  
 \*\*\*\*\*

+30:00

2+00

WATER DUM 01+53  
01+57

#### NPC ADVISORY

| PERFORM TASK            | YES | NO |
|-------------------------|-----|----|
| STDN UPLINK Pg. 1-7     |     | ✓  |
| NPC PAD AND BURN Pg 1-7 |     | ✓  |

⊗ IF NO NPC - PREP LUNCH - FOOD AT SPOONS L3

#### BMAG 1 GDC/IMU COMPARISON RESULTS

|            |    |   |  |  |  |  |  |  |         |
|------------|----|---|--|--|--|--|--|--|---------|
| N20        | R  | + |  |  |  |  |  |  |         |
|            | P  | + |  |  |  |  |  |  |         |
|            | Y  | + |  |  |  |  |  |  |         |
| ASCP<br>tw | R  |   |  |  |  |  |  |  |         |
|            | P  |   |  |  |  |  |  |  |         |
|            | Y  |   |  |  |  |  |  |  |         |
| DET        | ΔT |   |  |  |  |  |  |  | (30:00) |

2+00

(11002)  
(01111)**P00**

V49E (MNVR TO STAR ACQ PAD ATT) (Pg 1-a)  
 PERFORM IMU CHECK  
 P52(AUTO OPTICS TO 1ST PAD STAR)  
 (IF STAR IN SXT FOV-IMU IS GO)

\*\*\*\*\*IF STAR NOT IN SXT FOV\*\*\*\*\*  
 \* PERFORM CHECK USING GDC, OHC, AND SCT\*  
 \*\*\*\*\*IF STAR>5° FROM CENTER OF SCT\*\*\*\*\*  
 \* USE 12° MARK ON WINDOW \*  
 \*\*\*\*\*

P31; BYPASS AUTO MNVR  
 LOAD N95 PAD VALUE  
 SET DET  
 P31 (FINAL COMP: RECORD)

2+10

-12:00

GO TO SPS BURN CUE CARD

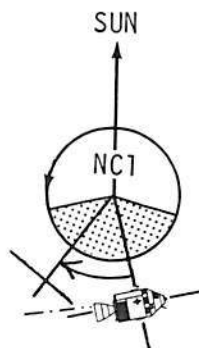
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**NC1** (180,0/9,2) (TRIM VGX  $\pm 0.2$ )  
 RECORD BURN STATUS  
 V56E  
 P52(OPT 3)(RECORD)(IF STARS NOT  
 AVAILABLE REF STAR ACQ PAD (Pg 1-a)

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## NC1 PAD DATA

|                  |              | NOMINAL     | PRELIMINARY | FINAL       |
|------------------|--------------|-------------|-------------|-------------|
| N95              | HR           | + 0 0 2     | + 0 0 2     | + 0 0 2     |
| TIG NC1          | MIN          | + 2 3       | + 2 3       | + 2 3       |
|                  | SEC          | + 3 3 0 0   | + 3 0 1 4   | + 3 7 3 5   |
| N81              | $\Delta V_X$ | + 2 1 5 5   | + 2 0 7 4   | + 2 0 6 3   |
| $\Delta V_{NC1}$ | $\Delta V_Y$ | + 0 0 0 0   | + 0 0 0 0   | + 0 0 0 0   |
|                  | $\Delta V_Z$ | + 0 0 0 0   | + 0 0 0 0   | + 0 0 0 0   |
| N22              | R            | + 1 8 0 0 0 | + 1 8 0 0 0 | + 1 8 0 0 0 |
| NC1              | P            | + 0 0 9 0 0 | + 0 0 9 0 0 | + 0 0 9 0 0 |
|                  | Y            | + 0 0 2 0 0 | + 0 0 2 0 0 | + 0 0 2 0 0 |
|                  | $\Delta V_C$ | + 2 0 0 8   | + 1 9 4 0   | + 1 9 3 0   |
| BT               |              | + 0 0 1 0   | + 0 0 0 9   | + 0 0 0 9   |

CSM WGT 30852 PTRIM 4107 YTRIM +0.35

## BURN ATT CHECK

|      |           |
|------|-----------|
| STAR | 3 3       |
| SA   | 7 3 6 7 0 |
| TA   | 0 . 0     |

2588

XXXXXXXXXXXXXXXXXXXXXXXXXXXX  
 X N81 COMP LIMITS  $\pm 0.5$  FPS X  
 X X  
 X CMC/STDN WITHIN LIMITS GO CMC X  
 X NO AGREEMENT GO STDN X  
 XXXXXXXXXXXXXXXXXXXXXXXXXXXX

## NC1 ONBOARD DATA

## P31 FINAL COMP

|                  |                  |             |
|------------------|------------------|-------------|
| N84              | $\Delta V_{NC2}$ | + 0 0 4 1 8 |
|                  | $\Delta H_{NC2}$ | + 0 0 3 3 6 |
|                  | $\Delta V_{NCC}$ | + 0 0 3 2 9 |
| N81              | $\Delta V_X$     | + 0 2 0 5 9 |
| $\Delta V_{NC1}$ | $\Delta V_Y$     | + 0 0 0 0 0 |
|                  | $\Delta V_Z$     | + 0 0 0 0 0 |

## SPS BURN STATUS

|              |     |   |     |  |       |
|--------------|-----|---|-----|--|-------|
| $\Delta TIG$ | 0   |   |     |  |       |
| AFTER TRIM   |     |   |     |  |       |
| $\Delta VC$  | -   |   |     |  | 1 5 1 |
| FDAI         | R   | + |     |  |       |
|              | P   | + |     |  |       |
|              | Y   | + |     |  |       |
| N85          | VGX | - | 0 0 |  | 2     |
|              | VGX | + | 0 0 |  | 2     |
|              | VGZ | + | 0 0 |  | 1     |

## P52 (OPTION 3)

|               |     |             |
|---------------|-----|-------------|
| N71 1ST STAR  |     | 0 0 0 2 5   |
| N71 2ND STAR  |     | 0 0 0 3 3   |
| N05(R1) } ERR |     | 0 0 0 0 0   |
| N93           | X   | + 0 0 0 0 9 |
| GYRO          | Y   | + 0 0 0 6 6 |
| TORQUING      | Z   | + 0 0 0 5 0 |
| ANGLES        |     |             |
| TIME OF       | HR  | + 0 0 0 2 7 |
| GYRO          | MIN | + 0 0 0 2 9 |
| TORQUE        | SEC | + 0 0 0 0 0 |

NOMINAL M=5

2+30

(11002)  
(01111)

A  
R  
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A

H2 PURGE LINE HTR-ON(up)

P20 (OPT 5), (N78=+0,+9000,+18000)(N70=47)

CONFIGURE CAMERAS(B3): SWS APPROACH

~~CM2/DAC10/18/CX01-BRKT (B8)~~

~~MIR (T8,1/250,∞)/1FPS (1MAG)~~

~~CM4/HDC01/CX04 (f8,1/250,∞)~~

FOR FLYAROUND/INSPECTION AND SEVA PHOTOS,

GET FOLLOWING AND PLACE IN TSB:

~~16MM MAG CX24 (A9)~~ **CX02 (B8)** SUN →

~~70MM MAG CX23 (B8)~~

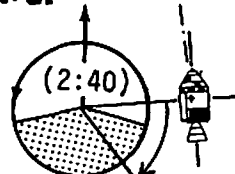
~~35MM CAM NK03 (A9)~~

~~300MM LENS (A2)~~

~~35MM CASSETTES CX21 (A9)~~

~~BW07(A9) → CX22 (A9)~~

~~LOAD NK03 WITH CX21~~



~~UNSTOW PORTABLE TIMER (B3)~~

~~UNSTOW TV CAMERA, ZOOM LENS, CABLES (2),~~

~~MONITOR, AND CAMERA MOUNT (A5)~~

~~ASSEMBLE AND MOUNT TV CAMERA IN~~

~~RH RNDZ WINDOW~~

~~RESTOW BAGS AND DUST CAPS (A5)~~

~~FUEL CELL PURGE (S/1-3)(20 MIN~~

~~AFTER LINE HTR-ON)~~

CONFIGURE NK WITH 300 mm LENS; STOW 55 mm LENS  
CX FILM-ASA 64  
BW FILM-ASA 80  
FOR 300 mm LENS - 1/250 OR FASTER TO PREVENT BLUR

2+40

2+50

3+00

~~H2 PURGE LINE HTR-OFF (10 MIN~~  
~~AFTER PURGE)~~

**3+30**

~~PERFORM TV PREP CHECKLIST (Pg 1-7)~~  
~~S-BAND AUX TV-TV (2 MIN PRIOR STDN AOS)~~

TRANSMIT NC1 N81 APPLIED AND  
SPS BURN STATUS (Pg 1-5)  
TRANSMIT GYRO TORQUE ANGLES  
AND TIME OF GYRO TORQUE (Pg 1-5)

```
*****IF NPC REQUIRED*****
* STDN UPLINK (P27) [CSM & SWS SV'S]*
* NPC PAD (COPY) *
*****
```

✓ NC2 PRELIMINARY PAD (COPY Pg 1-9)  
S BAND AUX TV-OFF (CTR) WHEN CHECKOUT COMPLETE

```
*****IF NPC REQUIRED*****
*
* V48E (LOAD CSM WGT & TRIMS, IF
* REQ'D)(LOAD 2°/SEC) (11003)
* ENTER P38 14 MIN PRIOR TO TIG
* BYPASS AUTO MNVR
* LOAD PAD VALUES IN N33 & N81
* SET DET
* GDC ALIGN BEFORE IMU TORQUE
*
*
* GO TO SPS BURN CUE CARD
*
*
* P40
***
```

\*\*\*\*\*  
\* GYRO COMP UPDATE \*  
\*\*\*\*\*

✓PNL 7 DIRECT 02 VLV-CLOSE(CW)  
✓STDN UPLINK (P27) [CSM & SWS SV'S]

|                  |              |   |   |   |   |   |   |   |   |
|------------------|--------------|---|---|---|---|---|---|---|---|
| N83              | HR           | + | X | X | X |   |   |   |   |
| TIG NPC          | MIN          | + | X | X | X |   |   |   |   |
|                  | SEC          | + | X | X | X |   |   |   |   |
| N81              | $\Delta V_X$ | + |   |   | 0 | 0 | 0 | 0 | 0 |
| $\Delta V_{NPC}$ | $\Delta V_Y$ |   |   |   |   |   |   |   |   |
|                  | $\Delta V_Z$ | + |   |   | 0 | 0 | 0 | 0 | 0 |
|                  | $\Delta V_C$ |   |   |   |   |   |   |   |   |
|                  | BT           |   | X | X | X |   |   |   |   |

CSM WGT \_\_\_\_\_ PTRIM \_\_\_\_\_ YTRIM \_\_\_\_\_

## TV PREPARATION CHECKLIST

TV SOURCE - CM  
REMOVE & STOW LENS CAP  
LINEAR SW - GAM 1/2  
AVG SW - AVG  
F-STOP - 22  
ZOOM - 25  
FOCUS - ∞  
MONITOR ON - ON  
REFINE SETTINGS AS REQ'D

**NOMINAL M=5**

NOMINAL M=5

3+30

(11002)  
(01111)

C  
Y  
I  
M  
A  
D

\*\*\*  
\* NPC (TRIM VGY +0.2) \*  
\* RECORD BURN STATUS \*  
\* TRANSMIT NPC N81 APPLIED AND \*  
\* SPS BURN STATUS \*  
\* GDC ALIGN AFTER MNVR AND \*  
\* IMU TORQUE \*  
\* V48E (LOAD 0.5°/SEC) (11002) \*  
\* P20 (OPT 5) \*  
\*\*\*\*\*

3+40

3+50

4+00

SPS BURN STATUS

|            |     |   |   |   |  |  |  |  |  |
|------------|-----|---|---|---|--|--|--|--|--|
| ΔTIG       |     |   |   |   |  |  |  |  |  |
| AFTER TRIM |     |   |   |   |  |  |  |  |  |
| ΔVC        |     |   |   |   |  |  |  |  |  |
| FDAI       | R   | + |   |   |  |  |  |  |  |
|            | P   | + |   |   |  |  |  |  |  |
|            | Y   | + |   |   |  |  |  |  |  |
| N85        | VGX |   | 0 | 0 |  |  |  |  |  |
|            | VGY |   | 0 | 0 |  |  |  |  |  |
|            | VGZ |   | 0 | 0 |  |  |  |  |  |

P52 (OPTION 3)

|                      |     |   |   |   |   |    |
|----------------------|-----|---|---|---|---|----|
| N71 1ST STAR         |     |   | 0 | 0 | 0 | 37 |
| N71 2ND STAR         |     |   | 0 | 0 | 0 | 45 |
| N05(R1) } ERR        |     |   | 0 | 0 | 0 | 0  |
| N93                  | X   | - |   |   |   |    |
| GYRO TORQUING ANGLES | Y   | + | 0 | 0 | 0 | 75 |
|                      | Z   | + | 0 | 0 | 0 | 33 |
|                      | HR  | + | 0 | 0 | 0 |    |
| TIME OF GYRO TORQUE  | MIN | + | 0 | 0 | 0 |    |
|                      | SEC | + | 0 |   |   |    |

P52 (OPTION 3)  
(FOR M=5)

|                      |     |   |   |   |   |  |
|----------------------|-----|---|---|---|---|--|
| N71 1ST STAR         |     |   | 0 | 0 | 0 |  |
| N71 2ND STAR         |     |   | 0 | 0 | 0 |  |
| N05(R1) } ERR        |     |   |   |   |   |  |
| N93                  | X   |   |   |   |   |  |
| GYRO TORQUING ANGLES | Y   |   |   |   |   |  |
|                      | Z   |   |   |   |   |  |
|                      | HR  | + | 0 | 0 | 0 |  |
| TIME OF GYRO TORQUE  | MIN | + | 0 | 0 | 0 |  |
|                      | SEC | + | 0 |   |   |  |

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5/22/73

PAGE 1-8



4+00

(11002)  
(01111)

P52 (OPT 3)(RECORD Pg 1-8)(IF STARS NOT  
AVAILABLE REF STAR ACQ PAD (Pg 1-a)

4+10

NC2 FINAL PAD (COPY)  
V48E IF REQ'D  
NCC PRELIMINARY PAD (COPY Pg 1-11)  
NSR PRELIMINARY PAD (COPY Pg 1-12)  
TRANSMIT GYRO TORQUE ANGLES AND  
TIME OF GYRO TORQUE (Pg 1-8)

P32; BYPASS AUTO MNVR  
LOAD N28 PAD VALUES,  
SET DET  
V32E (P32 RECYCLE: RECORD)

4+20

-12:00

4+30

P32 (FINAL COMP: RECORD)

R=300.00 NM  
R=-279.1 FPS

NOMINAL M=5

## NC2 PAD DATA

|                  |              | NOMINAL |   |   |   |           | PRELIMINARY |   |   |   |           | FINAL |   |   |   |           |
|------------------|--------------|---------|---|---|---|-----------|-------------|---|---|---|-----------|-------|---|---|---|-----------|
| N28              | HR           | +       | X | X | X | 0 0 4     | +           | X | X | X | 0 0 4     | +     | X | X | X | 0 0 4     |
| TIG NC2          | MIN          | +       | X | X | X | 4 1       | +           | X | X | X | 4 1       | +     | X | X | X | 4 1       |
|                  | SEC          | +       | X | X | X | 1 6 5 0   | +           | X | X | X | 1 9 2 7   | +     | X | X | X | 1 9 2 4   |
| N81              | $\Delta V_X$ | +       | X | X | X | 0 4 7 0   | +           | X | X | X | 0 4 3 7   | +     | X | X | X | 0 4 4 5   |
| $\Delta V_{NC2}$ | $\Delta V_Y$ | +       | X | X | X | 0 0 0 0   | +           | X | X | X | 0 0 0 0   | +     | X | X | X | 0 0 0 0   |
|                  | $\Delta V_Z$ | +       | X | X | X | 0 0 0 0   | +           | X | X | X | 0 0 0 0   | +     | X | X | X | 0 0 0 0   |
| N22              | R            | +       | X | X | X | 1 8 0 0 0 | +           | X | X | X | 1 8 0 0 0 | +     | X | X | X | 1 8 0 0 0 |
| NC2              | P            | +       | X | X | X | 1 8 9 0 0 | +           | X | X | X | 1 8 9 0 0 | +     | X | X | X | 1 8 9 0 0 |
|                  | Y            | +       | X | X | X | 0 0 1 0 0 | +           | X | X | X | 0 0 1 0 0 | +     | X | X | X | 0 0 1 0 0 |
|                  | $\Delta V_C$ | +       | X | X | X | 0 3 2 4   | +           | X | X | X | 0 3 0 3   | +     | X | X | X | 0 3 1 1   |
|                  | BT           | +       | X | X | X | 0 0 0 2   | +           | X | X | X | 0 0 0 2   | +     | X | X | X | 0 0 0 2   |

CSM WGT 50225 PTRIM +1.08 YTRIM +0.26

## SINGLE BANK

## NC2 ONBOARD DATA

|                  |                  | P32 RECYCLE |   |   |   |     | P32 FINAL COMP |   |   |   |         |
|------------------|------------------|-------------|---|---|---|-----|----------------|---|---|---|---------|
| N84              | $\Delta V_{NCC}$ | +           | 0 |   |   |     | +              | 0 |   |   | 38.7    |
|                  | $\Delta H_{NCC}$ | +           | 0 | 0 |   |     | +              | 0 | 0 |   | 20.0    |
|                  | $\Delta V_{NSR}$ | +           | 0 | 0 |   |     | +              | 0 | 0 |   | 12.4    |
| N81              | $\Delta V_X$     | +           | 0 |   |   |     | +              | 0 |   |   | 0 4 4 5 |
| $\Delta V_{NC2}$ | $\Delta V_Y$     | +           | 0 | 0 | 0 | 0 0 | +              | 0 | 0 | 0 | 0 0     |
|                  | $\Delta V_Z$     | +           | 0 | 0 | 0 | 0 0 | +              | 0 | 0 | 0 | 0 0     |

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
X N81 COMP LIMITS  $\pm 0.5$  FPS X  
X X  
X CMC/STON WITHIN LIMITS GO CMC X  
X NO AGREEMENT GO STON X  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXX



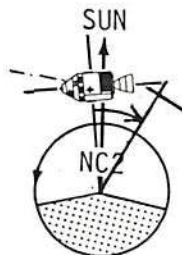
NOMINAL M=5

4+30

(11002)  
(01111)

GO TO SPS BURN CUE CARD

P40



4+40

0:00

**NC2** (180,0/189,1) (TRIM VGX  $\pm 0.2$ )  
RECORD BURN STATUS  
TRANSMIT NC2 N81 APPLIED (Pg 1-9) AND  
SPS BURN STATUS  
P33  
SET DET

-40:00

4+50



XXXXXXXXXXXXXXXXXXXXX  
X UNREASONABLE UPDATE X  
X 1.00 NM, 6.0 FPS X  
XXXXXXXXXXXXXXXXXXXXX

5+00

SPS BURN STATUS

| AFTER TRIM   |     |                                     |                                     |                                     |                                     |
|--------------|-----|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| $\Delta TIG$ |     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |                                     |
| $\Delta VC$  | -   | <input checked="" type="checkbox"/> |                                     | 1.10                                |                                     |
| FDAI         | R   | +                                   |                                     |                                     | <input checked="" type="checkbox"/> |
|              | P   | +                                   |                                     |                                     | <input checked="" type="checkbox"/> |
|              | Y   | +                                   |                                     |                                     | <input checked="" type="checkbox"/> |
| N85          | VGX | +                                   | 0                                   | 0                                   | 1.6                                 |
|              | VGY |                                     | 0                                   | 0                                   |                                     |
|              | VGZ |                                     | 0                                   | 0                                   | -2                                  |

DATE 5/21/73  
5/22/73

PAGE 1-10

5+00

(11002)  
(01111)S  
X  
TC  
Y  
I
 XXXXXXXXXXXXXXXXXXXX  
 X UNREASONABLE UPDATE X  
 X 1.00 NM, 6.0 FPS X  
 XXXXXXXXXXXXXXXXXXXX

5+10

A  
C  
N
 NCC FINAL PAD (COPY)  
 V48E IF REQ'D  
 NSR FINAL PAD (COPY Pg 1-12)

 DO NOT OVERWRITE  
 NCC TIG (N11)  
 UNLESS ADVISED  
 BY STON

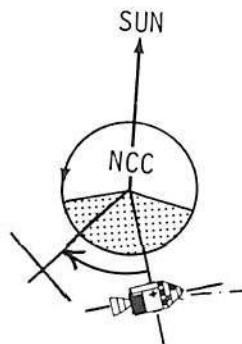
-12:00

29

P33 (FINAL COMP: RECORD)

GO TO SPS BURN CUE CARD

P40



5+20

0:00

 (180,2/11,359) (TRIM VG's  $\pm 0.2$ )  
 RECORD BURN STATUS

5+30

 TRY P34; PERFORM VHF SWITCH LIST (Pg 1-11)  
 ACQUIRE VHF RANGING (R=133.00NM)

NOMINAL M=5

## NCC PAD DATA

|                  |              | NOMINAL     | PRELIMINARY | FINAL       |
|------------------|--------------|-------------|-------------|-------------|
| N11              | HR           | + 0 0 5     | + 0 0 5     | + 0 0 5     |
| TIG              | NCC MIN      | + 2 7       | + 2 7       | + 2 7       |
|                  | SEC          | + 2 6.5 0   | + 26.1 0    | + 26.3 0    |
| N81              | $\Delta V_X$ | + 0 2 9.6   | + 0 38.7    | + 0 36.3    |
| $\Delta V_{NCC}$ | $\Delta V_Y$ | + 0 0 0.0   | + 0 0 7.3   | + 0 0 6.5   |
|                  | $\Delta V_Z$ | + 0 0 0.9   | + 0 0 1.0   | + 0 1 5.0   |
| N22              | R            | + 1 8 0.0 0 | + 1 8 0.0 0 | + 1 7 6.0 0 |
| NCC              | P            | + 0 1 1.0 0 | + 0 0 7.0 0 | + 0 3 1.0 0 |
|                  | Y            | + 3 5 9.0 0 | + 0 1 2.0 0 | + 0 1 1.0 0 |
|                  | $\Delta V_C$ | + 0 1 6.1   | + 0 2 6.6   | + 0 2 6.4   |

 CSM WGT 30090 PTRIM +1.08 YTRIM +0.24

## NCC ONBOARD DATA

 XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
 X N81 COM'P LIMITS  $\pm 1.5, \pm 5.0, \pm 15.0$  FPS X  
 X CMC/CHART WITHIN LIMITS GO CMC X  
 X IF NOT CMC/STON WITHIN LIMITS GO CMC X  
 X IF NOT CHART/STON WITHIN LIMITS GO STON X  
 X NO AGREEMENT-NORMALLY GO STON X  
 XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

## P33 FINAL COMP

|                  |              |             |
|------------------|--------------|-------------|
| N82              | $\Delta V_X$ | + 0 2 1.1   |
| $\Delta V_{NSR}$ | $\Delta V_Y$ | + 0 4.3     |
|                  | $\Delta V_Z$ | - 0 1 5.2   |
| N81              | $\Delta V_X$ | + 0 0 3 7.5 |
| $\Delta V_{NCC}$ | $\Delta V_Y$ | + 0 0 0 6.0 |
|                  | $\Delta V_Z$ | - 0 0 1 3.2 |

## VHF SWITCH LIST

 VHF AM A-OFF(ctr)  
 VHF AM B-DUPLEX  
 VHF RANGING-ON(up)  
 VHF ANTENNA-RIGHT  
 E'MS FUNC- $\Delta V$  SET/VHF RNG  
 EMS MODE-BACKUP/VHF RNG  
 VHF RNG-RESET  
 PANEL 6,9,10:  
 VHF AN-RCV

## SPS BURN STATUS

|              |     |   |   |   |  |
|--------------|-----|---|---|---|--|
| ATIG         |     |   |   |   |  |
|              |     |   |   |   |  |
|              |     |   |   |   |  |
| $\Delta V_C$ |     |   |   |   |  |
| FDAI         | R   |   |   |   |  |
|              | P   |   |   |   |  |
|              | Y   |   |   |   |  |
| N85          | VGX | 0 | 0 | 0 |  |
|              | VGX | 0 | 0 | 0 |  |
|              | VGZ | 0 | 0 | 0 |  |

NOMINAL M=5

117.08

DATE 5/21/73  
5/22/73

NSR PAD DATA

TIG NSR(N13)=TIG NCC(N11) + 00:37:00= 00:04:26.30

|                  |              | NOMINAL     | PRELIMINARY | FINAL       |
|------------------|--------------|-------------|-------------|-------------|
| NB1              | $\Delta V_X$ | + 0 1 8 8   | + 0 1 6 7   | + 0 2 1 0   |
| $\Delta V_{NSR}$ | $\Delta V_Y$ | + 0 0 0 0   | + 0 0 4 6   | + 0 0 4 3   |
|                  | $\Delta V_Z$ | + 0 0 0 0   | - 0 0 4 2   | - 0 1 7 5   |
| N22              | R            | + 1 8 0 0 0 | + 1 7 7 0 0 | + 1 7 3 0 0 |
| NSR              | P            | + 2 2 6 0 0 | + 2 3 9 0 0 | + 2 6 5 0 0 |
|                  | Y            | + 0 0 0 0 0 | + 0 1 6 0 0 | + 0 1 0 0 0 |
|                  | $\Delta V_C$ | 0 0 4 8     | 0 0 5 1     | 0 1 4 2     |

29970 +1.08 +1.1 TAIL -12.7  
-10.22 NSR ONBOARD DATA

|                  |                     | P34 RECYCLE | P34 FINAL COMP |
|------------------|---------------------|-------------|----------------|
| N75              | $\Delta H_{NSR}$    | + 0 0 9.8   | + 0 0 10.1     |
|                  | $\Delta T(NSR/TPI)$ | + 0 0 B 16  | + 5 7 B 53     |
|                  | $\Delta T(TPI/TPI)$ | + 6 1 B 46  | - B 37         |
| NB1              | $\Delta V_X$        | + 0 20.9    | + 0 21.3       |
| $\Delta V_{NSR}$ | $\Delta V_Y$        | + 0 3.9     | + 0 4.4        |
|                  | $\Delta V_Z$        | - 0 14.7    | - 0 16.5       |

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
X NB1 COMP LIMITS NA,NA,NA FPS X  
X USE P34 FINAL COMP  $\Delta V_Y$  FOR CMC SOLN X  
X OR X  
X BURN P34 RECYCLE, STOP OR CHART SOLN X  
X IF NCC WAS BURNED CMC, STOP X  
X OR CHART, RESPECTIVELY OR X  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

| NSR CHART SOLN |   |   |   |   |   |
|----------------|---|---|---|---|---|
| $\Delta V_X$   | X |   |   |   |   |
| $\Delta V_Y$   | * | S | T | D | H |
| $\Delta V_Z$   | X |   |   |   |   |

V32E (P34 RECYCLE: RECORD)  
SET DET

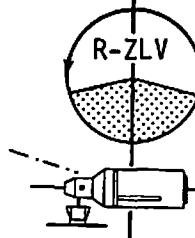
IF NCC EXECUTION IS QUESTIONABLE  
START RECORDING CHART DATA AT NSR-28

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
X UNREASONABLE UPDATE X  
X 1.00 NM, 6.0 FPS X  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

R CHART DATA POINTS LOCATED  
AT ~~USE~~ MINUS  
28, 24, 20, 12.

SWS IN R-ZLV

SUN



LOSS VHT

TRANSMIT NCC N81 APPLIED AND  
SPS BURN STATUS (Pg 1-11)

~~SET HDSUPFLG (V25N7E, 106E, 2000E, 1E)~~  
~~P34 (FINAL COMP: RECORD)~~  
LOAD N81 WITH CORRECT VALUES

IF NCC EXECUTION IS QUESTIONABLE  
THEN CMC FINAL COMP IS PRIME

GO TO SPS BURN CUE CARD

P40

5+30

(11002)  
(01111)

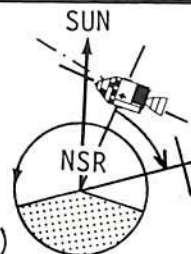
-31:00

5+40

5+50

-12:00

6+00



6+00

(11002)  
(01111)

0:00

(11003)  
(11003)  
(01111)

6+10

-50:00

6+20

6+30

**NSR** (180,0/226,0) (TRIM VG's  $\pm 0.2$ )  
RECORD BURN STATUS

P35; BYPASS AUTO MNVR  
V48E (LOAD 4 JET, 2°/SEC)  
KEY V23N78E, +E  
KEY V58E; MNVR HEADS UP  
TRANSMIT NSR N81 APPLIED (Pg 1-12) AND  
SPS BURN STATUS  
V32E (1ST P35 RECYCLE)  
REACQUIRE VHF RANGING (R=76.00 NM)  
SET DET

XXXXXXXXXXXXXXXXXXXXX  
X UNREASONABLE UPDATE X  
X 0.50 M1, 3.0 FPS X  
XXXXXXXXXXXXXXXXXXXXX

V76E (LOAD N72 WITH TIG TPI-32)

S  
X  
T

\*\*AFTER STDN GO (UNLESS NO COMM)-  
PERFORM PSM ACT SWITCH LIST (Pg 1-13)

TPI PRELIMINARY PAD (COPY Pg 1-14)  
DOCKING ATTITUDE (COPY Pg 1-17)

## SPS BURN STATUS

|            |     |   |   |   |  |  |  |  |  |
|------------|-----|---|---|---|--|--|--|--|--|
| ΔTIG       | 0   |   |   |   |  |  |  |  |  |
| AFTER TRIM |     |   |   |   |  |  |  |  |  |
| ΔVC        |     |   |   |   |  |  |  |  |  |
| FDA1       | R   | + |   |   |  |  |  |  |  |
|            | P   | + |   |   |  |  |  |  |  |
|            | Y   | + |   |   |  |  |  |  |  |
| N85        | VGX | + | 0 | 0 |  |  |  |  |  |
|            | VGX | - | 0 | 0 |  |  |  |  |  |
|            | VGZ | - | 0 | 0 |  |  |  |  |  |

|                 |  |  |   |   |  |  |  |  |  |
|-----------------|--|--|---|---|--|--|--|--|--|
| CHART TPI TIG 1 |  |  |   |   |  |  |  |  |  |
| -               |  |  | 3 | 2 |  |  |  |  |  |
| TPI-32          |  |  |   |   |  |  |  |  |  |

## PSM ACTIVATION SWITCH LIST

SM RCS PSM 1 He-OPEN, tb-gray  
SM RCS QUAD PRPLNT B-CLOSE, tb(2)-bp  
SM RCS PSM PRPLNT B-OPEN, tb-gray  
SM RCS QUAD PRPLNT C-CLOSE, tb(2)-bp  
SM RCS PSM PRPLNT C-OPEN, tb-gray  
SM RCS QUAD PRPLNT A-CLOSE, tb(2)-bp  
SM RCS PSM PRPLNT A-OPEN, tb-gray  
SM RCS QUAD PRPLNT D-CLOSE, tb(2)-bp  
SM RCS PSM PRPLNT D-OPEN, tb-gray  
SM RCS QUAD He(4)-CLOSE, tb(4)-bp

## TPI TIG (N37)

|                  |             |  |  |  |  |  |  |  |  |
|------------------|-------------|--|--|--|--|--|--|--|--|
| PRELAUNCH Pg 1-a | 07:02       |  |  |  |  |  |  |  |  |
| RECYCLES         | 07:02:18.97 |  |  |  |  |  |  |  |  |
|                  | 06:58:56.33 |  |  |  |  |  |  |  |  |
| CHART TPI TIG 2  |             |  |  |  |  |  |  |  |  |

NOMINAL M=5



6+30

-32:00  
(11103)  
(01111)

-28:00

-24:00

6+40

-16:00

6+50

-12:00

7+00

S  
X  
T

N  
F  
D

M  
I  
L

A  
C  
N

38

XXXXXXXXXXXXXXXXXXXXX  
Y UNREASONABLE UPDATE X  
X 0.50 MI, 3.0 FPS X  
XXXXXXXXXXXXXXXXXXXXX

R CHART DATA FOR TPI -32 (GET)

R CHART DATA FOR TPI -28 (GET)

R CHART DATA FOR TPI -24 (GET)

V32E (2ND P35 RECYCLE: RECORD)

\*\*\*\*\*TPI EARLY/LATE LOGIC\*\*\*\*\*  
\* 2nd RECYCLE  
\* IF CMC SOLUTION INDICATES TIG SLIP > +10 MIN. 5  
\* USE CMC TIG OPTION; N37=PRELAUNCH N37+10 MIN  
\* CONTINUE CHART SOLUTION FOR FINAL ΔV COMPARISON  
\* AT FINAL COMP-USE NOMINAL COMPARISON LOGIC  
\* IF ALL COMPARISONS DISAGREE-BURN THE  
\* SOLUTION WHOSE TIG (CMC 2ND RECYCLE,  
\* STDN PREL. PAD) COMPARES CLOSEST WITH  
\* CHART TPI TIG 2  
\*\*\*\*\*

R CHART DATA FOR TPI -16 (GET)

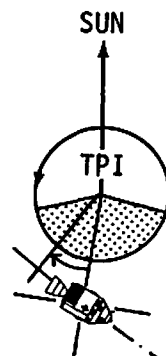
V77E; V83E, SET ORDEAL (FDAI 1)  
TPI FINAL PAD (COPY)

H=231

P35 (FINAL COMP: RECORD)

GO TO SPS BURN CUE CARD  
ULLAGE AT -14 SEC

P40



TPI PAD DATA

|                     |                     | NOMINAL |   |   |   |           | PRELIMINARY |   |   |   |           | FINAL |   |   |   |           |
|---------------------|---------------------|---------|---|---|---|-----------|-------------|---|---|---|-----------|-------|---|---|---|-----------|
| N37                 | HR                  | +       | X | X | X | 0 0 7     | +           | X | X | X | 0 0 7     | +     | X | X | X | 0 0 7     |
| TIG TPI             | MIN                 | +       | X | X | X | 0 2       | +           | X | X | X | 0 4       | +     | X | X | X | 0 3       |
|                     | SEC                 | +       | X | X | X | 5 3 0 0   | +           | X | X | X | 4 6 0 0   | +     | X | X | X | 5 0 0 0   |
| N81                 | ΔV <sub>X</sub>     | +       | X | X | X | 1 8 9     | +           | X | X | X | 1 8 3     | +     | X | X | X | 1 8 4     |
| ΔV <sub>TPI</sub>   | ΔV <sub>Y</sub>     | +       | X | X | X | 0 0 0     | +           | X | X | X | 0 0 7     | +     | X | X | X | 0 1 3     |
|                     | ΔV <sub>Z</sub>     | -       | X | X | X | 0 8 1     | -           | X | X | X | 0 6 7     | -     | X | X | X | 0 7 2     |
| N59                 | ΔV <sub>F</sub> /BT | +       | X | X | X | 2 0 5 4 8 | +           | X | X | X | 1 9 4 4 7 | +     | X | X | X | 1 9 6 4 8 |
| ΔV <sub>R</sub> /BT |                     | +       | X | X | X | 0 0 0 0 0 | +           | X | X | X | 0 0 0 0 2 | +     | X | X | X | 0 1 2 0 5 |
| (LOS)               | ΔV <sub>D</sub> /BT | +       | X | X | X | 0 1 1 0 5 | +           | X | X | X | 0 2 3 1 2 | +     | X | X | X | 0 1 9 1 0 |
|                     | ΔV <sub>C</sub>     | X       | X | X | X | 0 0 6 6   | X           | X | X | X |           | X     | X | X | X | 0 0 7 1   |

TPI ONBOARD DATA

CSM WGT \_\_\_\_\_  
PTRIM \_\_\_\_\_  
YTRIM \_\_\_\_\_

6.9  
.8  
-12.6

20.8  
28.8  
- 9+23  
+ 19.2  
+ .7  
- 6.5  
+ 20.0  
+ .7  
+ 2.9

|                   |                   | P35 RECYCLE |   |   |       |     | P35 FINAL COMP |   |         |       |     |
|-------------------|-------------------|-------------|---|---|-------|-----|----------------|---|---------|-------|-----|
| N37               | HR                | +           | 0 | 0 | 0     | 0 7 | +              | 0 | 0       | 0     | 0 7 |
| TIG TPI           | MIN               | +           | 0 | 0 | 0     | 0 2 | +              | 0 | 0       | 0     | 0 3 |
|                   | SEC               | +           | 0 | 2 | 2 9 5 |     | +              | 0 | 4       | 7 3 3 |     |
| N58               | ΔV <sub>TPI</sub> | +           | 0 | 0 | 2 0 1 |     | +              | 0 | 0       | 1 9 7 |     |
|                   | ΔV <sub>TPF</sub> | +           | 0 | 0 | 2 8 1 |     | +              | 0 | 0       | 2 2 1 |     |
|                   | ΔT(TPI/TPI)       | +           | 0 | 0 | 8 0 9 |     |                |   | 1 8 2 9 |       |     |
| N81               | ΔV <sub>X</sub>   | +           | 0 | 0 | 1 8 2 |     | +              | 0 | 0       | 1 8 1 |     |
| ΔV <sub>TPI</sub> | ΔV <sub>Y</sub>   | +           | 0 | 0 |       | 5   | +              | 0 | 0       | 0 0 6 |     |
|                   | ΔV <sub>Z</sub>   | -           | 0 | 0 | 8 7   |     | -              | 0 | 0       | - 7 8 |     |
| N59               | ΔV <sub>F</sub>   | +           | 0 | 0 | 2 0 1 |     |                | 0 | 0       |       |     |
| ΔV <sub>R</sub>   |                   | +           | 0 | 0 | 0 0 6 |     |                | 0 | 0       |       |     |
| (LOS)             | ΔV <sub>D</sub>   | +           | 0 | 0 | 0 0 5 |     |                | 0 | 0       |       |     |

XXXXXXXXXXXXXXXXXXXXX  
X N81 COMP LIMITS ±1.0, ±3.0, ±3.0 FPS X  
X  
X CMC/CHARTS WITHIN LIMITS GO CMC X  
X IF NOT CMC/STDN WITHIN LIMITS GO CMC X  
X IF NOT CHART/STDN WITHIN LIMITS GO CHART X  
X NO AGREEMENT GO STDN X  
X  
X FOR CHART/STDN: RECALL P35-TIG OPT. X  
X USE CHART/STDN TIG X  
XXXXXXXXXXXXXXXXXXXXX

TPI CHART SOLN

|                 |   |   |   |       |
|-----------------|---|---|---|-------|
| ΔV <sub>X</sub> | + | X |   | 1 8 6 |
| ΔV <sub>Y</sub> | * | S | T | D     |
| ΔV <sub>Z</sub> | - | X |   | 3 9   |

7+00

(11103)  
(01111)

0:00

+3:00

+4:30

+7:30

7+10

+8:30

+9:00

+12:00

+15:00

+16:30

7+20

+19:30

+20:30

+21:00

+24:00

7+30

S  
X  
T  
/  
V  
H  
F  
6S  
X  
T  
/  
V  
H  
F  
6V  
H  
FTPM1 (0,25/25,0) (TRIM VG's  $\pm 0.2$ )

P36

REACQUIRE VHF RANGING (R=21.00 NM)  
V87E, FDAI 1-ORB RATECENTER SWS IN SXT WHEN  
READING CHART DATA  $\theta_p$  AND TA $\theta_p$  AND TA CHART DATA FOR TPI +4:30

R CHART DATA FOR TPI +7:30

SWS MNVR TO SI  
AT 7+10 GETR,  $\theta_p$  AND TA CHART DATA FOR TPI +8:30  
PRO (P36 FINAL COMP: RECORD)

P41

XXXXXXXXXXXXXXXXXXXXX  
X UNREASONABLE UPDATE X  
X 0.50 MI, 3.0 FPS X  
XXXXXXXXXXXXXXXXXXXXX

P36

V25N33E (LOAD TPI TIG)

07 03 47.

CENTER SWS IN SXT WHEN  
READING CHART DATA  $\theta_p$  AND TA $\theta_p$  AND TA CHART DATA FOR TPI +16:30

GET OUT SUNGLASSES

R CHART DATA FOR TPI +19:30

R,  $\theta_p$  AND TA CHART DATA FOR TPI +20:30  
PRO (P36 FINAL COMP: RECORD)

P41

TPM2

P37  
CMC MODE-HOLD (MONITOR LOS RATES AND R)  
PERFORM TV PREP CHECKLIST (Pg 1-15)  
PERFORM PRE-BRAKING SWITCH LIST (Pg 1-15)  
PERFORM EVAP DEACT SWITCH LIST (Pg 1-15)  
PNL 252 WASTE STOW VENT VLV-CLOSE (CW)

## TPM1 ONBOARD DATA

| P36 FINAL COMP     |              |             | TPM1 CHART SOLN |  |  |  |
|--------------------|--------------|-------------|-----------------|--|--|--|
| N59                | $\Delta V_F$ | + 0 0 0 0.5 |                 |  |  |  |
| $\Delta V_{(LOS)}$ | $\Delta V_R$ | - 0 0 0 0.0 |                 |  |  |  |
|                    | $\Delta V_D$ | + 0 0 0 0.3 |                 |  |  |  |

+ ( )  
0  
+ /

## TPM2 ONBOARD DATA

| P36 FINAL COMP     |              |             | TPM2 CHART SOLN |  |  |  |
|--------------------|--------------|-------------|-----------------|--|--|--|
| N59                | $\Delta V_F$ | + 0 0 0 1.1 |                 |  |  |  |
| $\Delta V_{(LOS)}$ | $\Delta V_R$ | - 0 0 0 0.8 |                 |  |  |  |
|                    | $\Delta V_D$ | - 0 0 0 4.5 |                 |  |  |  |

## PRE-BRAKING SWITCH LIST

|                     |                             |
|---------------------|-----------------------------|
| FDAI SCALE-5/1      | LARC PWR NORMAL(2)-AC/DC    |
| FDAI SELECT-1/2     | LARC PWR DIRECT(2)-MNA/MNB  |
| FDAI SOURCE-ATT SET | SC CONT/MODE-CMC/HOLD       |
| ATT SET-GDC         | CMC MODE(3)-ATT 1/RATE 2    |
| MAN ATT(3)-RATE CMD | AUTO MCS SELECT(16)-MNA/MNB |
| LIMIT CYCLE-OFF     | THC-ARMED                   |
| DBD/RATE-MIN/LOW    | RHC-2-ARMED                 |
| THC PWR-ON(up)      |                             |

## GLYCOL EVAPORATOR DEACTIVATION SWITCH LIST

|                                           |
|-------------------------------------------|
| GLYCOL EVAP H2O FLOW-OFF                  |
| GLYCOL EVAP STEAM PRESS-MAN               |
| GLYCOL EVAP STEAM PRESS-INCR (for 58 sec) |

## TV PREPARATION CHECKLIST

|                            |
|----------------------------|
| TV SOURCE-CM               |
| REMOVE & STOW LENS CAP     |
| LINEAR SW-LINEAR GAM 1/2   |
| AVG SW-PEAK AVG            |
| F-STOP-16 22               |
| ZOOM-25 as required        |
| FOCUS-∞                    |
| MONITOR ON-ON              |
| (REFINE SETTINGS AS REQ'D) |

NOMINAL M=5

7+30

(11/03)  
(11102)  
(01111)

7+40

(10/02)  
(11111)

7+50

8+00

S BAND AUX TV-TV (2 MIN PRIOR STDN AOS)

PRO (P48)

~~V40E (LOAD 0.5°/SEC)~~

NULL LOS RATES & ADJUST R TO BRAKING GATE

N83 (ΔVX, ΔVY, ΔVZ CONT) (OPTIONAL)

DAC-ON (1000 FT), HDC -30 FRAMES (1000 FT)

~~CONFIGURE NK WITH 300 MM LENS, STOP~~

~~55 LENS NK-10 FR (IF DESIRED)~~

STATION KEEP ON SWS +X AXIS (R ~100 FT)

(MAINTAIN ROLL ~0° WITH SUN ON BOTTOM

OF S/C)

POO

SPOTLIGHT - ON (DOWN)

S BAND AUX TV-OFF (CTR) AT STDN LOS

CONFIGURE COMM (PANEL 9)

MODE - VOX

VOX SENS - AS RECD (~6)

S-BD - T/R (VERIFY)

AUDIO CONT - NORM (VERIFY)

INTERCOM - T/R (VERIFY)

REMOVE TV CAMERA FROM BRKT AND HANDHOLD

CM4/NK-60 FRAMES/HDC - 100 FRAMES (F8, 1/250)

V48E (LOAD 2 JET (BD), 0.5°/SEC, AC ROLL)

CMC MODE - FREE

MNVR TO WING #1 (AVOID ATM SOLAR ARRAYS

AND 45' DISCONE ANTENNAS)

USE METERING SYSTEM FOR NK PHOTOS

PHOTO DAMAGED AREAS USING NK FOR CLOSEUPS

AND HDC FOR WIDE-ANGLES (IF VERY HIGH

RESOLUTION IS DESIRED, LOAD BNOT IN NK)

REPLACE TV CAMERA ON BRKT

S BAND AUX TV-TV (2 MIN PRIOR STDN AOS)

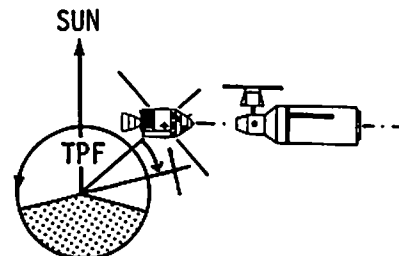
SAS WING #1 TV (R ~60 ft)

F-STOP-FC (IF TV CAMERA IMAGES SUN)

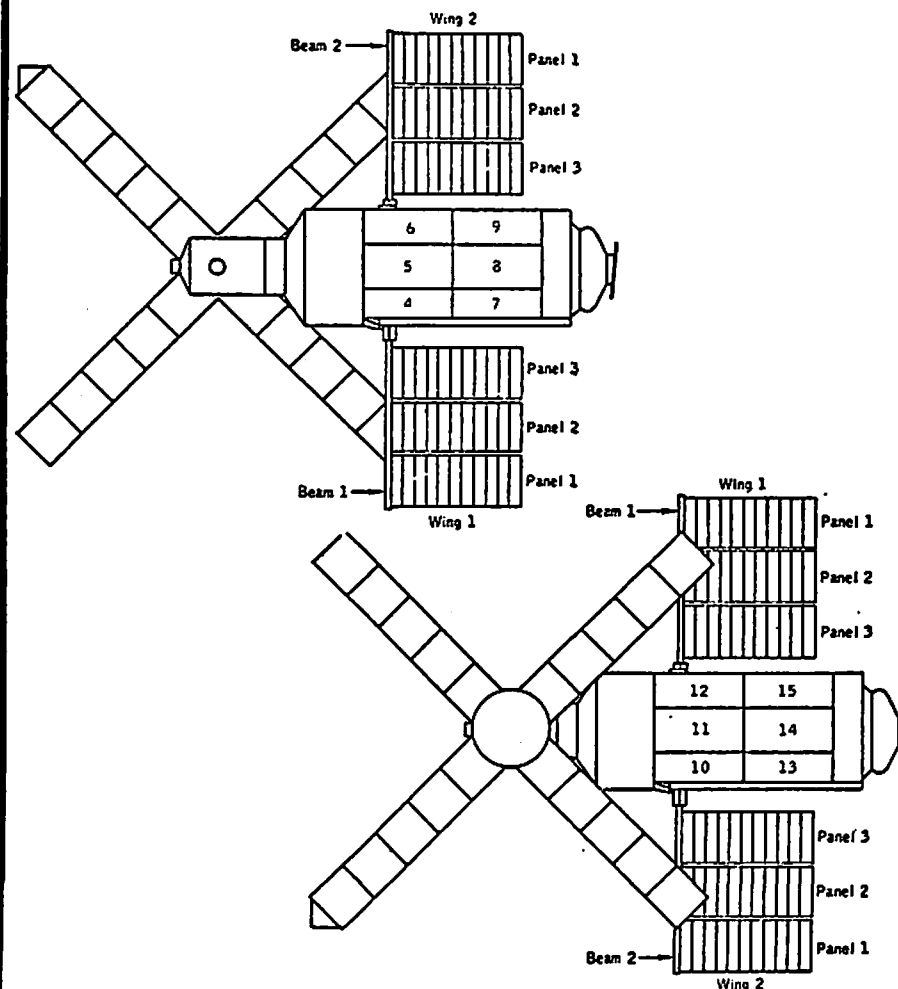
TAKE HDC PICTURES OUT CM4 BY TV

CAMERA OR CM3

TAKE NK PICTURES OUT CM3 IF POSSIBLE



| BRAKING GATES       |         |             |     |              |        |
|---------------------|---------|-------------|-----|--------------|--------|
| RETICLE ANGLE (DEG) |         | SOLAR PANEL |     |              |        |
| R (MI)              | R (FPS) | SWS         | DIA | (BETA - 10°) | R (FT) |
| 1.00                | 30      | .2          |     | .8           | 6000   |
| .50                 | 20      | .4          |     | .7           | 3000   |
| .25                 | 10      | .8          |     | .5           | 1500   |
| .08                 | 5       | 2.5         |     | 10.0         | 500    |
| .05                 |         | 4.2         |     |              | 300    |
| .03                 |         | 6.3         |     |              | 200    |
| .02                 |         | 10.0        |     |              | 130    |



8+00

(11102)  
(01111)  
(10102)  
(11111)G  
D  
S  
T  
E  
X  
M  
I  
L  
B  
D  
A

WHEN SAS WING #1 TV COMPLETE -  
MNVR TO WING #2 (AVOID ATM SOLAR ARRAYS)

SAS WING #2 TV (R ~60 FT)

8+10

WHEN SAS WING #2 TV COMPLETE -  
MNVR TO SWS +X AXIS

(11102)  
(11111)

S BAND AUX TV-OFF (CTR) AT STDN LOS  
CMC MODE - AUTO; ~~V49E (LOAD 4 JET)~~  
BMAG MODE (3) - RATE 2  
V49E (MNVR TO DOCKING ATTITUDE)  
DISASSEMBLE AND STOW TV CAMERA, ZOOM LENS,  
CABLES, MONITOR, CAMERA MOUNT, NK, AND  
HDC (TSB)  
RECONFIGURE DAC - 6 FPS FOR SOFT DOCKING  
PERFORM PRE-SOFT DOCKING SWITCH LIST  
(Pg 1-17)  
CSM/SWS SOFT DOCKING; PERFORM SOFT  
DOCKING CHECKLIST (Pg 1-17)

8+20

**EAT SUPPER**  
GO TO LAUNCH CHECKLIST FOR  
SEVA PROCEDURES

REMOVE CUE CARD FROM BACK OF  
RENDEZVOUS BOOK AND STOW  
IN R3

8+30

## DOCKING ATTITUDE

FINAL

|     |   | NOMINAL   | UPDATE    |       |
|-----|---|-----------|-----------|-------|
| N22 | R | + 191 0 0 | + 185 0 0 | 199   |
|     | P | + 035 0 0 | + 039 0 0 | 034.4 |
|     | Y | + 006 0 0 | + 009 0 0 | 008   |

## PRE-SOFT DOCKING SWITCH LIST

|                                   |                                  |
|-----------------------------------|----------------------------------|
| BMAG MODE(3)-ATT 1/RATE 2         | DOCKING PROBE EXT/REL tb(2)-gray |
| cb DOCK PROBE(2)-CLOSE            | SPOT LIGHT-ON(down)              |
| DOCKING PROBE RETRACT(2)-OFF(ctr) | SEGS-PYRO-ARM(2)-ON(up)          |
| DOCKING PROBE EXT/REL-RETRACT     | FC REACS VALVES-LATCH            |

## SOFT DOCKING CHECKLIST

## TO INITIATE CAPTURE

VERIFY STATION KEEPING ON SWS +X AXIS  
(RANGE=40 TO 100 FT) (DOCK TGT=1.8 TO 0.8° IN COAS)  
LOAD EMS WITH +00.0 FPS (-100.0 FPS)  
E'IS FUNC/MODE-ΔV/NORMAL  
DAC-ON  
THC-INITIATE 0.4 +0.1 FPS CLOSING RATE  
RHC-MAINTAIN MINIMUM RELATIVE ALIGNMENT ANGLES

AT CAPTURE (IF NO CAPTURE-SEE BACKUP DOCKING PROCEDURES (S/2-1))

DOCKING PROBE EXT/REL tb(either)-bp (IF GRAY-SEE DOCKING MALFUNCTION  
PROCEDURES #2, Pg 1-3 IN CSM MALFUNCTION PROCEDURES BOOK)  
CMC MODE-FREE  
RHC PHR DIRECT(2)-OFF(ctr)  
REPORT CAPTURE TO STDN  
ALLOW PROBE TO DAMP SC MOTION (10 SEC)  
THC-NUL PITCH AND YAW MISALIGNMENT<3°  
RHC-NUL ROLL MISALIGNMENT<5°  
DOCKING PROBE RETRACT PRIN 1 (SEC 2 IF REQ)

## AT DOCK LATCH

DOCKING PROBE EXT/REL tb(2)-gray

## AFTER SOFT DOCK

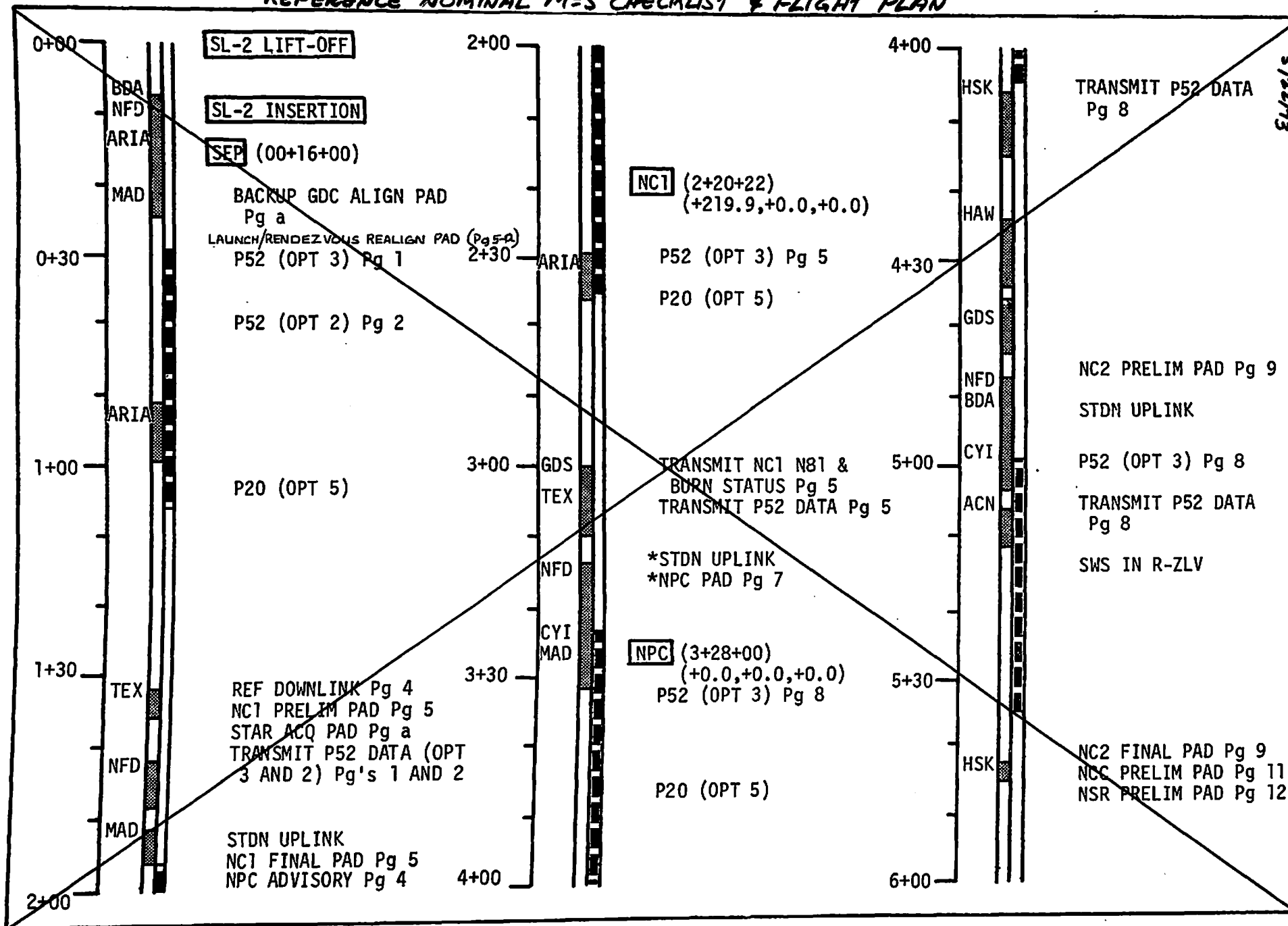
|                                   |                                        |
|-----------------------------------|----------------------------------------|
| V45E                              | E'IS FUNC/MODE-OFF/STBY                |
| SEGS-PYRO-ARM(2)-SAFE             | DBD/RATE-MIN/LOW                       |
| SEGS-LOGIC(2)-OFF                 | THC PHR-OFF                            |
| cb-SEGS-ARM(2)-OPEN               | RHC PHR NORMAL#1-OFF(ctr)              |
| cb DOCK PROBE(2)-OPEN             | RHC PHR DIRECT(2)-OFF(ctr)             |
| DOCKING PROBE EXT/REL-OFF         | SC CONT/MODE-CMC/FREE                  |
| DOCKING PROBE RETRACT(2)-OFF(ctr) | BMAG MODE(3)-RATE 2                    |
| DAC-OFF                           | AUTO RCS-SELECT(A1,B2,A3,B4,B3,B4)-OFF |
| SPOT LIGHT-OFF(up)                | cb-SCS A/G & B/D ROLL(4)-OPEN          |
| FC-REACS VALVES-NORM              | cb-SCS PITCH & YAW(4)-OPEN             |
| VHF AM B-OFF(ctr)                 | THC-LOCKED                             |
| VHF RANGING-OFF                   | RHC(2)-LOCKED                          |
| COAS PHR-OFF                      |                                        |

NOMINAL M=5'

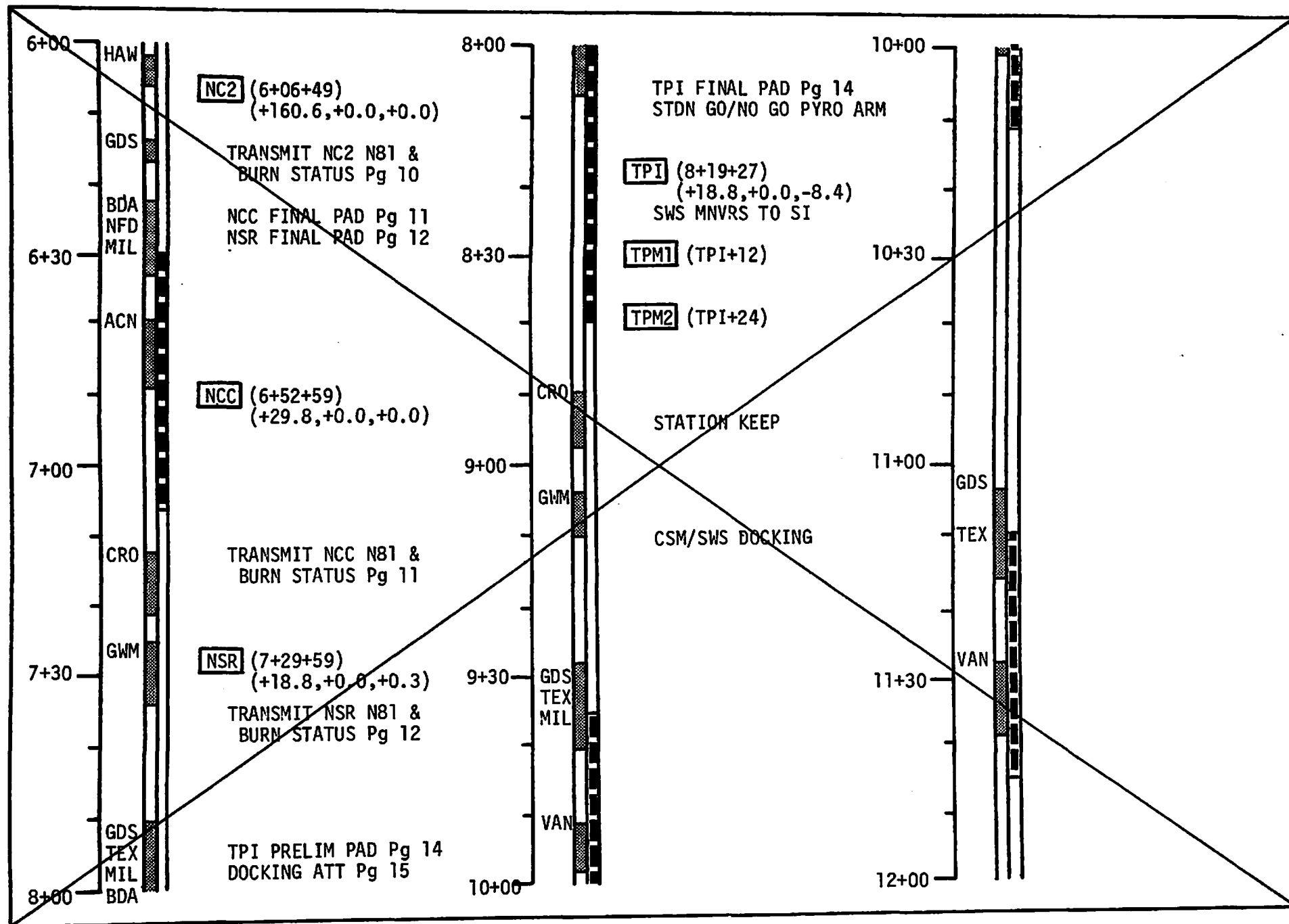


NOMINAL M=6

# REFERENCE NOMINAL M=5 CHECKLIST & FLIGHT PLAN



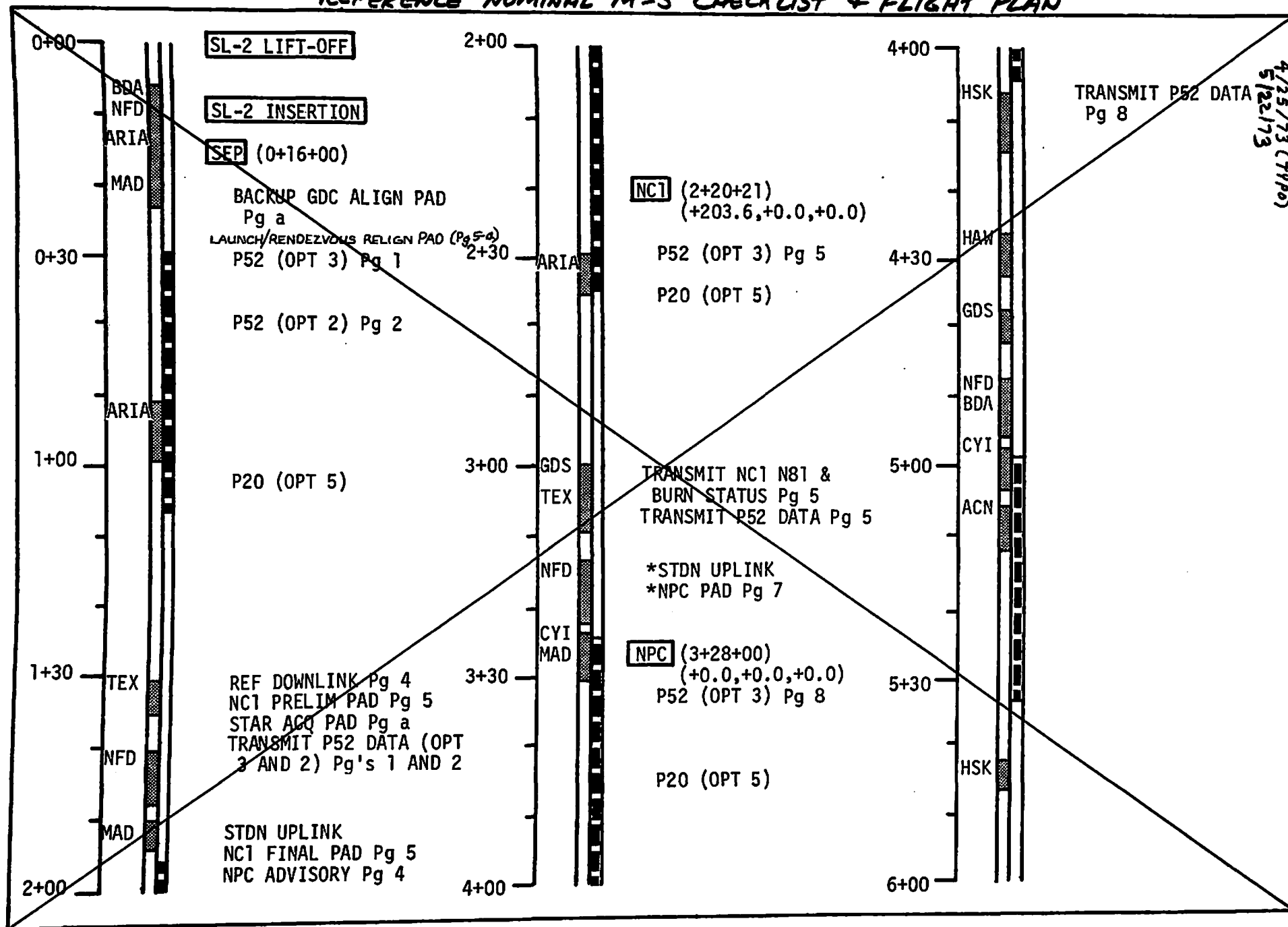
DATE 4/6/73  
4/20/73 (P1)  
4/25/73 (P2)  
5/22/73



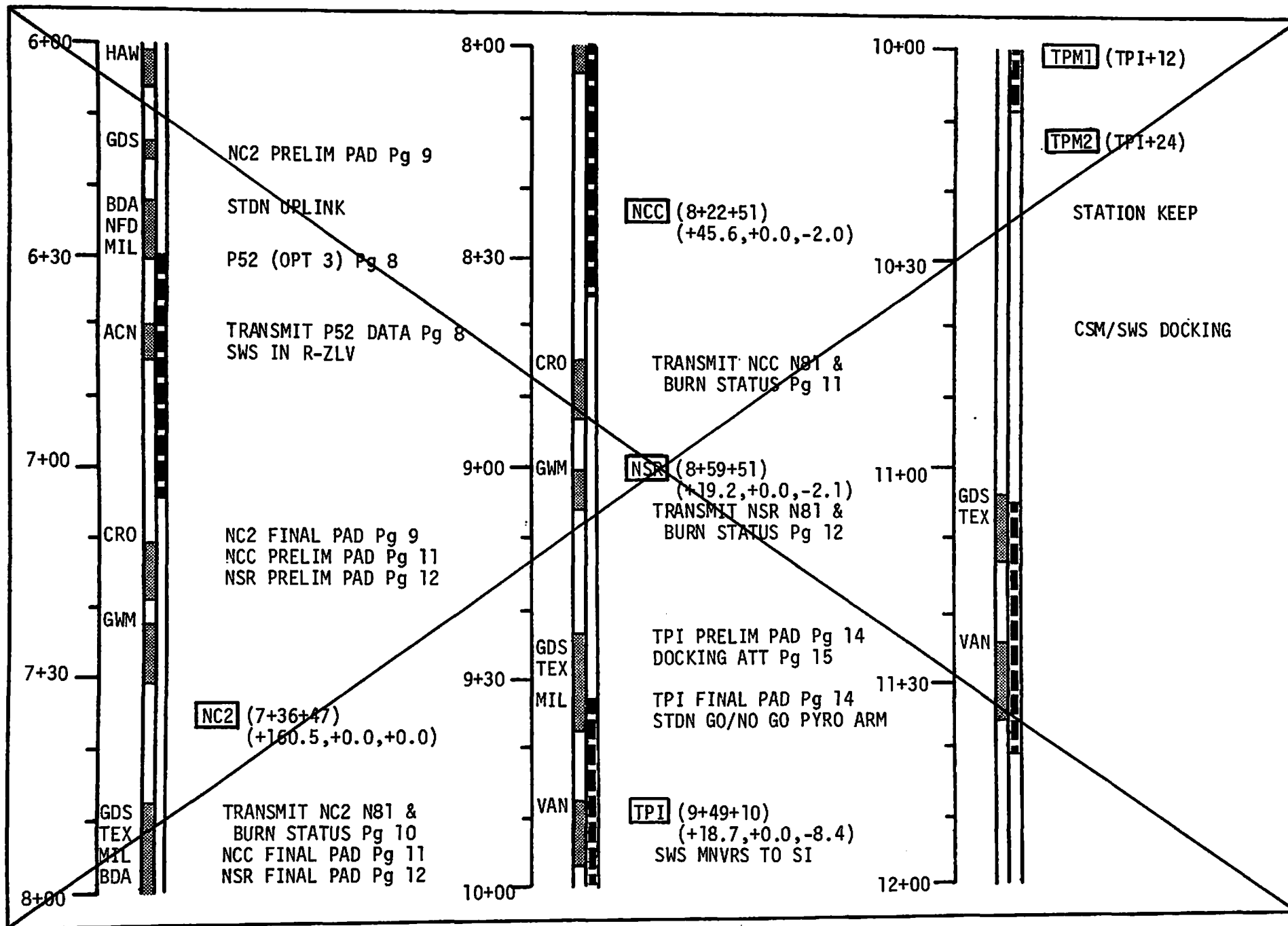
NUMINAL M=6

NOMINAL M=7

# REFERENCE NOMINAL M=5 CHECKLIST & FLIGHT PLAN



DATE 4/6/73  
4/20/73 (P.E.I.)

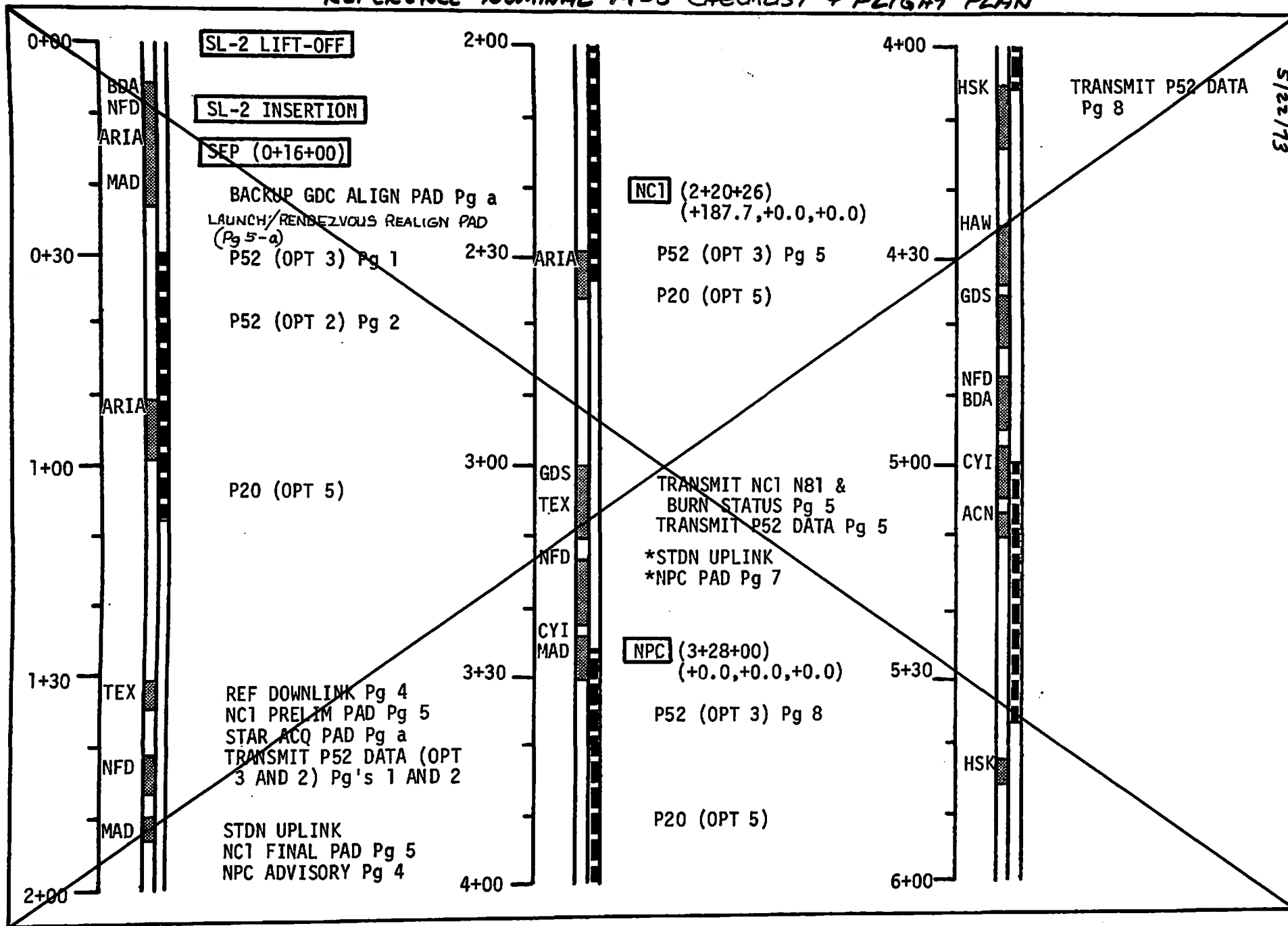


NOMINAL M=7

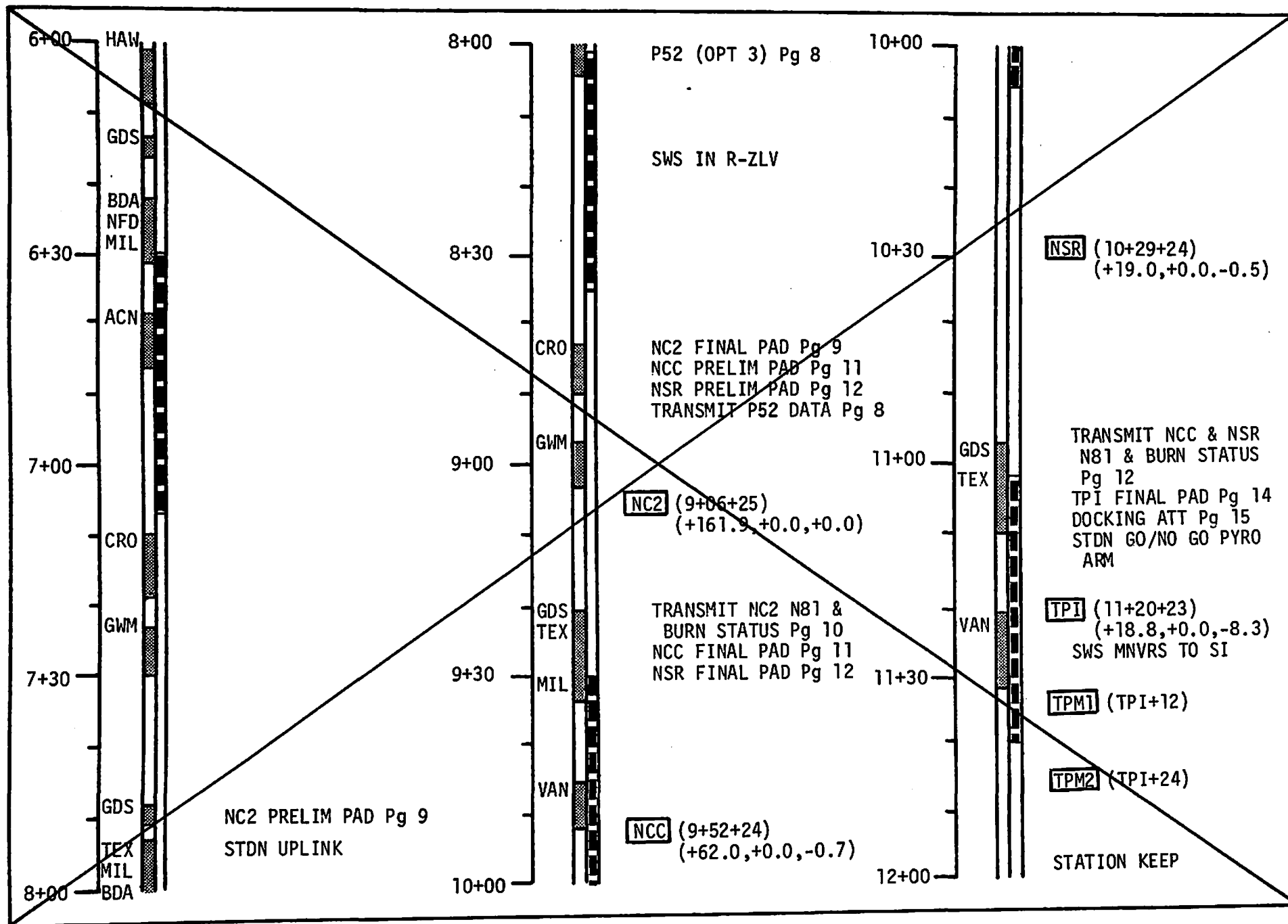
DATE 4/6/73  
5/24/73

NOMINAL M=8

REFERENCE NOMINAL M=5 CHECKLIST & FLIGHT PLAN



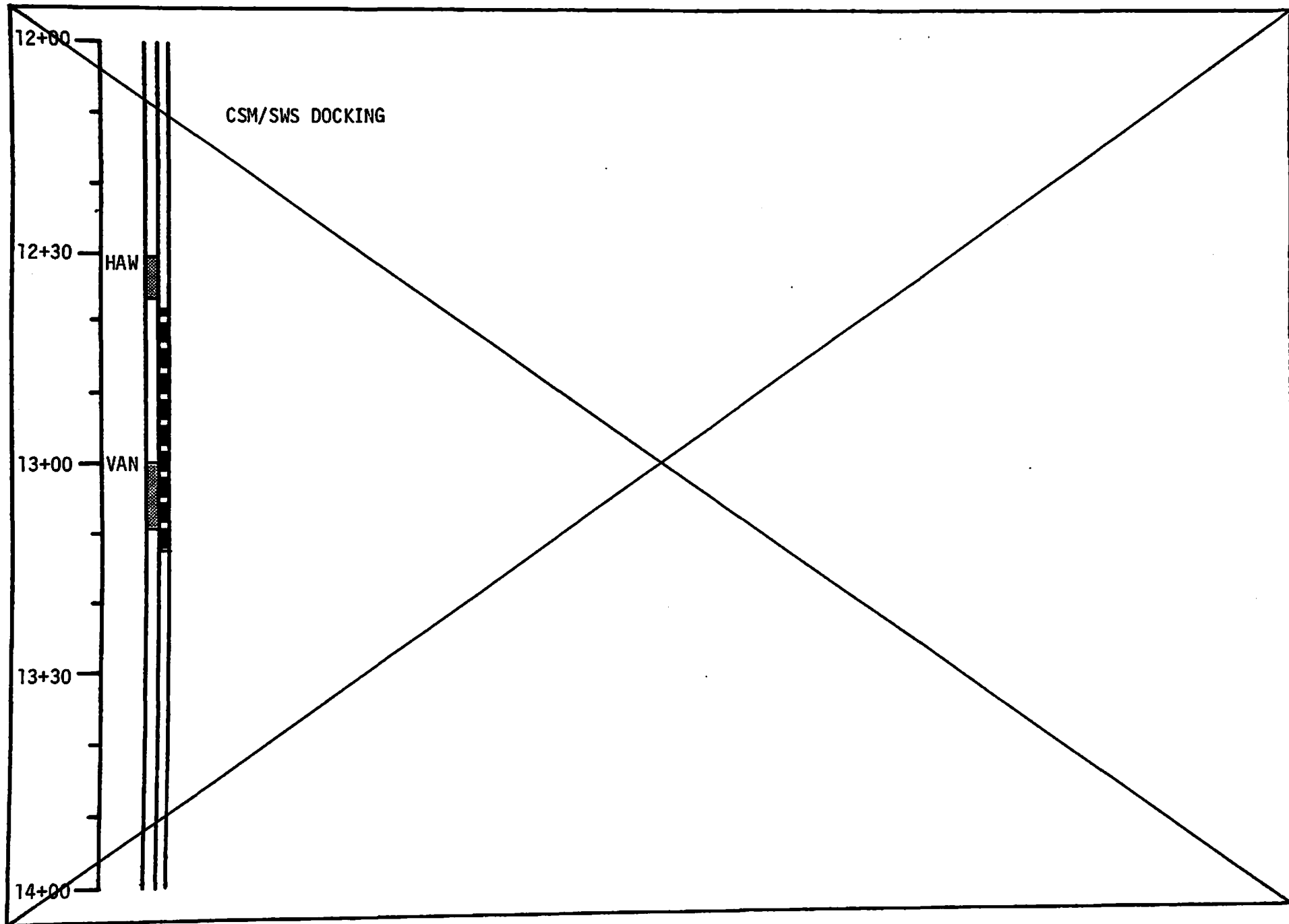
DATE 4/6/73  
4/20/73 (P&I)  
4/25/73 (TYPO)  
5/22/73



NOMINAL M=8

DATE 4/6/73  
5/22/73

NOMINAL M=8



DATE ~~4/6/73~~  
5/22/73

PAGE 4-3



# CMC FAILURE

## (I) WHEN CMC FAILURE OCCURS

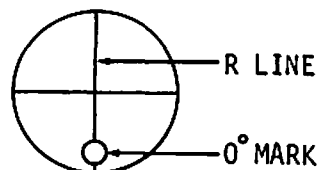
- 1.) PERFORM STEP (II)
- 2.) IF CMC FAILED AFTER NC1, PERFORM STEP (III)
- 3.) IF CMC FAILED AFTER NC2, NEED TO SET ORDEAL BEFORE NEXT TIG
- 4.) PROCEED TO APPROPRIATE POINT IN TIMELINE (DO STDN/MET CLOCK CHECK)

## (II) CMC FAILURE SWITCH LIST

SC CONT-SCS  
DBD/RATE-MIN/HIGH  
BMAG MODE(3)-ATT 1/RATE 2  
MAN ATT(3)-RATE CMD  
FDAI SELECT-1  
FDAI SOURCE-ATT SET  
ATT SET-IMU  
LIMIT CYCLE-OFF

### NOTES

- 1.) AN IMU GO/NO GO CHECK (STEP III) AND/OR A LAUNCH/RNDZ REALIGN (Pg 5-2) IS REQUIRED BEFORE OBTAINING INITIAL IMU DRIFT CHECK DATA
- 2.) TWO ADDITIONAL OPPORTUNITIES TO OBTAIN IMU DRIFT DATA ARE INCLUDED IN THE TIMELINE. THESE ARE PROVIDED FOR CASES WHERE DATA CANNOT BE OBTAINED AT NOMINAL TIMES. A COMPLETE DRIFT CHECK REQUIRES TWO SETS OF DRIFT DATA.
- 3.) RECOMMENDED TRACKING OF HORIZON AND SHS: DBD/RATE-MIN/HIGH MAN ATT PITCH-MIN IMP
- 4.) BACKUP GDC ALIGN STAR PAIR IS VISIBLE FROM 55+6 MIN TO 58-55 TO 58



## (III) IMU GO/NO GO CHECK

ATT SET-IMU (VERIFY)  
SET ASCP tw TO STAR ACQ PAD  
ATT (Pg 5-a)  
RHC-MNVR TO STAR ACQ PAD ATT  
BMAG MODE(3)-RATE 2  
RATE-LOW  
BMAG MODE(3)-ATT 1/RATE 2  
RHC-MNVR (TRIM) TO ZERO  
FDAI 1 ERROR NEEDLES  
OHC-SET IN TPAC ANGLES  
OF 1ST PAD STAR  
IF STAR<5° FROM CTR  
OF SCT-IMU IS GO  
\*IF STAR>5° FROM CTR OF SCT  
\* PERFORM BACKUP GDC & IMU  
\* ALIGN (G/3-18),  
\* REF PAD (Pg 5-a)  
RATE-HIGH

### DEFINITIONS

0° = ORDEAL ANGLES TO PLACE EARTH  
HORIZON IN CENTER OF COAS  
H = AVERAGE ALTITUDE  
0 = ORDEAL FDAI PITCH

## NC1 TARGETING PAD

|               |     | NOMINAL |   |   |       | NOMINAL UPDATE |   |   |  | PRELAUNCH UPDATE |   |   |  |
|---------------|-----|---------|---|---|-------|----------------|---|---|--|------------------|---|---|--|
| N95           | HR  | +       | + | + | 002   | +              | + | + |  | +                | + | + |  |
| TIG NC1       | MIN | +       | + | + | 23    | +              | + | + |  | +                | + | + |  |
|               | SEC | +       | + | + | 33.00 | +              | + | + |  | +                | + | + |  |
| N57 HALF REVS |     |         |   |   | 03    |                |   |   |  |                  |   |   |  |
| N37           | HR  | +       | + | + | 007   | +              | + | + |  | +                | + | + |  |
| TIG TPI       | MIN | +       | + | + | 02    | +              | + | + |  | +                | + | + |  |
|               | SEC | +       | + | + | 53.00 | +              | + | + |  | +                | + | + |  |

## BACKUP GDC ALIGN PAD

|                 |   | NOMINAL |   |   |       | NOMINAL UPDATE |   |  |  | UPDATE |   |  |  |
|-----------------|---|---------|---|---|-------|----------------|---|--|--|--------|---|--|--|
| 0°/R STARS      |   | 2       | 5 | / | 34    |                | / |  |  |        | / |  |  |
| ASCP tw         | R |         |   |   | 2018  |                |   |  |  |        |   |  |  |
|                 | P |         |   |   | 3581  |                |   |  |  |        |   |  |  |
| (R:0Z REFSIMAT) | Y |         |   |   | 359.5 |                |   |  |  |        |   |  |  |

## LAUNCH / RNDZ REALIGN PAD

|                 |   | NOMINAL |  |  |       | NOMINAL UPDATE |  |  |  | UPDATE |  |  |  |
|-----------------|---|---------|--|--|-------|----------------|--|--|--|--------|--|--|--|
| ASCP tw         | R |         |  |  | 359.5 |                |  |  |  |        |  |  |  |
|                 | P |         |  |  | 206.2 |                |  |  |  |        |  |  |  |
| (R:0Z REFSIMAT) | Y |         |  |  | 000.9 |                |  |  |  |        |  |  |  |

## STAR ACQUISITION PAD

|                 |    | NOMINAL |  |  |        | NOMINAL UPDATE |  |  |   | UPDATE |  |  |   |
|-----------------|----|---------|--|--|--------|----------------|--|--|---|--------|--|--|---|
| ASCP tw (NC1)   | R  |         |  |  | 180    |                |  |  |   |        |  |  |   |
|                 | P  |         |  |  | 009    |                |  |  |   |        |  |  |   |
| (R:0Z REFSIMAT) | Y  |         |  |  | 002    |                |  |  |   |        |  |  |   |
| 1ST STAR        |    |         |  |  | 25     |                |  |  |   |        |  |  |   |
| TPAC            | SA |         |  |  | 227.90 |                |  |  | 0 |        |  |  | 0 |
| ANGLES          | TA |         |  |  | 029.40 |                |  |  | 0 |        |  |  | 0 |
| 2ND STAR        |    |         |  |  | 33     |                |  |  |   |        |  |  |   |
| TPAC            | SA |         |  |  | 074.60 |                |  |  | 0 |        |  |  | 0 |
| ANGLES          | TA |         |  |  | 025.70 |                |  |  | 0 |        |  |  | 0 |
| 3RD STAR        |    |         |  |  | 34     |                |  |  |   |        |  |  |   |

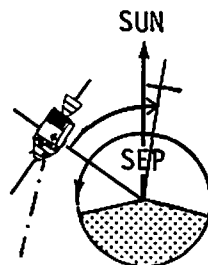
DATE 5/21/73  
5/22/73

0+00

SL-2 LIFT OFF

0+10

SL-2 INSERTION



SL-2 SEPARATION (180,0/317,0)  
COMPLETE LAUNCH C/L THRU STEP 2, Pg L/3-1

VERIFY PRELAUNCH NC1 TARGETING  
PAD COPIED (Pg 5-a)

0+20

BACKUP GDC ALIGN PAD (COPY Pg 5-a)  
LAUNCH/RNDZ REALIGN PAD (COPY Pg 5-a)

PERFORM OPTICS DUST COVER  
JETTISON CHECKLIST (Pg 5-1)

UNSTOW & MOUNT ORDEAL BOX (U3)

0+30

## PLT SYSTEMS CHECKS

MAIN REG CHECK  
SEC RAD LEAK CHECK  
ECS POST INSERTION CONFIG  
SM RCS MONITORING CHECK  
CM RCS MONITORING CHECK  
C/W OPERATIONAL CHECK  
~~PG PURGE CHECK~~  
EPS MONITORING CHECK  
ECS MONITORING CHECK  
SECONDARY GLYCOL LOOP CHECK  
SPS MONITORING CHECK  
EXTEND DOCKING PROBE

## OPTICS DUST COVER JETTISON CHECKLIST

INSTALL OPTICS EYEPIECES (LEB)  
OPTICS PHR-ON(up)  
OPT ZERO-OFF, THEN ZERO (15 SEC)  
OPT ZERO-OFF  
OPT MODE-MAN  
OPT COUPLING CONT-DIRECT  
OPT SPEED CONT-HI  
OHC-MAX RIGHT (OBS EJECT THRU SCT)  
(SXT 40°, SCT 150° SHAFT ANGLE)

CMC FAILURE

## CMC FAILURE

DATE ~~4/20/73~~  
5/22/73 (P&I)

PAGE 5-2

0+30

0+40

0+50

0:00

7+00

PERFORM GDC & IMU LAUNCH/RNDZ  
REALIGN CHECKLIST (Pg 5-2)

ROLL 180° TO HEADS DOWN ATT

MNVR TO STAR ACQ PAD ATT (Pg 5-a)  
PERFORM IMU DRIFT CHECK (RATE-LOW)  
RECORD SA'S AND TA'S REQUIRED  
TO CENTER STARS IN SXT.  
RECORD GET WHEN FINISHED

START SCS DRIFT CHECK (BMAG 2)  
ALIGN GDC  
RESET AND START DET COUNTING UP  
RATE-HIGH

### GDC & IMU LAUNCH/RNDZ REALIGN CHECKLIST

ASSUMPTIONS: 1) GDC & IMU ARE ALIGNED  
TO LAUNCH REFSMMAT.  
2) GDC & IMU COMPARE WITHIN  
5° (IF NOT, PERFORM  
BACKUP GDC & IMU  
ALIGNMENT (6/3-18)).

FDAI SELECT-1  
FDAI SOURCE-ATT SET  
ATT SET-IMU.

SET ASCP tw TO 0,0,0  
RHC-MNVR TO 0,0,0 ON FDAI 1(IMJ)  
BMAG MODE(3)-RATE 2  
DBD/RATE-MIN/LOW  
BMAG MODE(3)-ATT 1/RATE 2  
RHC-TWEAK MNVR TO ZERO FDAI 1 ERR NEEDLES

```
SET ASCP tw to LAUNCH/RNDZ REALIGN
PAD VALUES (Pg 5-a)
ATT SET-GDC
GDC ALIGN pb-push
SET ASCP tw TO 0,0,0
```

IMU CAGE-ON(up) & HOLD (until MNVR complete)  
RHC-MNVR TO 0.0.0 ON FDAI 1(GDC)  
BMAG MODE(3)-RATE 2  
DBD/RATE-MIN/LOW  
BMAG MODE(3)-ATT 1/RATE 2  
RHC-TWEAK MNVR TO ZERO FDAI 1 ERR NEEDLES  
IMU CAGE-RELEASE

ATT SET-IMU  
RATE-HIGH

### IMU DRIFT CHECK

|                           |     | CHECK |   |   |   |   |
|---------------------------|-----|-------|---|---|---|---|
| ASCP<br>tw                | R   | X     | X | X |   |   |
|                           | P   |       |   |   |   |   |
|                           | Y   |       |   |   |   |   |
| 1ST STAR                  |     | X     | X | X | X |   |
| TPAC<br>ANGLES            | SA  | X     | X | X |   |   |
|                           | TA  |       |   |   |   |   |
| 2ND STAR                  |     | X     | X | X | X |   |
| TPAC<br>ANGLES            | SA  | X     | X | X |   |   |
|                           | TA  |       |   |   |   |   |
| TIME OF<br>DRIFT<br>CHECK | HR  | +     | 0 | 0 | 0 |   |
|                           | MIN | +     | 0 | 0 | 0 |   |
|                           | SEC | +     | 0 | X | X | X |

|            |            | RESULTS |   |  |  |   |
|------------|------------|---------|---|--|--|---|
| H2O        | R          | +       |   |  |  |   |
|            | P          | +       |   |  |  |   |
|            | Y          | +       |   |  |  |   |
| ASCP<br>tw | R          | X       |   |  |  | X |
|            | P          | X       |   |  |  | X |
|            | Y          | X       |   |  |  | X |
| DET        | $\Delta T$ | X       | X |  |  |   |

(30:00)

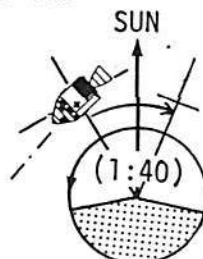
# CMC FAILURE

DATE 4/6/73  
5/22/73

1+30

TEX

NC1 PRELIMINARY PAD (COPY Pg 5-5)  
STAR ACQ PAD (COPY Pg 5-a)  
TRANSMIT IMU DRIFT DATA (Pg 5-2)  
TRANSMIT EMS ΔV TEST RESULTS



1+40

NFD

1+50

MAD

+30:00

NC1 FINAL PAD (COPY Pg 5-5)   
NC1 BURN ATT CHECK PAD (COPY Pg 5-5)  
\*\*\*\*\*IF BMAG 2>10°/HR/AXIS\*\*\*\*\*  
\* END SCS DRIFT CHECK (BMAG 1) \*  
\* GDC/IMU COMPARISON CHECKLIST (Pg 5-3) \*  
\*\*\*\*\*

\* IF NO NPC- PREP LUNCH - FOOD AT SPOONS L 3

BMAG 1 GDC/IMU COMPARISON

|            |     | RESULTS |                                                                                       |                                                                                       |                                                                                       |
|------------|-----|---------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| ASCP<br>tw | IMU | R       |  |                                                                                       |  |
|            |     | P       |  |                                                                                       |  |
|            |     | Y       |  |                                                                                       |  |
| ASCP<br>tw | GDC | R       |  |                                                                                       |  |
|            |     | P       |  |                                                                                       |  |
|            |     | Y       |  |                                                                                       |  |
| DET        | ΔT  |         |  |  | (30:00)                                                                               |

2+00

~~MNVR TO STAR ACQ PAD ATT (Pg 5 a)~~  
~~PERFORM IMU CHECK (RATE LOW)~~  
~~OHC-SET IN TPAC ANGLES OF 1ST PAD STAR~~  
~~(IF STAR<5° FROM CTR OF SCT-IMU IS GO)~~

2+00

\*\*\*\*\*IF STAR>5°FROM CTR OF SCT\*\*\*\*\*  
 \* ~~TIG>15 MIN: PERFORM BACKUP GDC ALIGN~~ \*  
 \* ~~(4/3-16) REF CHECKLIST (Pg 5-2)~~ \*  
 \* ~~AND PAD (Pg 5-5)~~ \*  
 \* ~~TIG<15 MIN: PERFORM CHECK USING~~ \*  
 \* ~~GDC, OHC AND SCT~~ \*  
 \*\*\*\*\*IF STAR>5°FROM CENTER OF SCT\*\*\*\*\*  
 \* USE 12° MARK ON WINDOW \*  
 \*\*\*\*\*

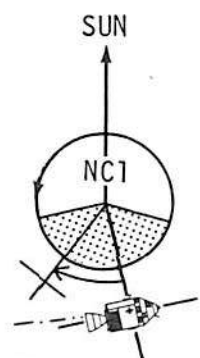
SET DET

GO TO SPS BURN CUE CARD (SCS Δ)  
 SET ASCP tw TO NC1 BURN ATT

-12:00

2+10

PERFORM BURN ATT CHECK  
 USING IMU, OHC, SCT



2+20

0:00

**NC1** (180,0/339,2)  
 RECORD BURN STATUS

PITCH UP TO TRACK EARTH HORIZON

\*\*\*\*\*IF REQUIRED\*\*\*\*\*  
 \* MNVR TO STAR ACQ PAD ATT (Pg 5-a) \*  
 \* PERFORM IMU DRIFT CHECK (RATE-LOW) \*  
 \* RECORD SA'S AND TA'S REQUIRED \*  
 \* TO CENTER STARS IN SXT. \*  
 \* RECORD GET WHEN FINISHED \*  
 \* TRACK EARTH HORIZON \*  
 \*\*\*\*\*

2+30

A  
R  
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A

# NC1 PAD DATA

|                   |                 | NOMINAL     | PRELIMINARY | FINAL |
|-------------------|-----------------|-------------|-------------|-------|
| N95               | HR              | + 0 0 2     |             |       |
| TIG NC1           | MIN             | + 2 0       |             |       |
|                   | SEC             | + 1 1 9 0   |             |       |
| N81               | ΔV <sub>X</sub> | + 2 1 9 0   |             |       |
| ΔV <sub>NC1</sub> | ΔV <sub>Y</sub> | + 0 0 0 0   |             |       |
|                   | ΔV <sub>Z</sub> | + 0 0 0 0   |             |       |
| N22               | R               | + 1 8 0 0 0 |             |       |
| NC1               | P               | + 3 3 9 0 0 |             |       |
|                   | Y               | + 0 0 2 0 0 |             |       |
|                   | ΔV <sub>C</sub> | + 2 0 4 1   |             |       |
|                   | BT              | + 0 0 1 0   |             |       |

REFER PG.1-5

## BURN ATT CHECK

|      |   |  |  |  |   |
|------|---|--|--|--|---|
| STAR |   |  |  |  |   |
| SA   |   |  |  |  | 0 |
| TA   | 0 |  |  |  | 0 |

## SPS BURN STATUS

|            |  |  |  |  |  |
|------------|--|--|--|--|--|
| ΔTIG       |  |  |  |  |  |
| AFTER TRIM |  |  |  |  |  |
| ΔVC        |  |  |  |  |  |

## IMU DRIFT CHECK

|          |     |         |  |  |  |
|----------|-----|---------|--|--|--|
| ASCP     | R   |         |  |  |  |
| tw       | P   |         |  |  |  |
|          | Y   |         |  |  |  |
| 1ST STAR |     |         |  |  |  |
| TPAC     | SA  |         |  |  |  |
| ANGLES   | TA  |         |  |  |  |
| 2ND STAR |     |         |  |  |  |
| TPAC     | SA  |         |  |  |  |
| ANGLES   | TA  |         |  |  |  |
| TIME OF  | HR  | + 0 0 0 |  |  |  |
| DRIFT    | MIN | + 0 0 0 |  |  |  |
| CHECK    | SEC | + 0     |  |  |  |

CMC FAILURE

# CMC FAILURE

2+30

H2 PURGE LINE HTR-ON(up)

CONFIGURE CAMERAS(B3): SWS APPROACH

CM2/DAC10/75/CX01-BRKT(B8)

MIR (T8,1/250,∞)/8FPS (1MAG)

CM4/HDC01/CX03 (f8, 1/250,∞) (CX04)

UNSTOW PORTABLE TIMER (B3)

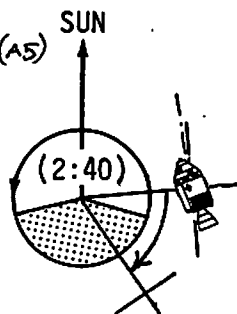
UNSTOW TV CAMERA, ZOOM LENS, CABLES (2),

MONITOR, AND CAMERA MOUNT (A5)

ASSEMBLE AND MOUNT TV CAMERA IN

RH RNDZ WINDOW

RESTOW BAGS AND DUST CAPS(A5)



2+40

2+50

FUEL CELL PURGE (S/1-3)(20 MIN  
AFTER LINE HTR-ON)

3+00

H2 PURGE LINE HTR-OFF (10 MIN  
AFTER PURGE)

FOR FLYAROUND/INSPECTION AND SEVA PHOTOS  
GET FOLLOWING AND PLACE IN TSB:

16 mm MAG CX02(B8)

70 mm MAG CX23(B8)

35 mm CAM NK03(A9)

300 mm LENS(A2)

35 mm CASSETTES CX21(A9)

CX22(A9)

BW07(A9)

LOAD NK03 WITH CX21

CONFIGURE NK WITH 300 mm LENS

STOW 55 mm LENS

CX FILM-ASA 64

BW FILM-ASA 80

FOR 300 mm LENS-1/250 OR FASTER TO PREVENT BLUR



PERFORM TV PREP CHECKLIST (Pg 5-7)  
S BAND AUX TV-TV (2 MIN PRIOR STDN AOS)

\*\*\*\*\*IF NPC REQUIRED\*\*\*  
\* NPC PAD (COPY) \*  
\*\*\*\*\*

NC2 PRELIMINARY PAD (COPY Pg 5-9)  
S BAND AUX TV-OFF (CTR) WHEN CHECKOUT COMPLETE

\*\*\*\*\*IF NPC REQUIRED\*\*\*\*\*

```
*
* SET DET
* GO TO SPS BURN CUE CARD (SCS Δ)
* SET ACSP tw TO NPC BURN ATT
* AND MNVR TO NPC BURN ATT
*
*
*
*
*
*
*
*
*
*
*
*
*
*
*
*
*
*
*
```

PNL 7 DIRECT O<sub>2</sub> VLV-CLOSE (CW)

```

***
*
*
*
*
* NPC
* RECORD BURN STATUS
* TRANSMIT NPC ΔVC LOADED AND
* SPS BURN STATUS
***

```

|                  |              |   |   |   |   |   |   |   |
|------------------|--------------|---|---|---|---|---|---|---|
| N33              | HR           | + | X | X | X |   |   |   |
| TIG NPC          | MIN          | + | X | X | X |   |   |   |
|                  | SEC          | + | X | X |   |   |   |   |
| N81              | $\Delta V_X$ | + | X |   | 0 | 0 | 0 | 0 |
| $\Delta V_{NPC}$ | $\Delta V_Y$ |   | X |   |   |   |   |   |
|                  | $\Delta V_Z$ |   | X |   |   |   |   |   |
| N22              | R            | + |   |   |   | 0 | 0 |   |
| NPC              | P            | + |   |   |   | 0 | 0 |   |
|                  | Y            | + |   |   |   | 0 | 0 |   |
|                  | $\Delta V_C$ |   | X | X |   |   |   |   |
|                  | BT           |   | X | X |   |   |   |   |

## TV PREPARATION CHECKLIST

TV SOURCE - CM  
REMOVE & STOW LENS CAP  
LINEAR SW - GAM 1/2  
AVG SW - AVG  
F-STOP - 22  
ZOOM - 25  
FOCUS - ∞  
MONITOR ON - ON  
(REFINE SETTINGS AS REQ'D)

SPS BURN STATUS

|                   |   |   |  |  |   |   |
|-------------------|---|---|--|--|---|---|
| $\Delta TIG$      | X | X |  |  | . |   |
| <b>AFTER TRIM</b> |   |   |  |  |   |   |
| $\Delta VC$       |   | X |  |  |   | . |

CMC FAILURE



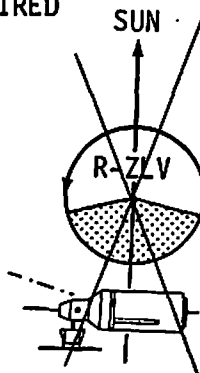
CY  
M  
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\*

1

MNVR TO STAR ACQ PAD ATT (Pg 5-a)  
PERFORM IMU DRIFT CHECK (RATE-LOW)  
RECORD SA'S AND TA'S REQUIRED  
TO CENTER STARS IN SXT.  
RECORD GET WHEN FINISHED

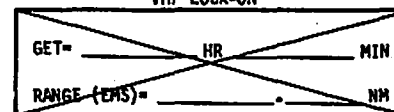


4+00

### VHF SWITCH LIST

VHF AM A-OFF(ctr)  
VHF A4 B-DUPLEX  
VHF RANGING-ON(up)  
VHF ANTENNA-RIGHT  
E4S FUNC-AV SET/VHF RNG  
EMS MODE-BACKUP/VHF RNG  
VHF RNG-RESET  
PANEL 6,9,10:  
VHF A11-RCV

### VHF LOCK-ON



## IMU DRIFT CHECK

|                           |     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |
|---------------------------|-----|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| ASCP<br>tw                | R   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
|                           | P   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
|                           | Y   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| 1ST STAR                  |     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| TPAC<br>ANGLES            | SA  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
|                           | TA  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| 2ND STAR                  |     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| TPAC<br>ANGLES            | SA  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
|                           | TA  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| TIME OF<br>DRIFT<br>CHECK | HR  | +                                   | 0                                   | 0                                   | 0                                   | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
|                           | MIN | +                                   | 0                                   | 0                                   | 0                                   | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
|                           | SEC | +                                   | 0                                   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |

4+00

4+10

4+20

-12:00

4+30

NC2 FINAL PAD (COPY)  
NCC PRELIMINARY PAD (COPY Pg 5-11)  
NSR PRELIMINARY PAD (COPY Pg 5-12)  
TRANSMIT IMU DRIFT DATA (Pg 5-8)

SET DET

~~R=300.00 NM~~  
~~R=-520.7 FPS~~

GO TO SPS BURN CUE CARD (SCS Δ)  
SET ASCP tw TO NC2 BURN ATT

~~R=300.00 NM~~  
~~R=-279.1 FPS~~

## NC2 PAD DATA

|                  |              | NOMINAL |   |   |   |       | PRELIMINARY |   |   |   |     | FINAL |   |   |   |     |
|------------------|--------------|---------|---|---|---|-------|-------------|---|---|---|-----|-------|---|---|---|-----|
| N28              | HR           | +       | X | X | X | 0 0 4 | +           | X | X | X |     | +     | X | X | X |     |
| TIG NC2          | MIN          | +       | X | X | X | 3 6   | +           | X | X | X |     | +     | X | X | X |     |
|                  | SEC          | +       | X | 0 | 3 | 3 0   | +           | X |   |   |     | +     | X |   |   |     |
| N81              | $\Delta V_X$ | +       | X | 1 | 6 | 0 7   | +           | X |   |   |     | +     | X |   |   |     |
| $\Delta V_{NC2}$ | $\Delta V_Y$ | +       | X | 0 | 0 | 0 0   | +           | X | 0 | 0 | 0 0 | +     | X | 0 | 0 | 0 0 |
|                  | $\Delta V_Z$ | +       | X | 0 | 0 | 0 0   | +           | X |   |   |     | +     | X |   |   |     |
| N22              | R            | +       | 1 | 8 | 0 | 0 0   | +           |   |   | 0 | 0   | +     |   |   | 0 | 0   |
| NC2              | P            | +       | 1 | 5 | 9 | 0 0   | +           |   |   | 0 | 0   | +     |   |   | 0 | 0   |
|                  | Y            | +       | 0 | 0 | 1 | 0 0   | +           |   |   | 0 | 0   | +     |   |   | 0 | 0   |
|                  | $\Delta V_C$ | +       | X | 1 | 4 | 5 7   | +           | X |   |   |     | +     | X |   |   |     |
|                  | BT           | +       | X | 0 | 0 | 0 7   | +           | X |   |   |     | +     | X |   |   |     |

REFER PAGE 1-9

CMC FAILURE

## CMC FAILURE

 $4+30$ 

0:00

4+40

4+50

~~32-01~~

~~28.00~~

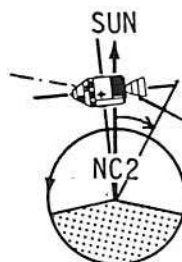
5+00

HAW

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D

B  
D  
A



189  
 NC2 (180,0/159,1)  
 RECORD BURN STATUS  
 TRANSMIT NC2 ΔVC LOADED (Pg 5-9) AND  
 SPS BURN STATUS  
 PITCH UP TO TRACK EARTH HORIZON  
~~REACQUIRE VHF RANGING (R-236.00 NM)~~

```

.SET DET

```

~~R CHART DATA FOR NCC-36~~  
SET ORDEAL (FDAI 1) WITH  $\theta_D=340, H=207$   
COAS BORESIGHTED ON HORIZON

~~R CHART DATA FOR NCC-32~~

~~R CHART DATA FOR NCC-28~~

```

*****IF REQUIRED*****
* MNVR TO STAR ACQ PAD ATT (Pg 5-a) *
* PERFORM IMU DRIFT CHECK (RATE-LOW) *
* RECORD SA'S AND TA'S REQUIRED *
* TO CENTER STARS IN SXT. *
* RECORD GET WHEN FINISHED *
* TRACK EARTH HORIZON *
*****

```

## SPS BURN STATUS

|              |   |   |  |  |   |   |
|--------------|---|---|--|--|---|---|
| $\Delta TIG$ | X | X |  |  | . |   |
| AFTER TRIM   |   |   |  |  |   |   |
| $\Delta VC$  |   | X |  |  |   | . |

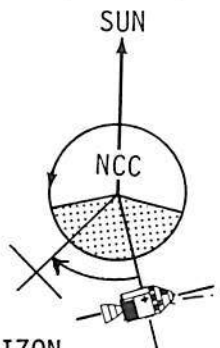
|               |  |   |   |   |   |   |   |
|---------------|--|---|---|---|---|---|---|
| N11 (TIG HCC) |  | . | . | . | . | . | . |
| -             |  | . | . | . | . | . | . |
| NCC-36        |  | . | . | . | . | . | . |

## IMU DRIFT CHECK

|                     |          |   |   |   |   |   |   |   |   |   |   |
|---------------------|----------|---|---|---|---|---|---|---|---|---|---|
| AJCP                | R        |   | X | X | X |   |   |   |   |   | . |
| tw                  | P        |   | X | X | X |   |   |   |   |   | . |
|                     | Y        |   | X | X | X |   |   |   |   |   | . |
|                     | 1ST STAR |   | X | X | X |   |   |   |   |   |   |
| TPAC ANGLES         | SA       |   | X | X | X |   |   |   |   | . |   |
|                     | TA       |   | X | X | X |   |   |   |   | . |   |
|                     | 2ND STAR |   | X | X | X |   |   |   |   |   |   |
| TPAC ANGLES         | SA       |   | X | X | X |   |   |   |   | . |   |
|                     | TA       |   | X | X | X |   |   |   |   | . |   |
|                     | HR       | + | 0 | 0 | 0 |   |   |   |   |   |   |
| TIME OF DRAFT CHECK | MIN      | + | 0 | 0 | 0 |   |   |   |   |   |   |
|                     | SEC      | + | 0 |   |   | X | X | X | X | X | X |

$5+30$ 

SET DET  
PERFORM VHF SWITCH LIST (Pg 5-8)  
ACQUIRE VHF RANGING (R=133.00NM)



|                  |              | NCC TIME |  |  |  |   |   | TIG TIME |   |   |   |  |  | TIME |  |  |   |   |  |  |  |  |  |  |  |  |
|------------------|--------------|----------|--|--|--|---|---|----------|---|---|---|--|--|------|--|--|---|---|--|--|--|--|--|--|--|--|
| N11              | HR           |          |  |  |  | 0 | 0 | 5        |   |   |   |  |  |      |  |  |   |   |  |  |  |  |  |  |  |  |
| TIG              | NCC MIN      | +        |  |  |  |   |   | 2        | 2 | + |   |  |  |      |  |  |   | + |  |  |  |  |  |  |  |  |
|                  | SEC          | +        |  |  |  | 1 | 2 | 5        | 0 | + |   |  |  |      |  |  |   | + |  |  |  |  |  |  |  |  |
| N81              | $\Delta V_X$ | +        |  |  |  | 0 | 3 | 0        | 5 |   |   |  |  |      |  |  |   |   |  |  |  |  |  |  |  |  |
| $\Delta V_{NCC}$ | $\Delta V_Y$ | +        |  |  |  | 0 | 0 | 0        | 0 |   |   |  |  |      |  |  |   |   |  |  |  |  |  |  |  |  |
|                  | $\Delta V_Z$ | -        |  |  |  | 0 | 0 | 0        | 9 |   |   |  |  |      |  |  |   |   |  |  |  |  |  |  |  |  |
|                  |              |          |  |  |  |   |   |          |   |   |   |  |  |      |  |  |   |   |  |  |  |  |  |  |  |  |
| N22              | R            | +        |  |  |  | 1 | 8 | 0        | 0 | 0 |   |  |  |      |  |  | + |   |  |  |  |  |  |  |  |  |
| NCC              | P            | +        |  |  |  | 3 | 4 | 0        | 0 | 0 | + |  |  |      |  |  |   | + |  |  |  |  |  |  |  |  |
|                  | Y            | +        |  |  |  | 0 | 0 | 1        | 0 | 0 | + |  |  |      |  |  |   | + |  |  |  |  |  |  |  |  |
|                  | $\Delta V_C$ |          |  |  |  |   |   |          |   |   |   |  |  |      |  |  |   |   |  |  |  |  |  |  |  |  |
|                  |              |          |  |  |  | 0 | 1 | 6        | 6 |   |   |  |  |      |  |  |   |   |  |  |  |  |  |  |  |  |

```

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
X N81 COMP LIMITS +1.0,NA,+15.0 FPS
X
X USE STDN AVV IN ALL CASES
X
X CHART/STDN WITHIN LIMITS GO STDN
X NO AGREEMENT-NORMALLY GO STDN
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

```

~~USING VECTOR CONVERSION CHART  
CHANGE BVX & BVZ TO  
ORDEAL BURN ATT & AVT~~[illegible]

|                                  |   |   |   |   |   |   |   |
|----------------------------------|---|---|---|---|---|---|---|
| $\Delta TIG$                     |   | X | X | X | X | X | . |
| FDAI<br>(FOR CHART<br>BURN ONLY) | R | X | X | X | X | X |   |
|                                  | P | X | X | X | X | X |   |
|                                  | Y | X | X | X | X | X |   |
| <b>AFTER TRAIN</b>               |   |   |   |   |   |   |   |
| $\Delta V_c$                     |   | X |   |   |   |   | . |

|               |  |   |   |  |  |  |
|---------------|--|---|---|--|--|--|
| N11 (TIG NCC) |  |   |   |  |  |  |
| +             |  | 0 | 9 |  |  |  |
| NSR-28        |  |   |   |  |  |  |

CMC FAILURE



# CMC FAILURE

5+30

5+40

5+50

6+00

R CHART DATA POINTS LOCATED  
AT ~~180~~ MINUS 28, 24, 20, 12  
NSR

SWS IN R-ZLV

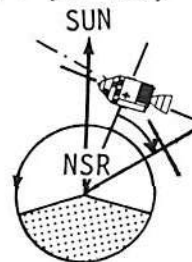
SET ORDEAL (FDAI 1) WITH  $\theta_D = 340, H = 217$   
COAS BORESIGHTED ON HORIZON

TRANSMIT NCC  $\Delta V_C$  LOADED AND  
SPS BURN STATUS (Pg 5-11)

IF NCC EXECUTION IS QUESTIONABLE  
THEN UNMATCHED CHART SOLN IS PRIME

GO TO SPS BURN CUE CARD (SCS  $\Delta$ )  
(FOR CHART SOLUTION)  
MNVR TO AND MAINTAIN ORDEAL BURN ATT  
(BURN  $\Delta V_T$  SPS)  
(BURN  $\Delta V_Y$  TIMED RCS), WITH  
AUTO RCS SELECT (16)-MNA/MNB  
RECORD FDAI 2 INRTL BURN ATT (AT TIG)

RCS X AXIS  
1FPS=2.35 SEC  
RCS Y&Z AXES  
1FPS=4.45 SEC



0/226,0  
NSR (180, 3/198, 2)  
RECORD BURN STATUS

## NSR PAD DATA

TIG NSR(N13)=TIG NCC(N11) + 00:37:00= : : .

|                  |              | NOMINAL     | PRELIMINARY | FINAL |
|------------------|--------------|-------------|-------------|-------|
| N81              | $\Delta V_X$ | + 0 1 9 7   |             |       |
| $\Delta V_{NSR}$ | $\Delta V_Y$ | + 0 0 0 0   |             |       |
|                  | $\Delta V_Z$ | - 0 0 0 9   |             |       |
| N22              | R            | + 1 8 0 0 0 | + 0 0       | + 0 0 |
| NSR              | P            | + 1 9 8 0 0 | + 0 0       | + 0 0 |
|                  | Y            | + 0 0 2 0 0 | + 0 0       | + 0 0 |
|                  | $\Delta V_C$ | + 0 0 5 1   |             |       |

REFER Pg 1-12

## NSR ONBOARD DATA

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
X N81 COMP LIMITS NA,NA,NA PPS X  
X USE STON  $\Delta V_Y$  IN ALL CASES X  
X BURN STON OR CHART SOLN X  
X IF NCC HAS BURNED STON X  
X OR CHART, RESPECTIVELY X  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

### NSR CHART SOLN

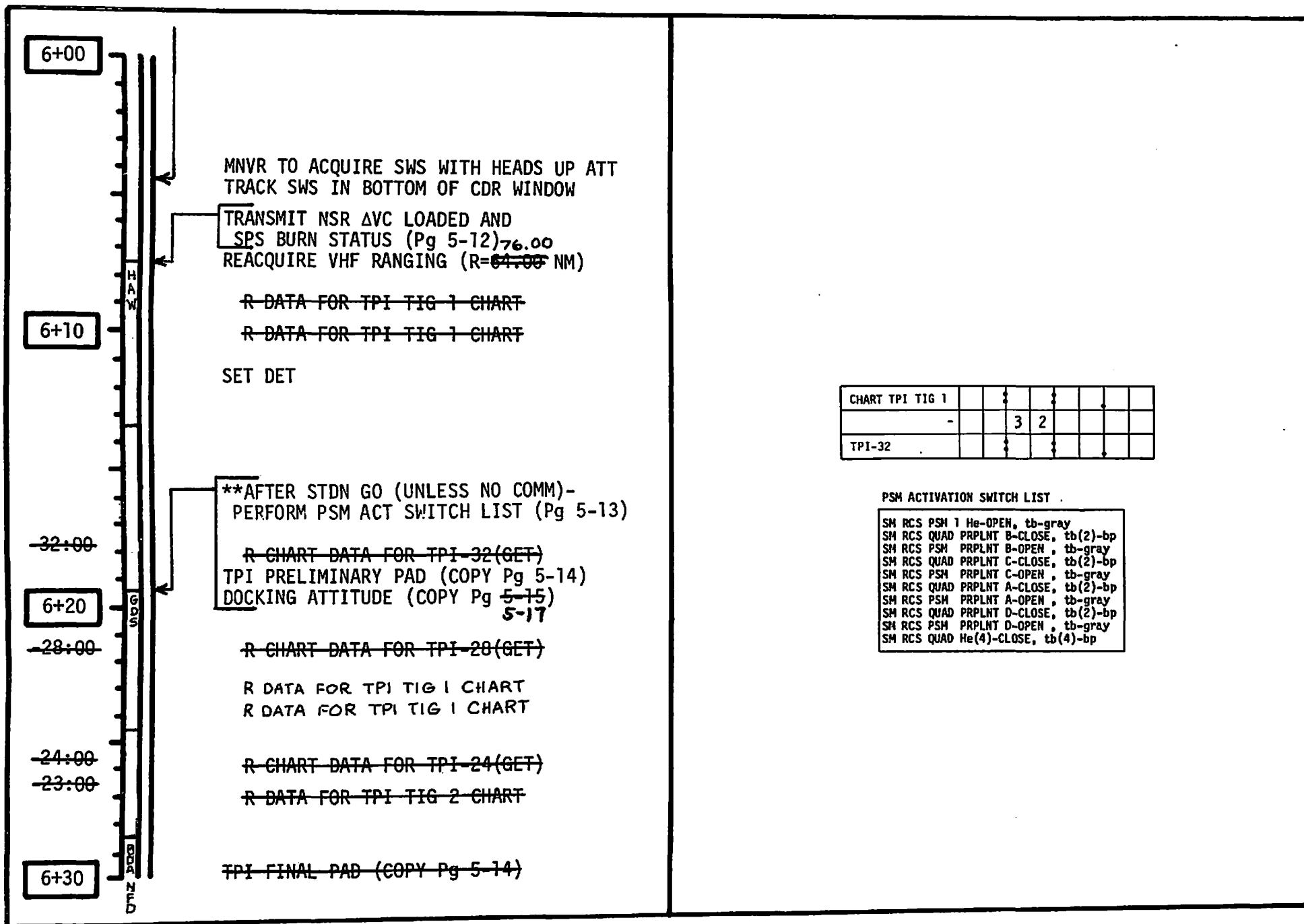
|              |   |   |   |   |   |
|--------------|---|---|---|---|---|
| $\Delta V_X$ |   |   |   |   |   |
| $\Delta V_Y$ | * | S | T | D | N |
| $\Delta V_Z$ |   |   |   |   |   |

|                         |   |  |   |   |   |
|-------------------------|---|--|---|---|---|
| ORDEAL BURN ATT         |   |  |   |   |   |
| $\Delta V_T$ (IN PLANE) | + |  |   |   |   |
| $\Delta V$ TAIL OFF     | - |  | 1 | 4 | 0 |
| $\Delta V_C$            | + |  |   |   |   |
| $\Delta V_R$ BT         |   |  |   |   |   |

USING VECTOR CONVERSION CHART  
CHANGE  $\Delta V_X$  &  $\Delta V_Z$  TO  
ORDEAL BURN ATT &  $\Delta V_T$

### SPS BURN STATUS

|                            |   |  |  |  |  |
|----------------------------|---|--|--|--|--|
| $\Delta TIG$               |   |  |  |  |  |
| FDAI (FOR CHART BURN ONLY) | R |  |  |  |  |
|                            | P |  |  |  |  |
|                            | Y |  |  |  |  |
| AFTER TRIM                 |   |  |  |  |  |
| $\Delta V_C$               |   |  |  |  |  |

DATE 4/6/73  
5/22/73

CMC FAILURE

6+30

-32:00

B  
D  
A  
N  
F  
D  
M  
I  
L

R CHART DATA FOR TPI -32 (GET)

-28:00

R CHART DATA FOR TPI -28 (GET)

-24:00

R CHART DATA FOR TPI -24 (GET)

6+40

R DATA FOR TPI TIG 2 CHART  
V32E (2ND P35 RECYCLE: RECORD)

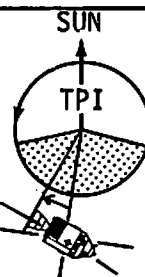
-16:00

R CHART DATA FOR TPI -16 (GET)  
MNVR TO BORESIGHT COAS ON HORIZON  
SET ORDEAL (FDAI 1)  
TPI FINAL PAD (COPY) $\theta_D = 339, H = 231$ 

6+50

-12:00

SET DET

GO TO SPS BURN CUE CARD (SCS  $\Delta$ )  
(FOR CHART SOLUTION)MNVR TO AND MAINTAIN ORDEAL BURN ATT  
(BURN  $\Delta VT$  SPS)(BURN  $\Delta VY$  TIMED RCS), WITH  
AUTO RCS SELECT (16)-MNA/MNBRCS X AXIS  
1FPS=2.35 SEC  
RCS Y&Z AXES  
1FPS=4.45 SEC

## TPI PAD DATA

|                  |                 | NOMINAL |   |   |   |         | PRELIMINARY |   |   |   |   | FINAL |   |   |   |   |
|------------------|-----------------|---------|---|---|---|---------|-------------|---|---|---|---|-------|---|---|---|---|
| N37              | HR              | +       | X | X | X | 0 0 6   | +           | X | X | X | X | +     | X | X | X | X |
| TIG TPI MIN      |                 | +       | X | X | X | 4 9     | +           | X | X | X | X | +     | X | X | X | X |
|                  | SEC             | +       | X | X | X | 3 8 7 0 | +           | X | X | X | X | +     | X | X | X | X |
| N81              | $\Delta V_X$    | +       | X | X | X | 1 8 8   |             | X | X | X | X |       | X | X | X | X |
| $\Delta V_{TPI}$ | $\Delta V_Y$    | +       | X | X | X | 0 0 0   |             | X | X | X | X |       | X | X | X | X |
|                  | $\Delta V_Z$    | -       | X | X | X | 0 8 3   |             | X | X | X | X |       | X | X | X | X |
| N59              | $\Delta V_F/BT$ | +       | 2 | 0 | 5 | 4 8     |             | X | X | X | X |       | X | X | X | X |
| $\Delta V$       | $\Delta V_R/BT$ | +       | 0 | 0 | 0 | 0 0     |             | X | X | X | X |       | X | X | X | X |
| (LOS)            | $\Delta V_D/BT$ | +       | 0 | 1 | 1 | 0 5     |             | X | X | X | X |       | X | X | X | X |
|                  | $\Delta V_C$    | +       | X | X | X | 0 0 6 8 |             | X | X | X | X |       | X | X | X | X |

REFER Pg 1-14

## TPI ONBOARD DATA

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
 X N81 COMP LIMITS  $\pm 1.0, NA, \pm 3.0$  FPS X  
 X X X  
 X USE STDN  $\Delta VY$  IN ALL CASES X  
 X X X  
 X CHART/STDN WITHIN LIMITS GO CHART X  
 X NO AGREEMENT GO STDN X  
 XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

## TPI CHART SOLN

|              |   |   |   |   |   |
|--------------|---|---|---|---|---|
| $\Delta V_X$ | X | X | X | X | X |
| $\Delta V_Y$ | * | S | T | D | N |
| $\Delta V_Z$ | X | X | X | X | X |

|                        |   |   |   |       |   |
|------------------------|---|---|---|-------|---|
| CHART TPI TIG 2        |   |   |   |       |   |
| ORDEAL BURN ATT        | X | X | X | X     | X |
| $\Delta VT$ (IN PLANE) | + | X | X | X     | X |
| $\Delta V$ TAIL OFF    | - | X | X | 1 4 0 | X |
| $\Delta V_C$           | + | X | X | X     | X |
| $\Delta V_R BT$        |   | X | X | X     | X |

USING VECTOR CONVERSION CHART  
 CHANGE  $\Delta V_X$  &  $\Delta V_Z$  TO  
 ORDEAL BURN ATT &  $\Delta VT$

7+00

7+00

0:00

**TP1** (0,25/25,0)

RATE-LOW

REACQUIRE VHF RANGING (R=21.00 NM)

CENTER SWS IN COAS WHEN  
READING CHART DATA  $\theta$ 

+4:30

 $\theta$  CHART DATA FOR TPI +4:30

+7:30

R CHART DATA FOR TPI +7:30

7+10

SWS MNVR TO SI  
AT 7+10 GET

+8:30

R AND  $\theta$  CHART DATA FOR TPI +8:30

THC PWR-ON(up)

**TPM1** BURN ASAP WITH SWS BORESIGHTED IN COAS  
( $\Delta V_F$ , RCS WITH EMS) ( $\Delta V_D$ , TIMED RCS)

REACQUIRE VHF RANGING (R=8.00 NM)

CENTER SWS IN COAS WHEN  
READING CHART DATA  $\theta$ 

+16:30

 $\theta$  CHART DATA FOR TPI +16:30

7+20

GET OUT SUNGLASSES

RCS X AXIS  
1FPS=2.35 SEC  
RCS Y&Z AXES  
1FPS=4.45 SEC

+19:30

R CHART DATA FOR TPI +19:30

+20:30

R AND  $\theta$  CHART DATA FOR TPI +20:30

THC PWR-ON(up)

**TPM2** BURN ASAP WITH SWS BORESIGHTED IN COAS  
( $\Delta V_F$ , RCS WITH EMS) ( $\Delta V_D$ , TIMED RCS)

REACQUIRE VHF RANGING

MONITOR LOS RATES AND CALCULATE AVE  $\dot{R}$ 

FOR R=1.5 NM

 $\dot{R} = (\Delta R(\text{nm}) / 1 \text{ MIN}) * 100$ 

PERFORM TV PREP CHECKLIST (Pg 5-15)

PERFORM PRE-BRAKING SWITCH LIST (Pg 5-15)

PERFORM EVAP DEACT SWITCH LIST (Pg 5-15)

PNL 252 WASTE STOW VENT VLV-CLOSE (CW)

## TPM1 ONBOARD DATA

## TPM1 CHART SOLN



## TPM2 ONBOARD DATA

## TPM2 CHART SOLN



## PRE-BRAKING SWITCH LIST

|                     |                             |
|---------------------|-----------------------------|
| FDAI SCALE-5/1      | RHC PWR NORMAL(2)-AC/DC     |
| FDAI SELECT-1       | RHC PWR DIRECT(2)-MNA/MNB   |
| FDAI SOURCE-ATT SET | SC CONT/MODE-SCS/FREE       |
| ATT SET-IMU         | B'MAG MODE(3)-ATT 1/RATE 2  |
| MAN ATT(3)-RATE C'D | AUTO RCS SELECT(16)-MNA/MNB |
| LIMIT CYCLE-OFF     | THC-ARMED                   |
| DBD/RATE-MIN/LOW    | RHC#2-ARMED                 |
| THC PWR-ON(up)      |                             |

## GLYCOL EVAPORATOR DEACTIVATION SWITCH LIST

|                                           |
|-------------------------------------------|
| GLYCOL EVAP H2O FLOW-OFF                  |
| GLYCOL EVAP STEAM PRESS-MAN               |
| GLYCOL EVAP STEAM PRESS-INCR (for 58 sec) |

## TV PREPARATION CHECKLIST

|                            |
|----------------------------|
| TV SOURCE-CM               |
| REMOVE & STOW LENS CAP     |
| LINEAR SW-LINEAR GAM 1/2   |
| AVG SW-PEAK AVG            |
| F-STOP-16 22               |
| ZOOM-25 as required        |
| FOCUS- $\infty$            |
| MONITOR ON-ON              |
| (REFINE SETTINGS AS REQ'D) |

CMC FAILURE



7+30

S BAND AUX TV-TV (2 MIN PRIOR STDN AOS)

NULL LOS RATES AND

ADJUST R TO BRAKING GATES (TIMED RCS)  
 DAC-ON (1000 FT), HDC -30 FRAMES (1000 FT)  
~~CONFIGURE NK WITH 300MM LENS, STOP~~  
~~55MM LENS NK-10 FR (IF DESIRED)~~

STATION KEEP ON SWS +X AXIS (R ~100 FT)  
 (MAINTAIN ROLL ~0° WITH SUN ON BOTTOM  
 OF S/C)

SPOT LIGHT - ON (DOWN)  
 S BAND AUX TV-OFF (CTR) AT STDN LOS  
 CONFIGURE COMM (PANEL 9)

MODE - VOX

VOX SENS - AS REQD (~6)

S-BD - T/R (VERIFY)

AUDIO CONT - NORM (VERIFY)

INTERCOM - T/R (VERIFY)

REMOVE TV CAMERA FROM BRKT AND HANDHOLD  
 CM4/NK-60 FRAMES/HDC - 100 FRAMES (FB, 1/250)

MAN ATT (3) - MIN IMP

MNVR TO WING #1 (AVOID ATM SOLAR ARRAYS  
 AND 45' DISCONE ANTENNAS)

USE METERING SYSTEM FOR NK PHOTOS  
 PHOTO DAMAGED AREAS USING NK FOR CLOSEUPS  
 AND HDC FOR WIDE-ANGLES (IF VERY HIGH  
 RESOLUTION IS DESIRED, LOAD BN67 IN NK)

REPLACE TV CAMERA ON BRKT  
 S BAND AUX TV-TV (2 MIN PRIOR STDN AOS)

SAS WING #1 TV (R~60ft)

~~E-STOP-PC (IF TV CAMERA IMAGES SUN)~~

TAKE HDC PICTURES OUT CM4 BY TV

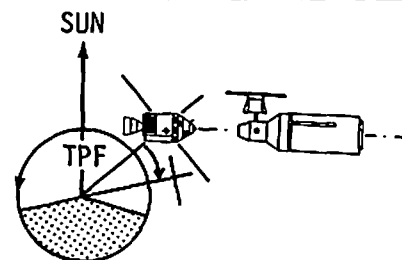
CAMERA OR CM3

TAKE NK PICTURES OUT CM3 IF POSSIBLE

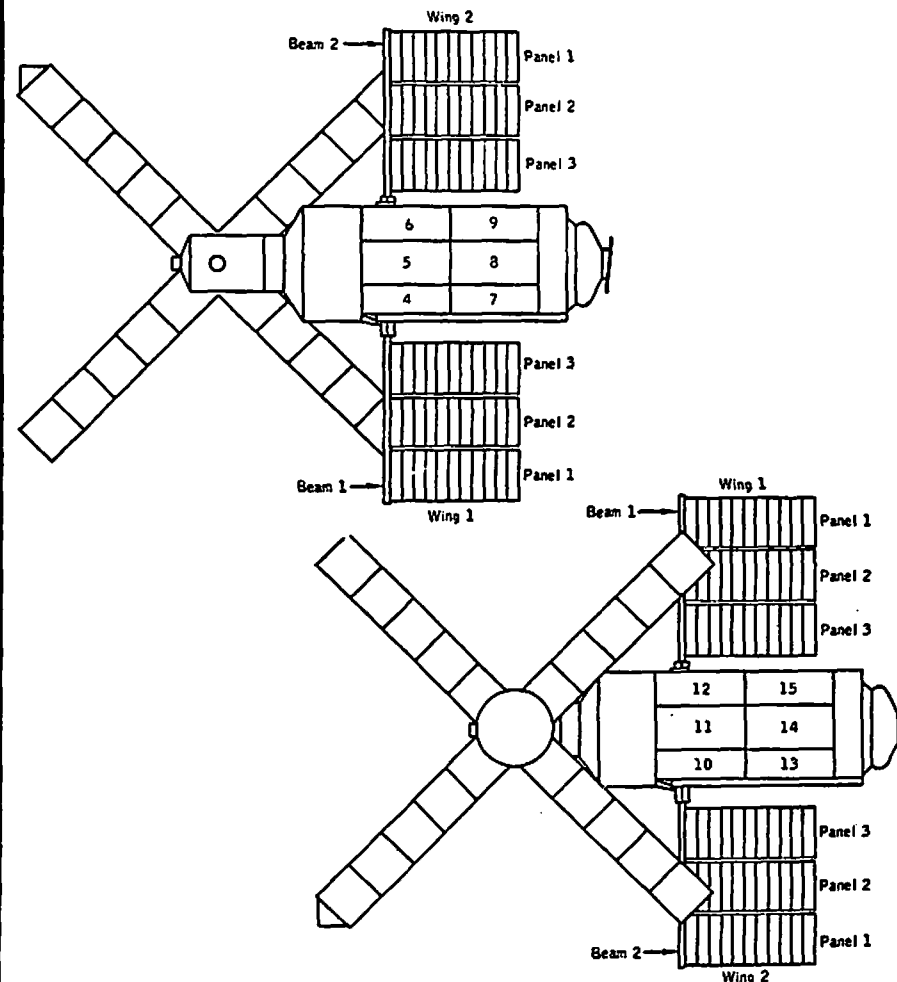
7+40

7+50

8+00



| BRAKING GATES |         |         |                         |        |
|---------------|---------|---------|-------------------------|--------|
| R (1000 FT)   | R (FPS) | SWS DIA | RETICLE ANGLE (DEG)     | R (FT) |
|               |         |         | SOLAR PANEL (BETA ~18°) |        |
| 1.00          | 30      | .2      | 8                       | 6000   |
| .50           | 20      | .4      | 7                       | 3000   |
| .25           | 10      | .8      | 5                       | 1500   |
| .08           | 5       | 2.5     | 10.0                    | 500    |
| .05           |         | 4.2     |                         | 300    |
| .03           |         | 6.3     |                         | 200    |
| .02           |         | 10.0    |                         | 130    |



GO TO LAUNCH CHECKLIST FOR  
SEVA PROCEDURES

### DOCKING ATTITUDE

**NOMINAL**

## UPDATE

|      |   |  |  |  |  |   |   |   |  |  |  |  |  |  |
|------|---|--|--|--|--|---|---|---|--|--|--|--|--|--|
| ASCP | R |  |  |  |  | 1 | 9 | 1 |  |  |  |  |  |  |
| tw   | P |  |  |  |  | 0 | 3 | 5 |  |  |  |  |  |  |
|      | Y |  |  |  |  | 0 | 0 | 6 |  |  |  |  |  |  |

### PRE-SOFT DOCKING SWITCH LIST

|                                   |                                    |
|-----------------------------------|------------------------------------|
| cb DOCK PROBE(2)-CLOSE            | SPOT LIGHT-ON(down)                |
| DOCKING PROBE RETRACT(2)-OFF(ctr) | <del>SECS PYRO ARM(2)-ON(up)</del> |
| DOCKING PROBE EXTD/REL-RETRACT    | FC REACS VALVES-LATCH              |
| DOCKING PROBE EXTD/REL tb(2)-gray |                                    |

## SOFT DOCKING CHECKLIST

TO INITIATE CAPTURE

VERIFY STATION KEEPING ON SWS +X AXIS  
(RANGE=40 TO 100 FT) (DOCK TGT=1.8 TO 0.8° IN COAS)  
LOAD E1'S WITH 100.0 FPS (-100.0 FPS)  
E1'S FUNC/MODE=ΔV/HORZ'L  
DAC-ON  
THC-INITIATE 0.4 ±0.1 FPS CLOSING RATE  
RHC-MAINTAIN MINIMUM RELATIVE ALIGNMENT ANGLES

AT CAPTURE (IF NO CAPTURE-SEE BACKUP DOCKING PROCEDURES (S/2-1))

DOCKING PROBE EXTD/REL tb(either)-bp (IF GRAY-SEE DOCKING MALFUNCTION  
PROCEDURES #2, Pg 12-3 IN CS4 MALFUNCTION PROCEDURES BOOK)  
B\*AG MODE(3)-RATE 2  
RHC PWR DIRECT(2)-OFF(ctr)  
REPORT CAPTURE TO STDN  
ALLOW PROBE TO DAMP SC MOTION (10 SEC)  
THC=NULL. PITCH AND YAW MISALIGNMENT<3°  
RHC=NULL ROLL MISALIGNMENT<5°  
DOCKING PROBE RETRACT PRIM-1 (SEC 2 IF REQ)

~~AT DOCK LATCH~~

~~DOCKING PROBE EXTD/REL tb(2)-gray~~

**AFTER SOFT DOCK**

```
SECS PYRO ARM(2) SAFE      EMS FUNC/MODE-OFF/STBY
SECS LOGIC(2) OFF          DBD/RATE-MIN/LOW
cb SECS ARM(2) OPEN        THC PHR-OFF
cb DOCK PROBE(2)-OPEN      RHC PHR NORVAL#1-OFF(ctr)
DOCKING PROBE EXTG/REL OFF  RHC PHR DIRECT(2)-OFF(ctr)
DOCKING PROBE RETRACT(2) OFF(ctr) SC CONT/MODE-SCS/FREE
DACK-OFF                  B'IAIG MODE(3)-RATE 2
SPOT LIGHT-OFF(up)         -AUTO RES-SELECT(A1,B2,A3,C4,B3,B4)-OFF
PG REACS-VALVES NORM      -cb SCS A/C & D/D ROLL(4) OPEN
VHF A4 B-OFF(ctr)         -cb SCS PITCH & YAW(4) OPEN
VHF RANGING-OFF           THC-LOCKED
GOAS PHR OFF              RHC(2)-LOCKED
```

# IMU FAILURE

- (I) WHEN IMU FAILURE OCCURS  
 1.) PERFORM STEPS (II), (III), (IV) AND (V)  
 2.) REQUEST EMP SL-50 UPLINK AT 1ST AVAILABLE OPPORTUNITY [MANUAL LOAD Pg 6-c]  
 3.) PROCEED TO APPROPRIATE POINT IN THE TIMELINE

- (II) IMU FAILURE SWITCH LIST  
 SC CONT-SCS  
 IMU PWR-OFF  
 DBD/RATE-MIN/HIGH  
 BMAG MODE(3)-ATT 1/RATE 2  
 MAN ATT(3)-RATE CMD  
 FDAI SELECT-1  
 FDAI SOURCE-ATT SET  
 ATT SET-GDC  
 LIMIT CYCLE-OFF

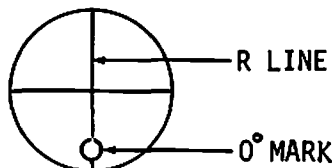
- (III) SET REFSMMAT FLAG  
 KEY V37E00E  
 V25N7E, 77E, 10000E, 1E

- (IV) INITIALIZE NO-DAP CONFIG  
 (VERIFY CSM-DAP HAS BEEN ACTIVATED PRIOR TO NO-DAP CONFIG)  
 KEY V48E, V21E, 00000E, PRO, PRO, PRO, V46E

- (V) SINGLE SENSOR POST TPI  
 KEY V57E, V22E, 1E, PRO

## NOTES

- 1.) N20 NEED NOT AGREE WITH CURRENT SC ATTITUDE EXCEPT WHEN TAKING AN OPTICS MARK, PERFORMING COAS LOS DETERMINATION, OR IN V83 FOR 9
- 2.) LOAD N81 DURING FINAL COMP TO AGREE WITH ΔV'S TO BE BURNED; LOAD N59 FOR TPI'S
- 3.) IF LOS ATTITUDE IS DESIRED (WHEN P20 IS RUNNING) KEY V22N78E, +E, V16N22E
- 4.) RECOMMEND TRACKING OF HORIZON AND SHS: DBD/RATE-MIN/HIGH MAN ATT PITCH-MIN I'P
- 5.) EMP SL-50 CONVERTS GYRO TORQUING ANGLES (H93) INTO RESULTANT ACTUAL COU ANGLES (N22) AND SETS N20=N22.
- 6.) USE VHF MARKING TO UPDATE CMC STATE VECTOR
- 7.) BACKUP GDC ALIGN STAR PAIR IS VISIBLE FROM SS+6 MIN TO -SR SS TO SR



## DEFINITIONS

- θ<sub>0</sub> = ORDEAL ANGLE TO PLACE EARTH HORIZON IN CENTER OF COAS  
 H = AVERAGE ALTITUDE  
 θ = ORDEAL FDAI PITCH

## NC1 TARGETING PAD

|               |     | NOMINAL |   |   |   | NOMINAL UPDATE |   |   |   | PRELAUNCH UPDATE |   |   |   |
|---------------|-----|---------|---|---|---|----------------|---|---|---|------------------|---|---|---|
| N95           | HR  | +       | + | + | + | +              | + | + | + | +                | + | + | + |
| TIG NC1       | MIN | +       | + | + | + | +              | + | + | + | +                | + | + | + |
|               | SEC | +       | + | + | + | +              | + | + | + | +                | + | + | + |
| N57 HALF REVS |     | +       | + | + | + | +              | + | + | + | +                | + | + | + |
| N37           | HR  | +       | + | + | + | +              | + | + | + | +                | + | + | + |
| TIG TPI       | MIN | +       | + | + | + | +              | + | + | + | +                | + | + | + |
|               | SEC | +       | + | + | + | +              | + | + | + | +                | + | + | + |

## TALIGN PAD

|       |     | NOMINAL |   |   |   | NOMINAL UPDATE |   |   |   | PRELAUNCH UPDATE |   |   |   |
|-------|-----|---------|---|---|---|----------------|---|---|---|------------------|---|---|---|
| N34   | HR  | +       | + | + | + | +              | + | + | + | +                | + | + | + |
| GET   | MIN | +       | + | + | + | +              | + | + | + | +                | + | + | + |
| ALIGN | SEC | +       | + | + | + | +              | + | + | + | +                | + | + | + |

## BACKUP GDC ALIGN PAD

|                  |   | NOMINAL |   |   |   | NOMINAL UPDATE |   |   |   | UPDATE |   |   |   |
|------------------|---|---------|---|---|---|----------------|---|---|---|--------|---|---|---|
| 0°/R STARS       |   | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| ASCP tw          | R | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| (R:IDZ REFSMMAT) | P | +       | + | + | + | +              | + | + | + | +      | + | + | + |
|                  | Y | +       | + | + | + | +              | + | + | + | +      | + | + | + |

## STAR ACQUISITION PAD

|                  |    | NOMINAL |   |   |   | NOMINAL UPDATE |   |   |   | UPDATE |   |   |   |
|------------------|----|---------|---|---|---|----------------|---|---|---|--------|---|---|---|
| N22 (NC1)        | R  | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| (R:IDZ REFSMMAT) | P  | +       | + | + | + | +              | + | + | + | +      | + | + | + |
|                  | Y  | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| N71 1ST STAR     |    | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| TPAC             | SA | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| ANGLES           | TA | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| N71 2ND STAR     |    | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| TPAC             | SA | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| ANGLES           | TA | +       | + | + | + | +              | + | + | + | +      | + | + | + |
| N71 3RD STAR     |    | +       | + | + | + | +              | + | + | + | +      | + | + | + |

# GDC REFSMMAT REALIGN (P52)

ASSUMPTIONS: ~~GDC IS ALIGNED TO IMU REFSMMAT BEFORE~~  
~~IMU FAILED. IF NOT, PERFORM GDC~~  
~~REFSMMAT DETERMINATION (G/3-20)~~  
 EMP SL-50 IS LOADED (FOR OPT 3)  
 CMC-ON, SCS OPERATING, IMU-OFF  
 OPTICS ARE ZEROED

1. RHC-Mnvr to acquire star pair (if required)  
 BMAG MODE(3)-RATE 2  
 DBD/RATE-MIN/LOW  
 BMAG MODE(3)-ATT 1/RATE 2  
 ZERO FDAI 1 err needles with ASCP tw  
 V25N20E  
 Load N20 with ASCP tw angles  
 V37E52E
2. F 04 06 R1 00001 ALIGN OPTION  
 R2 00002 NOM PRO to 3  
 3 REFSMMAT PRO to 6
3. F 06 34 GET ALIGN (0,0,0 initially)(hr, min, .01 sec)  
 Load GET ALIGN (Pg 6-a)  
 PRO
4. F 06 22 NEW ICDU ANGLES OG, IG, MG (.01°)  
 (If MG>±70°,MNVR and reload N20)  
 V32E to 4  
 Set ASCP tw to N22 values  
 GDC ALIGN pb-push  
 PRO
5. F 50 25 00013 GYRO TORQUE  
 PRO (NO ATT lt-on then off,  
 PROG ALM-ignore,  
 will take 5 to 6 minutes before  
 F 50 25 00015 is displayed)

6. F 50 25 00015 ACQ STARS  
 (opt 2) RATE-HIGH  
 V37E00E-procedures complete  
 (opt 3) LIMIT CYCLE-ON(up)  
 PRO  
 \*F 05 09 00405 NO PAIR\*  
 \*PRO to 7 \*
7. F 01 70 000DE STAR CODE  
 Load STAR CODE (if desired)  
 OPT MODE-CMC(verify)  
 OPT ZERO-OFF  
 PRO  
 \*F 05 09 00404 TA>90°\*  
 \*MNVR and reload N20 \*  
 \*PRO to 8 \*
8. 06 92 SHAFT,TRUN (.01,.001°)  
 (Mark Routine) OPT MODE-MAN  
 (3RD STAR CHECK) V37E to 17
9. F 51 PLEASE MARK  
 N20=current GDC attitude(verify)  
 MARK
10. F 50 25 00016 TERMINATE MARKS  
 PRO
11. F 01 71 000DE STAR CODE  
 Load STAR CODE (if necessary)  
 PRO  
 1ST MARK,to 7  
 2ND MARK,to 12
12. F 06 05 ANGULAR SEP. ERROR, ANGULAR SEP. (.01°)  
 (REJECT) V32E to 16 if ANG ERROR>.4°  
 (ACCEPT) PRO

# IMU FAILURE

13. F 06 93 TORQUING ANGLES OG,IG,MG (.001°)  
N93 is indicative of BMAG drift  
since last alignment  
V5N26E  
VERIFY LOAD (10001,1642,50006)  
KEY RLSE  
V30E  
  
\*F 01 70(RESTART TO 7)\*  
\*REDO 2ND MARK \*
14. F 06 22 NEW ICDU ANGLES OG,IG,MG (.01°)  
V21N26E, E  
(REJECT) ~~V32E to 15~~ V34E to 15  
(ACCEPT) Set ASCP tw to N22 values  
GDC ALIGN pb-push  
PRO (Sets N20=N22)
15. F 06 93 TORQUING ANGLES OG,IG,MG (.001°)  
V32E
16. F 50 25 00014 ALIGNMENT CHECK  
(3RD STAR CHECK) PRO to 6  
(BYPASS) ENTER
17. F 37 00E  
OHC-Drive trun<10°  
OPT ZERO-ZERO  
LIMIT CYCLE-OFF  
RATE-HIGH

## EMP SL-50 MANUAL LOAD

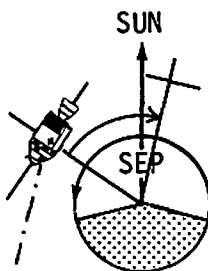
| DSKY<br>LOAD  | (REG 3<br>ADDRESS) |
|---------------|--------------------|
| V21NTE        | 77776E             |
| 3242E         | E                  |
| 6006E (3242)  | 34124E             |
| N15E          | E                  |
| 77775E        | 55237E             |
| E             | E                  |
| 2703E         | 35040E (3270)      |
| E             | E                  |
| 2661E         | 4647E              |
| E             | E                  |
| 77775E        | 24000E             |
| E             | E                  |
| 2711E         | 20456E             |
| E             | E                  |
| 2667E (3250)  | 5537E              |
| E             | E                  |
| 77775E        | 1677E              |
| E             | E                  |
| 2717E         | 5537E              |
| E             | E                  |
| 2675E         | 4E                 |
| E             | E                  |
| 77634E        | 6E (3300)          |
| E             | E                  |
| 24017E        | 31300E             |
| E             | E                  |
| 47471E        | 52033E             |
| E             | E                  |
| 77634E        | 31301E             |
| E             | E                  |
| 24017E (3260) | 54034E             |
| E             | E                  |
| 34747E        | 5537E (3305)       |
| E             | E                  |
| 77634E        | E                  |
| E             | V25N26E            |
| 24017E        | 10001E             |
| E             | 1642E              |
| 47303E        | 50006E             |
| E             |                    |

0+00

SL-2 LIFT OFF

0+10

SL-2 INSERTION



SL-2 SEPARATION (180,0/317,0)

COMPLETE LAUNCH C/L THRU STEP 2, Pg L/3-1  
 VERIFY PRELAUNCH NC1 TARGETING PAD AND  
 TALIGN PAD COPIED (Pg 6-a)

0+20

BACKUP GDC ALIGN PAD (COPY Pg 6-a)

PERFORM OPTICS DUST COVER  
 JETTISON CHECKLIST (Pg 6-1)  
 TWEAK MNVR IF REQ TO ~~180,170,2~~ 180,163,3  
 (LAUNCH REFSMMAT)  
 UNSTOW & MOUNT ORDEAL BOX (U3)

~~P52 (OPT 3) (Pg 6-b) (RECORD)~~  
~~REF STAR ACQ PAD STARS (Pg 6-a)~~

0+30

## PLT SYSTEMS CHECKS

MAIN REG CHECK  
 SEC RAD LEAK CHECK  
 ECS POST INSERTION CONFIG  
 SM RCS MONITORING CHECK  
 CM RCS MONITORING CHECK  
 C/W OPERATIONAL CHECK  
~~PLT CHECK~~  
 EPS MONITORING CHECK  
 ECS MONITORING CHECK  
 SECONDARY GLYCOL LOOP CHECK  
 SPS MONITORING CHECK  
 EXTEND DOCKING PROBE

## OPTICS DUST COVER JETTISON CHECKLIST

INSTALL OPTICS EYEPIECES (LEB)  
 OPTICS PWR-ON(up)  
 OPT ZERO-OFF, THEN ZERO (15 SEC)  
 OPT ZERO-OFF  
 OPT MODE-1A1  
 OPT COUPLING CONT-DIRECT  
 OPT SPEED CONT-HI  
 OHC-MAX RIGHT (OBS EJECT THRU SCT)  
 (SXT 40°, SCT 150° SHAFT ANGLE)

## P52 (OPTION 3)

|                |   |   |   |   |  |  |
|----------------|---|---|---|---|--|--|
| N71 1ST STAR   | X | 0 | 0 | 0 |  |  |
| N71 2ND STAR   | X | 0 | 0 | 0 |  |  |
| N05 (R1) 3 ERR | X |   |   |   |  |  |
| N93            | X |   |   |   |  |  |
| GYRO           | Y |   |   |   |  |  |
| TORQUING       | Z |   |   |   |  |  |
| ANGLES         |   |   |   |   |  |  |
| HR             | + | 0 | 0 | 0 |  |  |
| TIME OF        | + | 0 | 0 | 0 |  |  |
| GDC            | + | 0 |   |   |  |  |
| ALIGN          | + | 0 |   |   |  |  |
| SEC            |   |   |   |   |  |  |

IMU FAILURE

DATE ~~4/26/73~~  
 5/22/73 4/26/73 (99100)

# IMU FAILURE

0+30

0+40

0+50

1+00

P52 (OPT 2) (Pg 6-b) (RECORD)  
REF TALIGN PAD (Pg 6-a)  
(NOMINAL N22=~~180,339,2~~)  
180, 9, 2

P52 (OPT 3) (Pg 6-b) (RECORD Pg 6-1)  
REF STAR ACQ PAD STARS (Pg 6-b)

PERFORM COAS LOS DETERMINATION  
CHECKLIST (Pg 6-2)

BORESIGHT STAR

~~45-FQMAL HAUT~~  
(~~180,344,355~~)

37-NUNKI  
(180,26,347)

P00

P52 (OPTION 2)

|         |     |   |  |  |  |  |  |  |  |
|---------|-----|---|--|--|--|--|--|--|--|
| TIME OF | HR  | + |  |  |  |  |  |  |  |
| GDC     | MIN | + |  |  |  |  |  |  |  |
| ALIGN   | SEC | + |  |  |  |  |  |  |  |

COAS LOS DETERMINATION CHECKLIST

- COAS PWR-ON(up)  
MNR TO BORESIGHT ATTITUDE  
MAN ATT(3)-MIN IMP  
BORESIGHT ON STAR AND OBTAIN  
EXACT BORESIGHT ATTITUDE  
FROM ASCP tw (with FDAI 1  
ERROR NEEDLES ZEROED)  
LOAD N20 WITH ASCP tw VALUES  
MAN ATT(3)-RATE CMD
- F 04 06 V37E52E  
R1 00001  
R2 00003  
PRO

F 50 25 00015  
ENTER

F 01 70 0000E STAR CODE  
Load STAR CODE  
OPT MODE-CMC(verify)  
OPT ZERO-OFF  
PRO

06 92 SHAFT,TRUN  
RECORD SHAFT,TRUN  
(NOMINAL = 0°, +57.47°)
- V37E00E  
OPT MODE-MAN  
OHC-Drive trunk 10°  
OPT ZERO-ZERO

COAS LOS DETERMINATION

|     |    |   |  |  |  |  |  |  |  |
|-----|----|---|--|--|--|--|--|--|--|
| N92 | SA | + |  |  |  |  |  |  |  |
|     | TA | + |  |  |  |  |  |  |  |



1+00

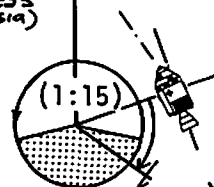
PERFORM EMS ΔV TEST & NULL  
BIAS CHECK CHECKLIST (Pg 6-3)  
PERFORM WASTE WATER  
DUMP CHECKLIST (Pg 6-3)  
P31; BYPASS MINKEY OPT  
LOAD NC1 TARGETING PAD (Pg 6-a) N57 R1  
P31 (FINAL COMP: RECORD) LOAD +3  
P00  
\*\* (LOAD +4 IF NC1 AT 1ST PERIGEE)\*\*  
PITCH UP TO TRACK EARTH HORIZON

1+10

PERFORM ~~PGA DOFFING &~~  
STOWAGE CHECKLIST (Pg 6-3)  
EMERGENCY CABIN PRESSURE SEL-OFF  
DIR O<sub>2</sub> - OPEN (UNTILL CABIN PRESS 5.7 psia)  
SIDE HATCH DUMP VLV - OPEN (UNTILL SUN  
CABIN PRESS 4.0 psia)  
DIR O<sub>2</sub> - OPEN UNTILL CABIN PRESS  
5.7 psia)  
EMERGENCY CABIN  
PRESSURE SEL- BOTH

1+20

PERFORM HOUSEKEEPING CHECKLIST (Pg 6-3)  
(IN PREPARATION FOR UNSTOWING CAMERA  
EQUIPMENT (Pg 6-6))



## WASTE WATER DUMP CHECKLIST

BAT VENT VLV-CLOSED (252)  
H2O QTY IND-WASTE (PANEL 2)  
WATER CONT PRESS REL VLV-DUMP A (352)  
MONITOR WASTE H2O QTY IND-DECREASING  
(5 % PER MIN)  
WHEN WASTE H2O QTY IND-15 %  
WATER CONT PRESS REL VLV-2  
BAT VENT VLV-VENT  
POT TK IN VLV-CLOSED (VERIFY) (352)

## P31 FINAL COMP

|        |        |   |   |   |   |   |   |   |   |
|--------|--------|---|---|---|---|---|---|---|---|
| N84    | ΔV NC2 | + | 0 |   |   |   |   |   |   |
|        | ΔH NC2 | + | 0 |   |   |   |   |   |   |
|        | ΔV NCC | + | 0 |   |   |   |   |   |   |
| N81    | ΔV X   | + | 0 |   |   |   |   |   |   |
| ΔV NC1 | ΔV Y   | + | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|        | ΔV Z   | + | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

## HOUSEKEEPING CHECKLIST

FOLD CENTER COUCH UP  
REMOVE UTILITY POLE RODS, STOW  
ON RIGHT WIRE RUN.  
STOW MSFC RODS ON LEFT WIRE RUN  
CUT LOOSE MSFC BASE PLATE &  
TEMP STOW  
CUT OFF JSC SAIL, STOW NEAR SOME  
STOW CENTER COUCH UNDER  
LEFT COUCH, STRAPS IN R5  
CLEAN OFF A1, A5, ETC FOR  
ACCESS TO STOWAGE LOCKERS  
STOW CLOTHES LINE UNDER STOWED  
COUCH  
CUT LOOSE MSFC SAIL, STOW  
BELOW CDR'S FEET  
CUT LOOSE T027, STOW  
UNDERNEATH PLT'S COUCH

## EVS ΔV TEST &amp; NULL BIAS CHECK CHECKLIST

EMS MODE-STBY  
EMS FUNC-ΔV SET/VHF RING  
SET ΔV ind to 1586.8 fps  
EMS MODE-NORMAL  
EMS FUNC-ΔV TEST  
SPS THRUST Lt-on/off (10 sec)  
ΔV ind stops at -0.1 to -41.5  
  
EMS MODE-STBY  
EMS FUNC-ΔV SET/VHF RING  
SET ΔV ind to -100.0 fps  
B'VAG MODE(3)-RATE 2  
EMS FUNC-ΔV (wait 5 sec)  
EMS MODE-NORMAL (for 1 min 40 sec)  
EMS MODE-STBY  
If ΔV<1 fps, do not bias  
If ΔV>1 fps but <10 fps,  
bias if desired  
If ΔV>10 fps, EMS is NO-GO  
B'VAG MODE(3)-ATT 1/RATE 2

## PGA DOFFING AND STOWAGE CHECKLIST

SNAP 2 TSB's (U1) ON AFT BKH & UEB  
STOW LIFE VESTS (F1)  
STOW WRIST DAMS & NECK DAMS (F2)  
STOW C/L POCKETS (F2)  
STOW SCISSORS, PROB'S, SUIT-POCKET EQUIP.,  
WATCHES & BANDS (TOP POCKET-TSB) SUN GLASSES  
STOW COMFORT GLOVES & WHISTLES (ACCESSORY BAGS)  
UNSTOW O2 UMBILICAL GREEN CABLE (B2)  
CONFIGURE CCU & O2 UMBILICALS (PER S-1-7, STEP 6)  
STOW CCA (ACCESSORY BAGS)  
  
DOFF PGA's (Ref EVA C/L (R1) 6004-3-2)  
INSTALL HELMETS & IV GLOVES (PGA BAG) ON PGA's  
CONNECT ELECTRICAL BUST COVERS (B2) TO PGA's  
STOW PGA's (UNDER COUCHES)  
  
DOFF UCTA's (WHEN FULL)  
PLACE UCTA CLAMPS (AG) ON UCTA's  
STOW UCTA's (T08 ON UEB)  
DOFF UCTA's (AG)  
STOW OBC/BIBBELTS (ACCESSORY BAGS)  
DOFF ENG HARNESSSES & LT WT HEADSETS (U2)  
DOFF TROUSERS, JACKETS & BOOTS (U3)  
UNSTOW CUE CARDS (DATA CARD KIT-R1)

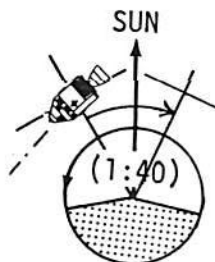
1+30

IMU FAILURE

1+30

T  
E  
X

NCI PRELIMINARY PAD (COPY Pg 6-5)  
 STAR ACQ PAD (COPY Pg 6-a)  
 TRANSMIT GYRO TORQUE ANGLES  
 AND TIME OF GDC ALIGN (Pgs 6-1 & 6-2)  
 TRANSMIT EMS ΔV TEST RESULTS



1+40

N  
F  
D

1+50

M  
A  
D

STDN UPLINK (P27) [CSM & SWS SV'S]  
 NCI FINAL PAD (COPY Pg 6-5) ~~Ⓢ~~  
 V48E LOAD NCI VALUES IF REQD.  
 NCI BURN ATT CHECK PAD (COPY Pg 6-5)

2+00

~~MNVR TO STAR ACQ PAD ATT (Pg 6-a)~~  
~~P52 (OPT 3) (Pg 6-b) (RECORD)~~

Ⓢ IF NO NPC - PREP LUNCH - FOOD AT SPOONS L3

P52 (OPTION 3)

|                   |                                     |   |   |   |   |  |
|-------------------|-------------------------------------|---|---|---|---|--|
| N71 1ST STAR      | <input checked="" type="checkbox"/> | 0 | 0 | 0 |   |  |
| N71 2ND STAR      | <input checked="" type="checkbox"/> | 0 | 0 | 0 |   |  |
| N05 (R1) } ERR    | <input checked="" type="checkbox"/> |   |   |   |   |  |
| N93 X             |                                     |   |   |   |   |  |
| GYRO TORQUING Y   |                                     |   |   |   |   |  |
| ANGLES Z          |                                     |   |   |   |   |  |
| TIME OF GDC ALIGN | HR                                  | + | 0 | 0 | 0 |  |
|                   | MIN                                 | + | 0 | 0 | 0 |  |
|                   | SEC                                 | + | 0 |   |   |  |

2+00

\*\*\*\*\* IF GDC DRIFT > 10°/HR/AXIS \*\*\*\*\*  
 \* USE 12° MARK ON WINDOW \*

P31  
 LOAD N95 PAD VALUE  
 SET DET

-12:00

2+10

P31 (FINAL COMP: RECORD)  
 V25N20E (LOAD PAD BURN ATT)

GO TO SPS BURN CUE CARD (SCS Δ)  
 IN P40 AT F 50 18, RECORD BURN ATT  
 SET ASCP tw TO BURN ATT  
 E, E TO F 16 40, RECORD ΔV<sub>T</sub>  
 P00

PERFORM BURN ATT CHECK  
 USING GDC, OHC, SCT

\*\*\* IF STAR > 5° FROM CTR OF SCT \*\*\*  
 \* USE 12° MARK ON WINDOW \*

\*\*\*\*\*

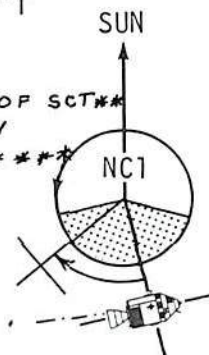
2+20

0:00

**NC1** (180, 0/339, 2)  
 RECORD BURN STATUS  
 P77  
 P00

~~PITCH UP TO TRACK EARTH HORIZON~~  
 P52 (OPT 3) (Pg 6-b) (RECORD Pg 6-4)

P00  
 PITCH UP TO TRACK EARTH HORIZON



2+30

IMU FAILURE

## NC1 PAD DATA

|                   |                 | NOMINAL     | PRELIMINARY | FINAL |
|-------------------|-----------------|-------------|-------------|-------|
| N95               | HR              | + 0 0 2     |             |       |
| TIG NC1           | MIN             | 2 0         |             |       |
|                   | SEC             | 1 1 9 0     |             |       |
| N81               | ΔV <sub>X</sub> | 2 1 9 0     |             |       |
| ΔV <sub>NC1</sub> | ΔV <sub>Y</sub> | 0 0 0 0     |             |       |
|                   | ΔV <sub>Z</sub> | 0 0 0 0     |             |       |
| N22               | R               | + 1 8 0 0 0 | + 0 0       | + 0 0 |
| NC1               | P               | + 3 3 9 0 0 | + 0 0       | + 0 0 |
|                   | Y               | + 0 0 2 0 0 | + 0 0       | + 0 0 |
|                   | ΔV <sub>C</sub> | 2 0 4 1     |             |       |
| BT                |                 | 0 0 1 0     |             |       |

REFER Pg 1-5

## BURN ATT CHECK

|      |   |  |  |  |   |
|------|---|--|--|--|---|
| STAR |   |  |  |  |   |
| SA   |   |  |  |  | 0 |
| TA   | 0 |  |  |  | 0 |

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
 X N81 COMP LIMITS ±0.5 FPS X  
 X X  
 X CMC/STON WITHIN LIMITS GO CMC X  
 X NO AGREEMENT GO STON X  
 XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

## NC1 ONBOARD DATA

P31 FINAL COMP

|                   |                   |             |  |  |  |
|-------------------|-------------------|-------------|--|--|--|
| N84               | ΔV <sub>NC2</sub> | + 0         |  |  |  |
|                   | ΔH <sub>NC2</sub> | + 0         |  |  |  |
|                   | ΔV <sub>NCC</sub> | + 0         |  |  |  |
| N81               | ΔV <sub>X</sub>   | + 0         |  |  |  |
| ΔV <sub>NC1</sub> | ΔV <sub>Y</sub>   | + 0 0 0 0 0 |  |  |  |
|                   | ΔV <sub>Z</sub>   | + 0 0 0 0 0 |  |  |  |

P40

|                 |                 |   |   |   |   |
|-----------------|-----------------|---|---|---|---|
| N18             | R               | + |   |   |   |
| BURN            | P               | + |   |   |   |
| ATT             | Y               | + |   |   |   |
| N40             | ΔV <sub>T</sub> | + |   |   |   |
| ΔV TAIL OFF     | -               |   | 1 | 4 | 0 |
| ΔV <sub>C</sub> | +               |   |   |   |   |

## SPS BURN STATUS

|            |  |  |  |  |  |
|------------|--|--|--|--|--|
| ΔTIG       |  |  |  |  |  |
| AFTER TRIM |  |  |  |  |  |
| ΔVC        |  |  |  |  |  |

# IMU FAILURE

2+30

A  
R  
I  
A

H2 PURGE LINE HTR-ON(up)

CONFIGURE CAMERAS(B3): SWS APPROACH

CM2/DAC10/75/CX01-BRKT(B8) ~~STOW 18 LENS IN PLT TSB~~

MIR (T8, 1/250, ∞) / 5FPS (1MAG)

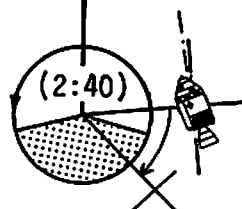
CM4/HDC01/CX03 (f8, 1/250, ∞)

← ~~CX04~~  
UNSTOW PORTABLE TIMER (B3)

UNSTOW TV CAMERA, ZOOM LENS, CABLES (2),  
MONITOR, AND CAMERA MOUNT (A5)

ASSEMBLE AND MOUNT TV CAMERA IN  
RH RNDZ WINDOW

RESTOW BAGS AND DUST CAPS (A5) SUN



2+40

2+50

FUEL CELL PURGE (S/1-3)(20 MIN  
AFTER LINE HTR-ON)

3+00

H2 PURGE LINE HTR-OFF (10 MIN  
AFTER PURGE)

FOR FLYAROUND / INSPECTION AND SEVA PHOTOS,  
GET FOLLOWING AND PLACE IN TSB:

16mm MAG CX02 (B8)

70mm MAG CX23 (B8)

35mm CAM NK03 (A9)

300mm LENS (A2)

35mm CASSETTES CX 21 (A9)

CX 22 (A9)

BW01 (A9)

LOAD NK03 WITH CX 21

CONFIGURE NK WITH 300 mm LENS

STOW 55mm LENS

CX FILM - ASA 64

BW FILM - ASA 80

FOR 300 mm LENS - 1/250 OR FASTER TO PREVENT BLUR



3+00

TV CHECKOUT;  
PERFORM TV PREP CHECKLIST (Pg 6-7)  
S BAND AUX TV-TV (2 MIN PRIOR STDN AOS)

TRANSMIT NC1 ATT,  $\Delta$ VC LOADED AND  
SPS BURN STATUS (Pg 6-5)  
TRANSMIT GYRO TORQUE ANGLES  
AND TIME OF GDC ALIGN (Pg 6-4)

\*\*\*\*\*IF NPC REQUIRED\*\*\*\*\*  
\* STDN UPLINK (P27) [CSM & SWS SV'S] \*  
\* NPC PAD (COPY) \*  
\*\*\*\*\*

NC2 PRELIMINARY PAD (COPY Pg 6-9)  
S BAND AUX TV - OFF (CTR) WHEN CHECKOUT COMPLETE

\*\*\*\*\*IF NPC REQUIRED\*\*\*\*\*  
\* \* \* \* \*  
\* SET DET \*  
\* GO TO SPS BURN CUE CARD (SCS  $\Delta$ ) \*  
\* SET ASCP tw TO NPC BURN ATT \*  
\* AND MNVR TO NPC BURN ATT \*  
\* \* \* \* \*  
\* P00 \*  
\* \* \* \* \*  
\* \* \* \* \*  
\* \* \* \* \*

PNL 7 DIRECT O<sub>2</sub> VLV-CLOSE(CW)  
STDN UPLINK (P27) [CSM & SWS SV'S]

\*\*\* \* \* \* \*  
\* \* \* \* \*  
\* NPC \* \* \* \* \*  
\* RECORD BURN STATUS \* \* \* \* \*  
\* TRANSMIT NPC ATT,  $\Delta$ VC LOADED AND \* \* \* \* \*  
\* SPS BURN STATUS \* \* \* \* \*

## NPC PAD DATA

|                           |                         |   |   |   |   |   |   |   |  |
|---------------------------|-------------------------|---|---|---|---|---|---|---|--|
| N33                       | HR                      | + | X | X |   |   |   |   |  |
| TIG NPC                   | MIN                     | + | X | X | X |   |   |   |  |
|                           | SEC                     | + | X | X |   |   |   |   |  |
| N81                       | $\Delta$ V <sub>X</sub> | + | X |   | 0 | 0 | 0 | 0 |  |
| $\Delta$ V <sub>NPC</sub> | $\Delta$ V <sub>Y</sub> |   | X |   |   |   |   |   |  |
|                           | $\Delta$ V <sub>Z</sub> |   | X |   |   |   |   |   |  |
| N22                       | R                       | + |   |   |   | 0 | 0 |   |  |
| NPC                       | P                       | + |   |   |   | 0 | 0 |   |  |
|                           | Y                       | + |   |   |   | 0 | 0 |   |  |
|                           | $\Delta$ V <sub>C</sub> |   | X | X |   |   |   |   |  |
|                           | BT                      |   | X | X |   |   |   |   |  |

## TV PREPARATION CHECKLIST

TV SOURCE - CM  
REMOVE & STOW LENS CAP  
LINEAR SW - GAM 1/2  
AVG SW - AVG  
F-STOP - 22  
ZOOM - 25  
FOCUS -  $\infty$   
MONITOR ON-ON  
(REFINE SETTINGS AS REQ'D)

## SPS BURN STATUS

|              |   |   |  |  |  |  |  |  |  |
|--------------|---|---|--|--|--|--|--|--|--|
| $\Delta$ TIG | X | X |  |  |  |  |  |  |  |
| AFTER TRIM   |   |   |  |  |  |  |  |  |  |
| $\Delta$ VC  |   | X |  |  |  |  |  |  |  |

IMU FAILURE

3+30

3+40

3+50

4+00

CYCLE

\* P77 (LOAD NPC PAD VALUES) \*

\* P00 \*

\* INITIATE TRACK OF EARTH HORIZON \*

\*\*\*\*\*

~~SWS IN R-ZLV~~

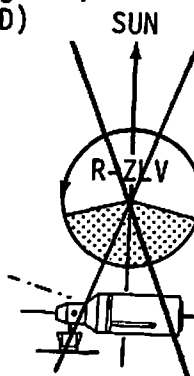
~~PERFORM VHF SWITCH LIST (Pg 6-8)~~

~~ACQUIRE VHF RANGING (R-619.00 NM)~~

~~(RECORD GET & RANGE OF VHF LOCK-ON)~~

MNVR TO STAR ACQ PAD ATT (Pg 6-a)

P52 (OPT 3) (Pg 6-b) (RECORD)



PITCH UP TO TRACK EARTH HORIZON

## VHF SWITCH LIST

VHF AM A-OFF(ctr)  
VHF AM B-DUPLEX  
VHF RANGING-ON(up)  
VHF ANTENNA-RIGHT  
EMS FUNC-ΔV SET/VHF RNG  
EMS MODE-BACKUP/VHF RNG  
VHF RNG-RESET  
PANEL 6,9,10:  
VHF AT-RCV

## VHF LOCK-ON

|              |    |     |
|--------------|----|-----|
| GET=         | HR | MIN |
| RANGE (EMS)= |    | NM  |

## P52 (OPTION 3)

|          |            |   |   |   |   |  |  |  |
|----------|------------|---|---|---|---|--|--|--|
| N71      | 1ST STAR   | X | 0 | 0 | 0 |  |  |  |
| N71      | 2ND STAR   | X | 0 | 0 | 0 |  |  |  |
| N05      | (R1) & ERR | X |   |   |   |  |  |  |
| N93      | X          |   |   |   |   |  |  |  |
| GYRO     | Y          |   |   |   |   |  |  |  |
| TORQUING | Z          |   |   |   |   |  |  |  |
| ANGLES   |            |   |   |   |   |  |  |  |
|          | HR         | + | 0 | 0 | 0 |  |  |  |
| TIME OF  | MIN        | + | 0 | 0 | 0 |  |  |  |
| GDC      | SEC        | + | 0 |   |   |  |  |  |
| ALIGN    |            |   |   |   |   |  |  |  |

4+00

P32

~~34:00~~~~V87; SET DET~~

XXXXXXXXXXXXXXXXXXXXX  
 X UNREASONABLE UPDATE X  
 X 1.00 NM, 6.0 FPS X  
 XXXXXXXXXXXXXXXXXXXXX

NC2 FINAL PAD (COPY) **Y48E IF RQD.**  
 NCC PRELIMINARY PAD (COPY Pg 6-11)  
 NSR PRELIMINARY PAD (COPY Pg 6-12)  
 TRANSMIT GYRO TORQUE ANGLES AND  
 TIME OF GDC ALIGN (Pg 6-8)

4+10

RECALL P32 (LOAD N28 PAD VALUE)  
 SET DET

~~R=300.00 NM~~  
~~R=520.7 FPS~~

4+20

-12:00

P32 (FINAL COMP: RECORD)  
 V25N20E (LOAD PAD BURN ATT)

~~R=300.00 NM~~  
~~R=-279.1 FPS~~

GO TO SPS BURN CUE CARD (SCS Δ)  
 IN P40 AT F 50 18, RECORD BURN ATT  
 SET ASCP tw TO BURN ATT  
 E,E TO F 16 40, RECORD ΔV<sub>T</sub>  
 P00

4+30

## NC2 PAD DATA

|                   |                 | NOMINAL     | PRELIMINARY | FINAL     |
|-------------------|-----------------|-------------|-------------|-----------|
| N28               | HR              | + 0 0 4     | + 0 0 0     | + 0 0 0   |
| TIG NC2           | MIN             | + 3 6       | + 0 0 0     | + 0 0 0   |
|                   | SEC             | + 0 3 0     | + 0 0 0     | + 0 0 0   |
| N81               | ΔV <sub>X</sub> | + 1 6 0 7   | + 0 0 0 0   | + 0 0 0 0 |
| ΔV <sub>NC2</sub> | ΔV <sub>Y</sub> | + 0 0 0 0   | + 0 0 0 0   | + 0 0 0 0 |
|                   | ΔV <sub>Z</sub> | + 0 0 0 0   | + 0 0 0 0   | + 0 0 0 0 |
| N22               | R               | + 1 8 0 0 0 | + 0 0 0 0   | + 0 0 0 0 |
| NC2               | P               | + 1 5 9 0 0 | + 0 0 0 0   | + 0 0 0 0 |
|                   | Y               | + 0 0 1 0 0 | + 0 0 0 0   | + 0 0 0 0 |
|                   | ΔV <sub>C</sub> | + 1 4 5 7   | + 0 0 0 0   | + 0 0 0 0 |
|                   | BT              | + 0 0 0 0   | + 0 0 0 0   | + 0 0 0 0 |

REFER Pg 1-9

## NC2 ONBOARD DATA

## P32 FINAL COMP

|                   |                   |           |
|-------------------|-------------------|-----------|
| N84               | ΔV <sub>NCC</sub> | + 0 0 0 0 |
|                   | ΔH <sub>NCC</sub> | + 0 0 0 0 |
|                   | ΔV <sub>NSR</sub> | + 0 0 0 0 |
| N81               | ΔV <sub>X</sub>   | + 0 0 0 0 |
| ΔV <sub>NC2</sub> | ΔV <sub>Y</sub>   | + 0 0 0 0 |
|                   | ΔV <sub>Z</sub>   | + 0 0 0 0 |

## P40

|             |                 |           |
|-------------|-----------------|-----------|
| N18         | R               | + 0 0 0 0 |
| BURN        | P               | + 0 0 0 0 |
| ATT         | Y               | + 0 0 0 0 |
| N40         | ΔV <sub>T</sub> | + 0 0 0 0 |
| ΔV TAIL OFF | -               | + 1 4 0   |
|             | ΔV <sub>C</sub> | + 0 0 0 0 |

IMU FAILURE



# IMU FAILURE

4+30

H  
A  
W

0:00

**NC2** (180,0/<sup>189</sup>~~159~~,1)

RECORD BURN STATUS

P77; P00

TRANSMIT NC2 ATT, ΔVC LOADED (Pg 6-9)

AND SPS BURN STATUS

~~P33; PITCH UP TO TRACK EARTH HORIZON~~

~~REACQUIRE VHF RANGING (R-236.00 NM)~~

~~V76E (LOAD N72 WITH TIG NCC-36)~~

~~V67E (LOAD +10000., +10.0, +1)~~

~~V93E; V87E; SET DET~~

SET DET

XXXXXXXXXXXXXXXXXXXXX  
X UNREASONABLE UPDATE X  
X 1.00 NM; 6.0 FPS X  
XXXXXXXXXXXXXXXXXXXXX

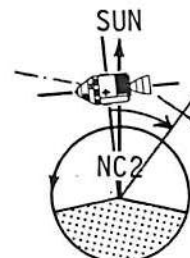
~~R CHART DATA FOR NCC-36~~

~~R CHART FOR DATA NCC-32~~

~~R CHART DATA FOR NCC-28~~

~~PITCH UP 15° (TO MAINTAIN VHF LOCK-ON)~~

~~P52 (OPT 3) (Pg 6-b) (RECORD)~~



## SPS BURN STATUS

|            |   |   |  |  |  |  |  |  |  |
|------------|---|---|--|--|--|--|--|--|--|
| ΔTIG       | × | × |  |  |  |  |  |  |  |
| AFTER TRIM |   |   |  |  |  |  |  |  |  |
| ΔVC        |   | × |  |  |  |  |  |  |  |

|               |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| N71 (TIG NCC) |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|               |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| N72           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

## P52 (OPTION 3)

|                      |     |   |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------------|-----|---|---|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| N71 1ST STAR         | ×   | 0 | 0 | 0 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| N71 2ND STAR         | ×   | 0 | 0 | 0 |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| N05 (R1) ERR         | ×   |   |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| N93                  | X   |   |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| GYRO TORQUING ANGLES | Y   |   |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                      | Z   |   |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| TIME OF GDR ALIGN    | HR  | + | 0 | 0 | 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                      | MIN | + | 0 | 0 | 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                      | SEC | + | 0 |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

DATE 4/6/73  
5/22/73

PAGE 6-10

5+00

20:00

16:00

5+10

12:00

5+20

0:00

5+30

~~R CHART DATA FOR NCC-20~~~~STDN UPLINK (P27) (IF REQ'D)~~~~R CHART DATA FOR NCC-16~~~~WHEN P52 COMPLETE: P33; V87E;  
CONTINUE TO TRACK EARTH HORIZON~~~~NCC FINAL PAD (COPY); V48E IF REQ'D~~~~NSR FINAL PAD (COPY Pg 6-12)~~~~TRANSMIT GYRO TORQUE ANGLES~~~~AND TIME OF GDC ALIGN (Pg 6-10)~~~~R CHART DATA FOR NCC-12~~~~V77E; P33 (FINAL COMP: RECORD)~~~~(RECORD  $\Delta V_Y$  FOR P77)~~~~V25N20E (LOAD PAD BURN ATT)~~GO TO SPS BURN CUE CARD (SCS  $\Delta$ )

IN P40 AT F 50 18, RECORD BURN ATT

SET ASCP tw TO BURN ATT

E,E TO F 16 40, RECORD  $\Delta V_T$ 

P00

DO NOT OVERWRITE  
NCC TIG (N11)  
UNLESS ADVISED  
BY STDN

11,359

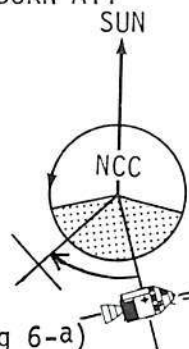
**NCC** (180,2/340,7)

RECORD BURN STATUS

P77 (LOAD CMC COMPUTED  $\Delta V_Y$ )

MNVR TO STAR ACQ PAD ATT (Pg 6-a)

P52 (OPT 3) (Pg 6-b) (RECORD)

IF NCC EXECUTION IS QUESTIONABLE  
START RECORDING CHART DATA AT NSR-28

## NCC PAD DATA

| NOMINAL          |              |   |   |   | PRELIMINARY |   |   |   |   | FINAL |   |  |   |   |
|------------------|--------------|---|---|---|-------------|---|---|---|---|-------|---|--|---|---|
| N11              | HR           | + |   |   | 0           | 0 | 5 |   |   | +     |   |  |   |   |
| TIG              | NCC MIN      | + |   |   | 2           | 2 |   |   |   | +     |   |  |   |   |
|                  | SEC          | + |   | 1 | 2           | 5 | 0 |   |   | +     |   |  |   |   |
| N81              | $\Delta V_X$ | + |   | 0 | 3           | 0 | 5 |   |   | +     |   |  |   |   |
| $\Delta V_{NCC}$ | $\Delta V_Y$ | + |   | 0 | 0           | 0 | 0 |   |   | +     |   |  |   |   |
|                  | $\Delta V_Z$ | - |   | 0 | 0           | 0 | 9 |   |   | +     |   |  |   |   |
| N22              | R            | + | 1 | 8 | 0           | 0 | 0 |   | 0 | 0     | + |  | 0 | 0 |
| NCC              | P            | + | 3 | 4 | 0           | 0 | 0 |   | 0 | 0     | + |  | 0 | 0 |
|                  | Y            | + | 0 | 0 | 1           | 0 | 0 |   | 0 | 0     | + |  | 0 | 0 |
| $\Delta V_C$     |              |   |   |   | 0           | 1 | 6 | 6 |   |       |   |  |   |   |

REFER Pg 1-11

## NCC ONBOARD DATA

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
 X N81 COMP LIMITS  $\pm 1.5 NA, \pm 15.0$  FPS X  
 X  $\pm 0.5 NA, \pm 3.0$  X  
 X USE STDN  $\Delta V_Y$  IN ALL CASES X  
 X CMC/CHART WITHIN LIMITS GO CMC X  
 X IF NOT CMC/STDN WITHIN LIMITS GO CMC X  
 X IF NOT CHART/STDN WITHIN LIMITS GO STDN X  
 X NO AGREEMENT-NORMALLY GO STDN X  
 XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

## SPS BURN STATUS

|              |  |  |  |  |  |  |  |  |  |
|--------------|--|--|--|--|--|--|--|--|--|
| $\Delta TIG$ |  |  |  |  |  |  |  |  |  |
| AFTER TRIM   |  |  |  |  |  |  |  |  |  |
| $\Delta VC$  |  |  |  |  |  |  |  |  |  |

## P52 (OPTION 3)

|         |                   |   |   |   |   |  |  |  |  |
|---------|-------------------|---|---|---|---|--|--|--|--|
| N71     | 1ST STAR          |   | 0 | 0 | 0 |  |  |  |  |
| N71     | 2ND STAR          |   | 0 | 0 | 0 |  |  |  |  |
| N05     | (R1) $\delta$ ERR |   |   |   |   |  |  |  |  |
| N93     | X                 |   |   |   |   |  |  |  |  |
|         | GYRO              |   |   |   |   |  |  |  |  |
|         | TORQUING          |   |   |   |   |  |  |  |  |
|         | ANGLES            |   |   |   |   |  |  |  |  |
|         | Z                 |   |   |   |   |  |  |  |  |
|         | HR                | + | 0 | 0 | 0 |  |  |  |  |
| TIME OF | MIN               | + | 0 | 0 | 0 |  |  |  |  |
| GDC     | SEC               | + | 0 |   |   |  |  |  |  |
| ALIGN   |                   |   |   |   |   |  |  |  |  |

## P33 FINAL COMP

|                  |              |   |   |   |   |   |   |  |  |
|------------------|--------------|---|---|---|---|---|---|--|--|
| N82              | $\Delta V_X$ |   | 0 |   |   |   |   |  |  |
| $\Delta V_{NSR}$ | $\Delta V_Y$ |   | 0 |   |   |   |   |  |  |
|                  | $\Delta V_Z$ |   | 0 |   |   |   |   |  |  |
| N81              | $\Delta V_X$ |   | 0 |   |   |   |   |  |  |
| $\Delta V_{NCC}$ | $\Delta V_Y$ | * | S | T | D | N | * |  |  |
|                  | $\Delta V_Z$ |   | 0 |   |   |   |   |  |  |

 $\Delta V_Y$  FOR P77

|                         |  |   |  |  |  |  |  |  |  |
|-------------------------|--|---|--|--|--|--|--|--|--|
| FINAL COMP $\Delta V_Y$ |  | 0 |  |  |  |  |  |  |  |
|-------------------------|--|---|--|--|--|--|--|--|--|

## NCC CHART SOLN

|              |   |   |   |   |   |   |  |  |  |
|--------------|---|---|---|---|---|---|--|--|--|
| $\Delta V_X$ |   |   |   |   |   |   |  |  |  |
| $\Delta V_Y$ | * | S | T | D | N | * |  |  |  |
| $\Delta V_Z$ |   |   |   |   |   |   |  |  |  |

## P40

|                     |              |   |  |  |   |   |   |  |  |
|---------------------|--------------|---|--|--|---|---|---|--|--|
| N18                 | R            | + |  |  |   |   |   |  |  |
| BURN                | P            | + |  |  |   |   |   |  |  |
| ATT                 | Y            | + |  |  |   |   |   |  |  |
| N40                 | $\Delta V_T$ | + |  |  |   |   |   |  |  |
| $\Delta V$ TAIL OFF | -            |   |  |  | 1 | 4 | 0 |  |  |
| $\Delta V_C$        | +            |   |  |  |   |   |   |  |  |

IMU FAILURE



5+30

PERFORM VHF SWITCH LIST (Pg 6-8)

PITCH UP TO TRACK EARTH HORIZON 127.00

ACQUIRE VHF RANGING (R=102.00 NM)

P34; V32E (P34 RECYCLE: RECORD)

-24.00

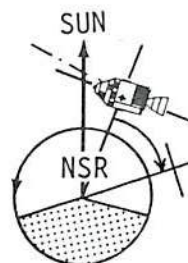
V87E; SET DET

SWS IN R-ZLV

R CHART DATA POINTS LOCATED  
AT ~~MINUS~~ MINUS 28, 24, 20, 12  
NSRXXXXXXXXXXXXXXXXXXXXX  
X UNREASONABLE UPDATE X  
X 1.00 NM, 6.0 FPS X  
XXXXXXXXXXXXXXXXXXXXX

5+40

VHF

TRANSMIT NCC ATT,  $\Delta V_C$  LOADED AND  
SPS BURN STATUS (Pg 6-11)TRANSMIT GYRO TORQUE ANGLES  
AND TIME OF GDC ALIGN (Pg 6-11)SET HDSUPFLG (V25N7E, 106E, 2000E, 1E)  
P34 (FINAL COMP: RECORD)  
(RECORD  $\Delta V_Y$  FOR P77)LOAD N81 WITH CORRECT VALUES  
V25N20E (LOAD PAD BURN ATT)IF NCC EXECUTION IS QUESTIONABLE  
THEN CMC FINAL COMP IS PRIMEGO TO SPS BURN CUE CARD (SCS  $\Delta$ )  
IN P40 AT F 50 18, RECORD BURN ATT  
SET ASCP tw TO BURN ATT  
E, E TO F 16 40, RECORD  $\Delta V_T$   
P00

0.00

6+00

0/226,0  
180,3/198,2  
RECORD BURN STATUS

## NSR PAD DATA

TIG NSR(N13)=TIG NCC(N11) + 00:37:00= : : .

|     |                  | NOMINAL | PRELIMINARY | FINAL |
|-----|------------------|---------|-------------|-------|
| N81 | $\Delta V_X$     | +       | +           | +     |
|     | $\Delta V_{NSR}$ | +       | +           | +     |
|     | $\Delta V_Y$     | +       | +           | +     |
|     | $\Delta V_Z$     | +       | +           | +     |
| N22 | R                | +       | +           | +     |
| NSR | P                | +       | +           | +     |
|     | Y                | +       | +           | +     |
|     | $\Delta V_C$     | +       | +           | +     |

REFER Pg 1-12

## NSR ONBOARD DATA

|     |                     | P34 RECYCLE | P34 FINAL COMP |
|-----|---------------------|-------------|----------------|
| N75 | $\Delta H_{NSR}$    | +           | +              |
|     | $\Delta T(NSR/TP1)$ | +           | +              |
|     | $\Delta T(TP1/TP1)$ | +           | +              |
| N81 | $\Delta V_X$        | +           | +              |
|     | $\Delta V_{NSR}$    | +           | +              |
|     | $\Delta V_Y$        | +           | +              |
|     | $\Delta V_Z$        | +           | +              |

XXXXXXXXXXXXXXXXXXXXX  
X N81 COMP LIMITS NA, NA, NA FPS X  
X USE STDN  $\Delta V_Y$  IN ALL CASES X  
X BURN P34 RECYCLE, STDN OR CHART SOLN X  
X IF NCC WAS BURNED CMC, STDN X  
X OR CHART, RESPECTIVELY X  
XXXXXXXXXXXXXXXXXXXXX $\Delta V_Y$  FOR P77FINAL COMP  $\Delta V_Y$  0

NSR CHART SOLN

|              |   |   |   |   |
|--------------|---|---|---|---|
| $\Delta V_X$ | + | + | + | + |
| $\Delta V_Y$ | + | + | + | + |
| $\Delta V_Z$ | + | + | + | + |

P40

|          |                     |   |   |   |   |
|----------|---------------------|---|---|---|---|
| N18      | R                   | + | + | + | + |
| BURN ATT | P                   | + | + | + | + |
|          | Y                   | + | + | + | + |
| N40      | $\Delta V_T$        | + | + | + | + |
|          | $\Delta V$ TAIL OFF | + | + | + | + |
|          | $\Delta V_C$        | + | + | + | + |

SPS BURN STATUS

|              |   |   |   |   |
|--------------|---|---|---|---|
| $\Delta TIG$ | + | + | + | + |
| AFTER TRIM   | + | + | + | + |
| $\Delta V_C$ | + | + | + | + |



6+30

-32:00

-28:00

-24:00

6+40

-16:00

6+50

-12:00

7+00

VHF

ACN

38

XXXXXXXXXXXXXXXXXXXXX  
X UNREASONABLE UPDATE X  
X 0.50 MI, 3.0 FPS X  
XXXXXXXXXXXXXXXXXXXXX

R CHART DATA FOR TPI -32 (GET)

R CHART DATA FOR TPI -28 (GET)

R CHART DATA FOR TPI -24 (GET)

V32E (2ND P35 RECYCLE: RECORD)

\*\*\*\*\*TPI EARLY/LATE LOGIC\*\*\*\*\*  
\* 2ND RECYCLE \*  
\* IF CMC SOLUTION INDICATES TIG SLIP > +10 MIN: 5 \*  
\* USE CMC TIG OPTION; N37=PRELAUNCH N37+10 MIN \*  
\* CONTINUE CHART SOLUTION FOR FINAL AV COMPARISON \*  
\* AT FINAL COMP-USE NOMINAL COMPARISON LOGIC \*  
\* IF ALL COMPARISONS DISAGREE-BURN THE \*  
\* SOLUTION WHOSE TIG (CMC 2ND RECYCLE, \*  
\* STDN PREL. PAD) COMPARES CLOSEST WITH \*  
\* CHART TPI TIG 2 \*  
\*\*\*\*\*

R CHART DATA FOR TPI -16 (GET)

V77E

MNVR TO BORESIGHT COAS ON HORIZON

SET ORDEAL (FDAI 1)

TPI FINAL PAD (COPY)

$\theta_D = 339$ , H=231

P35 (FINAL COMP: RECORD)

LOAD N81 R2 WITH STDN SOLN

PRO, N59E (RECORD)

P40; P00

GO TO SPS BURN CUE CARD (SCS  $\Delta$ )

BURN WITH SWS BORESIGHTED IN COAS

( $\Delta V_F$ , SPS)

( $\Delta V_R$  &  $\Delta V_D$ , TIMED RCS), WITH

AUTO RCS SELECT (16) - MNA/MNB

RCS X AXIS  
1FPS=2.35 SEC  
RCS Y&Z AXES  
1FPS=4.45 SEC

# TPI PAD DATA

|                  |                 | NOMINAL | PRELIMINARY | FINAL |
|------------------|-----------------|---------|-------------|-------|
| N37              | HR              | +       | +           | +     |
| TIG TPI          | MIN             | +       | +           | +     |
|                  | SEC             | +       | +           | +     |
| N81              | $\Delta V_X$    | +       | +           | +     |
| $\Delta V_{TPI}$ | $\Delta V_Y$    | +       | +           | +     |
|                  | $\Delta V_Z$    | +       | +           | +     |
| N59              | $\Delta V_F/BT$ | +       | +           | +     |
| $\Delta V$       | $\Delta V_R/BT$ | +       | +           | +     |
| (LOS)            | $\Delta V_D/BT$ | +       | +           | +     |
|                  | $\Delta V_C$    | +       | +           | +     |

REFEC Pg 1-14

# TPI ONBOARD DATA

|                  |                     | P35 RECYCLE | P35 FINAL COMP |
|------------------|---------------------|-------------|----------------|
| N37              | HR                  | +           | +              |
| TIG TPI          | MIN                 | +           | +              |
|                  | SEC                 | +           | +              |
| N58              | $\Delta V_{TPI}$    | +           | +              |
|                  | $\Delta V_{TPF}$    | +           | +              |
|                  | $\Delta T(TPI/TPI)$ | B           | B              |
| N81              | $\Delta V_X$        | +           | +              |
| $\Delta V_{TPI}$ | $\Delta V_Y$        | +           | +              |
|                  | $\Delta V_Z$        | +           | +              |
| N59              | $\Delta V_F$        | +           | +              |
| $\Delta V(LOS)$  | $\Delta V_R$        | +           | +              |
|                  | $\Delta V_D$        | +           | +              |

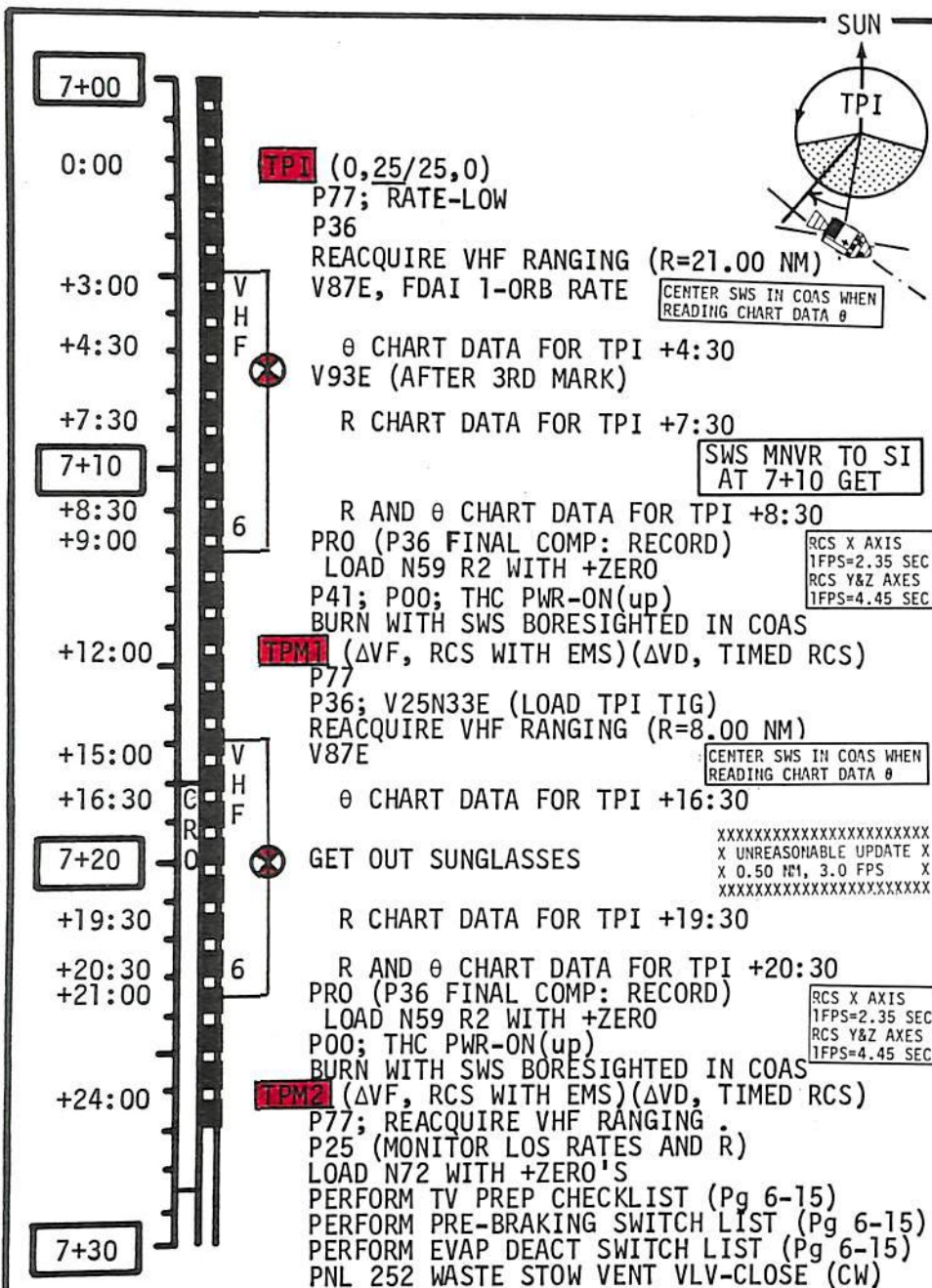
|                 |   |   |   |
|-----------------|---|---|---|
| $\Delta V_F$    |   |   |   |
|                 | - | 1 | 4 |
| $\Delta V_C$    |   |   |   |
| $\Delta V_R$ BT |   |   |   |
| $\Delta V_D$ BT |   |   |   |

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
X N81 COMP LIMITS  $\pm 1.0$  NA,  $\pm 3.0$  FPS X  
X USE STDN  $\Delta V_Y$  IN ALL CASES X  
X CMC/CHART WITHIN LIMITS GO CMC X  
X IF NOT CMC/STDN WITHIN LIMITS GO CHART X  
X IF NOT CHART/STDN WITHIN LIMITS GO CHART X  
X NO AGREEMENT GO STDN X  
X FOR CHART/STDN: RECALL P35-TIG OPT. X  
X USE CHART/STDN TIG X  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

# TPI CHART SOLN

|                                                                                                                                |              |   |   |   |   |
|--------------------------------------------------------------------------------------------------------------------------------|--------------|---|---|---|---|
| TO OBTAIN<br>LOS $\Delta V_F$ &<br>$\Delta V_D$ :<br>OVERWRITE<br>N81 $\Delta V_X$ &<br>$\Delta V_Z$ , PRO,<br>N59E,<br>RECORD | $\Delta V_X$ | + | + | + | + |
| $\Delta V_Y$                                                                                                                   | +            | + | + | + | + |
| $\Delta V_Z$                                                                                                                   | +            | + | + | + | + |
| $\Delta V_F$                                                                                                                   | +            | + | + | + | + |
| $\Delta V_R$                                                                                                                   | +            | + | + | + | + |
| $\Delta V_D$                                                                                                                   | +            | + | + | + | + |





## TPM1 ONBOARD DATA

| P36 FINAL COMP |                 |   |   | TPM1 CHART SOLN |  |  |  | ΔV <sub>D</sub> BT |  |  |  |
|----------------|-----------------|---|---|-----------------|--|--|--|--------------------|--|--|--|
| N59            | ΔV <sub>F</sub> | 0 | 0 |                 |  |  |  |                    |  |  |  |
| ΔV (LOS)       | ΔV <sub>R</sub> |   |   |                 |  |  |  |                    |  |  |  |
|                | ΔV <sub>D</sub> | 0 | 0 |                 |  |  |  |                    |  |  |  |

## TPM2 ONBOARD DATA

| P36 FINAL COMP |                 |   |   | TPM2 CHART SOLN |  |  |  | ΔV <sub>D</sub> BT |  |  |  |
|----------------|-----------------|---|---|-----------------|--|--|--|--------------------|--|--|--|
| N59            | ΔV <sub>F</sub> | 0 | 0 |                 |  |  |  |                    |  |  |  |
| ΔV (LOS)       | ΔV <sub>R</sub> |   |   |                 |  |  |  |                    |  |  |  |
|                | ΔV <sub>D</sub> | 0 | 0 |                 |  |  |  |                    |  |  |  |

## PRE-BRAKING SWITCH LIST

|                     |                             |
|---------------------|-----------------------------|
| FDAI SCALE-5/1      | RHC PWR NORMAL(2)-AC/DC     |
| FDAI SELECT-1       | RHC PWR DIRECT(2)-MMA/MHB   |
| FDAI SOURCE-ATT SET | SC CONT/MODE-SCS/FREE       |
| ATT SET-GDC         | BIMAG MODE(3)-ATT 1/RATE 2  |
| MAN ATT(3)-RATE CMD | AUTO RCS SELECT(16)-MMA/MHB |
| LIMIT CYCLE-OFF     | TIIC-ARMED                  |
| DBD/RATE-MIN/LOW    | RHC#2-ARMED                 |
| THC PWR-ON(up)      |                             |

## GLYCOL EVAPORATOR DEACTIVATION SWITCH LIST

|                                           |
|-------------------------------------------|
| GLYCOL EVAP H2O FLOW-OFF                  |
| GLYCOL EVAP STEAM PRESS-MAN               |
| GLYCOL EVAP STEAM PRESS-INCR (for 58 sec) |

## TV PREPARATION CHECKLIST

|                            |
|----------------------------|
| TV SOURCE-CM               |
| REMOVE & STOW LENS CAP     |
| LINEAR SW-LINEAR GAM 1/2   |
| AVG SW-PEAK AVG            |
| F-STOP-16 22               |
| ZOOM-25 as required        |
| FOCUS-∞                    |
| MONITOR ON-ON              |
| (REFINE SETTINGS AS REQ'D) |

IMU FAILURE

DATE 5/21/73  
5/21/73

7+30

G  
W  
M

S BAND AUX TV-TV (2 MIN PRIOR STDN AOS)

NULL LOS RATES AND

ADJUST R TO BRAKING GATES

V76E, PRO AFTER EACH THRUSTING

DAC-ON (1000 FT), HDC -30 FRAMES (1000 FT)

~~CONFIGURE NK WITH 300 MM LENS, STOW~~~~55 MM LENS NK-10 FR (IF DESIRED)~~STATION KEEP ON SWS +X AXIS (R ~100 FT)  
(MAINTAIN ROLL ~0° WITH SUN ON BOTTOM  
OF S/C

POO

SPOT LIGHT - ON (DOWN)

S BAND AUX TV-OFF (CTR) AT STDN LOS

CONFIGURE COMM (PANEL 9)

MODE - VOX

VOX SENS - AS RECD (~6)

S-BD - T/R (VERIFY)

AUDIO CONT - NORM (VERIFY)

INTERCOM - T/R (VERIFY)

REMOVE TV CAMERA FROM BRKT AND HANDHOLD  
CM4/NK-60 FRAMES/HDC - 100 FRAMES(F8, 1/250)

MAN ATT (3) - MIN IMP

MNVR TO WING #1 (AVOID ATM SOLAR ARRAYS  
AND 45' DISCONE ANTENNAS)USE METERING SYSTEM FOR NK PHOTOS  
PHOTO DAMAGED AREAS USING NK FOR CLOSEUPS  
AND HDC FOR WIDE-ANGLES (IF VERY HIGH  
RESOLUTION IS DESIRED, LOAD BN07 IN NK)

REPLACE TV CAMERA ON BRKT

S BAND AUX TV-TV (2 MIN PRIOR STDN AOS)

SAS WING #1 TV (R ~60 ft)  
~~STOP-PC (IF TV CAMERA IMAGES SUN)~~

TAKE HDC PICTURES OUT CM4 BY TV

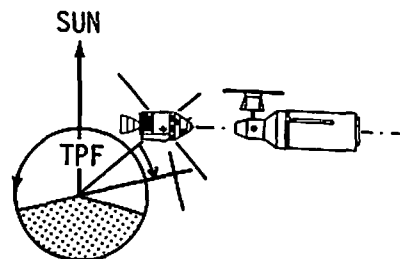
CAMERA OR CM3

TAKE NK PICTURES OUT CM3 IF POSSIBLE

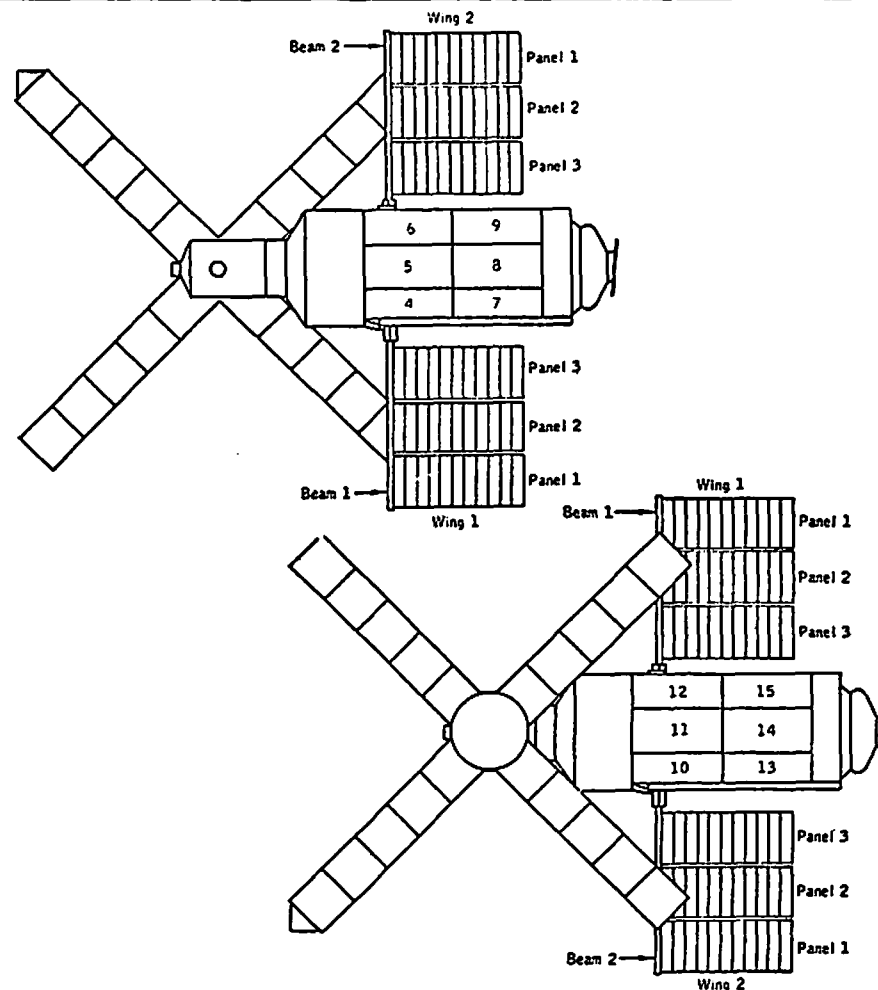
7+40

7+50

8+00

S  
G  
M

| BRAKING GATES |        |         |                    |              |       |
|---------------|--------|---------|--------------------|--------------|-------|
| R(M)          | R(FPS) | SWS DIA | RETICLE ANGLE(DEG) |              | R(FT) |
|               |        |         | SOLAR PANEL        | (BETA - 10°) |       |
| 1.00          | 30     | .2      | 8                  | 6000         |       |
| .50           | 20     | .4      | 17                 | 3000         |       |
| .25           | 10     | .8      | 35                 | 1500         |       |
| .08           | 5      | 2.5     | 10.8               | 500          |       |
| .05           |        | 4.2     |                    | 300          |       |
| .03           |        | 6.3     |                    | 200          |       |
| .02           |        | 10.0    |                    | 130          |       |





8+00

G  
D  
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T  
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L  
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A

WHEN SAS WING #1 TV COMPLETE -  
MNVR TO WING #2 (AVOID ATM SOLAR ARRAYS)

SAS WING #2 TV (R ~60 ft)

8+10

WHEN SAS WING #2 TV COMPLETE -  
MNVR TO SWS +X AXIS

8+20

S BAND AUX TV-OFF (CTR) AT STDN LOS  
MAN ATT (3) - RATE CMD  
MNVR TO DOCKING ATT (Pg 6-17)  
DISASSEMBLE AND STOW TV CAMERA, ZOOM LENS,  
CABLES, MONITOR, CAMERA MOUNT, NK, AND  
HDC (TSB)  
RECONFIGURE DAC - 6 FPS FOR SOFT DOCKING  
PERFORM PRE-SOFT DOCKING SWITCH LIST  
(Pg 6-17)  
CSM/SWS SOFT DOCKING; PERFORM SOFT  
DOCKING CHECKLIST (Pg 6-17)

GO TO LAUNCH CHECKLIST FOR  
SEVA PROCEDURES

8+30

IMU FAILURE

## DOCKING ATTITUDE

|            |   | NOMINAL |   |   |     | UPDATE |   |   |  |
|------------|---|---------|---|---|-----|--------|---|---|--|
| ASCP<br>tw | R | X       | X | X | 191 | X      | X | X |  |
|            | P | X       | X | X | 035 | X      | X | X |  |
|            | Y | X       | X | X | 006 | X      | X | X |  |

## PRE-SOFT DOCKING SWITCH LIST

|                                   |                                    |
|-----------------------------------|------------------------------------|
| cb DOCK PROBE(2)-CLOSE            | SPOT LIGHT-ON(down)                |
| DOCKING PROBE RETRACT(2)-OFF(ctr) | <del>SEGS PYRO ARM(2) ON(up)</del> |
| DOCKING PROBE EXT/REL-RETRACT     | FC REACS VALVES-LATCH              |
| DOCKING PROBE EXT/REL tb(2)-gray  |                                    |

## SOFT DOCKING CHECKLIST

## TO INITIATE CAPTURE

VERIFY STATION KEEPING ON SWS +X AXIS  
(RANGE=40 TO 100 FT) (DOCK TGT=1.8 TO 0.8° IN COAS)  
LOAD E'IS WITH ~~100.0 FPS~~ (-100.0 FPS)  
E'IS FUNC/MODE-ΔV/NORMAL  
DAC-ON  
THC-INITIATE 0.4 ±0.1 FPS CLOSING RATE  
RHC-MAINTAIN MINIMUM RELATIVE ALIGNMENT ANGLES

AT CAPTURE (IF NO CAPTURE-SEE BACKUP DOCKING PROCEDURES (S/2-1))

DOCKING PROBE EXT/REL tb(either)-bp (IF GRAY-SEE DOCKING MALFUNCTION  
PROCEDURES #2, Pg 7-3 IN CSM MALFUNCTION PROCEDURES BOOK)  
B'AG MODE(3)-RATE 2 ~~18-3~~  
RHC PWR DIRECT(2)-OFF(ctr)  
REPORT CAPTURE TO STDN  
ALLOW PROBE TO DAMP SC MOTION (10 SEC)  
THC-NULL PITCH AND YAW MISALIGNMENT<3°  
RHC-NULL ROLL MISALIGNMENT<5°  
~~DOCKING PROBE RETRACT PRIM 1 (SEC 2 IF REQ)~~

## AT DOCK LATCH

~~DOCKING PROBE EXT/REL tb(2)-gray~~

## AFTER SOFT DOCK

|                                              |                                              |
|----------------------------------------------|----------------------------------------------|
| SEGS PYRO ARM(2)-SAFE                        | E'IS FUNC/MODE-OFF/STBY                      |
| SEGS LOGIC(2)-OFF                            | DBO/RATE-MIN/LOW                             |
| cb SEGS ARM(2)-OPEN                          | THC PWR-OFF                                  |
| cb DOCK PROBE(2)-OPEN                        | RHC PWR NORMAL#1-OFF(ctr)                    |
| <del>DOCKING PROBE EXT/REL-OFF</del>         | RHC PWR DIRECT(2)-OFF(ctr)                   |
| <del>DOCKING PROBE RETRACT(2)-OFF(ctr)</del> | SC CONT/MODE-SCS/FREE                        |
| DAC-OFF                                      | B'AG MODE(3)-RATE 2                          |
| SPOT LIGHT-OFF(up)                           | AUTO RGS SELECT(A1,B2,A3,B4,B3,B4)-OFF       |
| FC REACS VALVES-NORM                         | <del>cb SCS A/C &amp; B/D ROLL(4)-OPEN</del> |
| VHF AM B-OFF(ctr)                            | <del>cb SCS PITCH &amp; YAW(4)-OPEN</del>    |
| VHF RANGING-OFF                              | THC-LOCKED                                   |
| COAS PWR-OFF                                 | RHC(2)-LOCKED                                |

# VHF, OPTICS & OTHER FAILURES

## OPTICS FROZEN

- NOTE: 1. USE P20 OPT 1 TO OBTAIN STAR ACQ ATT'S FOR P54'S AND PRE-NC1 IMU CHECK.  
 2. USE P54 (G/3-12) FOR IMU ALIGNMENTS.  
 3. USE SXT IN PLACE OF COAS WHEN USING THE ALTERNATE (COAS) MARKING ROUTINES.  
 4. USE V64 TO COMPUTE N78, REPEAT EACH TIME MINKEY IS INITIATED. USE N91 IF N91=TPAC USE TPAC'S IF N91≠TPAC'S  
 5. TO AVOID TAKING UNWANTED MARKS, BE SURE TO TERMINATE COAS MARKING ROUTINE (PRO ON F 53 45) BEFORE KEYING ANY Vxx AND/OR Nxx ENTER.

IMMEDIATELY- V16N91E, COMPARE WITH TPAC'S

OPTICS PWR-OFF

V64E, V24E, LOAD N94 WITH N91(TPAC) ANGLES, PRO-F 06 78, PRO.

SC CONT/MODE-CMC/FREE

CTR SWS(STAR) IN SXT WITH MIC (LEB). ENTER TO MARK.

BEFORE EACH MARKING PERIOD- V54E, PRO-F 53 45.  
 (V86E TO REJECT MARK)

## AC BUS 1 FAILED OR EMS RANGE DISPLAY BLANK

- NOTE: 1. CMC SOLUTIONS NOT AFFECTED BY BUS LOSS.  
 2. NO EMS CUTOFF FOR SCS BURNS.  
 FOR SCS BURNS- EMS FUNC-OFF, MAKE TIMED BURNS  
 SPS THRUST-DIRECT ON (TO INITIATE BURN)  
 ΔV THRUST-OFF (TO TERMINATE BURN)  
 FOR CHART RANGE DATA- V76E, V25E, LOAD TIME OF NEXT RANGE MARK, PRO, N77E, READ R.  
 FOR POST TPI CHART RANGE DATA- V76E, V25E, +0E, +0E, +0E, PRO, N77E, READ CURRENT R

## MARK BUTTON FAILED

- NOTE: FOR ALTERNATE PROCEDURE, SEE MARK BUTTON FAILED OPEN (EMP)(G/4-15).  
 IMMEDIATELY- OHC-DRIVE TRUN<10°, OPTICS ZERO-ZERO.  
 PERFORM OPTICS FROZEN PROCEDURE, EXCEPT LOAD N94 WITH ZEROS.

## CANNOT SEE THROUGH OPTICS

- NOTE: 1. USE P20 OPT 1 TO OBTAIN STAR ACQ ATT'S FOR P54'S AND TO PERFORM THE PRE-NC1 IMU CHECK. LOAD N78 REG 1 & REG 2 WITH ZERO'S.  
 2. USE P54 (G/3-12) FOR IMU ALIGNMENTS  
 3. VHF MARKS (V87E) REQUIRED TO UPDATE SV.  
 4. USE STDN ΔVY FOR ALL BURNS THRU TPI.  
 5. PERFORM VHF AND COAS MARKING POST TPI. TO AVOID TAKING UNWANTED MARKS, BE SURE TO TERMINATE COAS MARKING ROUTINE (PRO ON F 53 45) BEFORE KEYING ANY Vxx AND/OR Nxx ENTER.

FOR PRE-NC1 IMU CHECK (Pg ~~1-4~~ <sup>1-5</sup>)- IMU IS GO IF STAR<5° FROM CENTER OF COAS.

POST TPI (Pg ~~1-14~~ <sup>1-15</sup>)- BYPASS P36 AUTO MNVR, UPLINK ACTY LT-ON, V22N78E, +E, V58E (AUTO MNVR-SWS IN COAS). BEFORE PRE-TPM1 & TPM2 MARKING PERIODS- V54E, V24E, LOAD N94 WITH COAS LOS ANGLES RECORDED ON Pg 1-2, PRO-F 53 45, SC CONT/MODE-CMC/FREE, CTR SWS IN COAS, ENTER TO MARK (V86E TO REJECT).

READ θ FOR TPM CHART SOLN FROM ORDEAL FDAI WHILE CENTERING SWS IN COAS.

## NO VHF RANGING

- NOTE: CHART SOLUTIONS NOT POSSIBLE, DELETE V87'S & V76'S  
 IMMEDIATELY- V77E, V88E, V57E, V22E, 1E, PRO.  
 AFTER TPM1 (BEFORE 1ST MARK, Pg ~~1-15~~ <sup>1-16</sup>)- V93E  
 BEFORE THRUSTING (AFTER TPM2, Pg ~~1-15~~ <sup>1-16</sup>)- V37E47E, V83E, N83E AS REQD, CALC R & R USING COAS RANGING CHART (Pg 7-2).

## SWS TRACKER LIGHTS FAILED

- NOTE: 1. PERFORM NOMINAL RENDEZVOUS TIMELINE (SXT MARKS NOT POSSIBLE WHEN SWS IN DARKNESS).  
 2. CHART SOLN NOT POSSIBLE FOR TPM1 AND TPM2.  
 IMMEDIATELY- V57E, V22E, 1E, PRO.

4/25/73 (TYP0)  
 5/22/73 (P&I)

DATE 4/6/73  
 4/20/73 (P&I)

PAGE 7-1

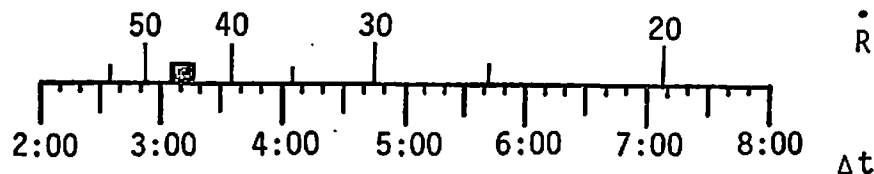
X 4 JET 1 FPS=2.5 SEC  
2 JET 1 FPS=5 SEC

# COAS RANGING CHART (USING ATM SOLAR ARRAYS)

$\alpha$  = ANGLE SUBTENDED BY THE TWO FORWARD TIPS OF THE ATM SOLAR ARRAYS

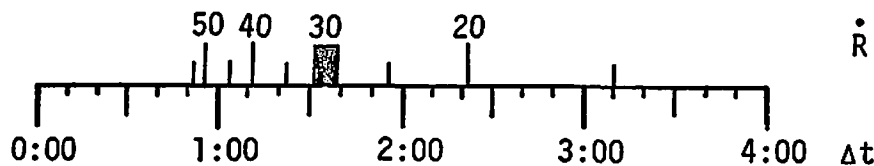
$\alpha$  .25° → .5°

R 2.8 → 1.4



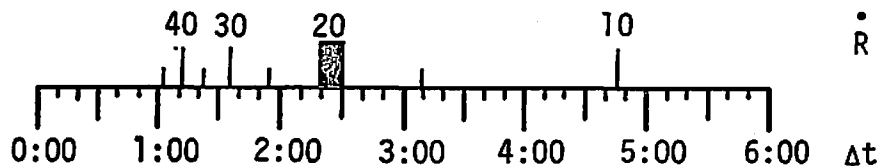
$\alpha$  .5° → .75°

R 1.4 → .94



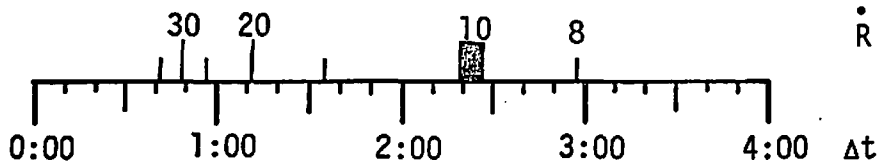
$\alpha$  .75° → 1.5°

R .94 → .47



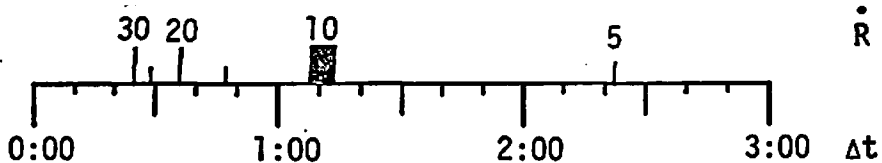
$\alpha$  1.5° → 3°

R .47 → .23



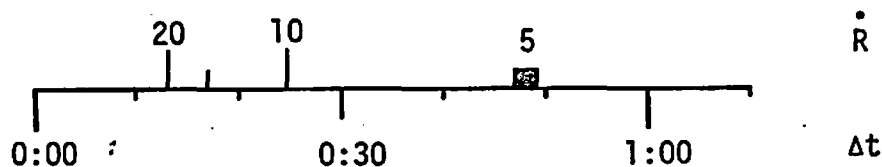
$\alpha$  3° → 6°

R .23 → .12



$\alpha$  6° → 9°

R .12 → .08



COAS RANGING CHART

NOTE: THIS CHART IS GOOD FOR BOTH THE SI ATTITUDE ( $\beta = -30^\circ$ ) AND THE BIASED SI ATTITUDE (i.e.  $40^\circ$  PITCH AND  $28^\circ$  ROLL). HOWEVER, VISIBILITY OF THE ATM SOLAR ARRAYS (PARTICULARLY THE UNDERSIDE) IS VERY SENSITIVE TO SWS LOOK ANGLES TO THE CSM AND MAY NOT BECOME USEFUL UNTIL LATE IN THE BRAKING GATES.